SahulatERP

Software Quality Assurance Test Report

Execution of the 2,000-case SQA pack across 20 feature areas

Product	SahulatERP, cloud ERP for Pakistani importer-distributors
Build tested	Branch `qa-execution`, commit `199931d` (product code identical to `main`)
Report generated	14 October 2026 at 20:02 (Asia/Karachi)
Test pack	`docs/qa/cases/*.md`, 20 files × 100 cases
Verdict	**Not ready for release**: 8 release-blocking (S1) defects open

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1. Executive summary

Not ready for release

All 2,000 cases in the pack were run or classified. 1,943 were executed automatically (97.2%). Of those, 1,829 passed and 114 failed. The remaining 57 cases need real phones, other browsers, external scanners or a second real account, and are listed in section 6. No case was left unrun.

2,000	1,829	114	57	0
Cases in the pack	Passed (94.1% of executed)	Failed (real defects)	Manual only	Not run

The 114 failing cases trace back to **76 distinct defects**. Every failing case is mapped to exactly one defect in section 4.

SEVERITY	MEANING	DEFECTS	FAILING CASES
S1	Wrong money, stock, permission or security, or data loss. Blocks release.	8	15
S2	Feature unusable, no workaround.	14	20
S3	Problem with a workaround.	26	47
S4	Cosmetic.	28	32

Release blockers (S1)

ID	DEFECT
D-01	Supplier statement adds payments to the balance
D-02	Supplier payment amount need not equal its allocations; the difference is booked as an FX gain/loss
D-03	Bill rate different from GRN rate leaves a GRNI residue
D-04	An opening-stock journal can be reversed and breaks the inventory control
D-05	Amount, quantity and FX precision is not enforced at input (the 4-dp gap)
D-06	Payroll attendance import accepts 31 days and overpays
D-07	RLS lets any member read every document in the tenant
D-08	Every non-Auditor role can change company settings, branding and modules

Features unusable without a fix (S2)

ID	DEFECT
D-09	Payroll cannot run on a real company
D-10	A stale draft payroll cannot be fixed
D-11	Imports screen is read-only
D-12	Aged balances for a past date include later receipts
D-13	Branch-limited Accountant sees an empty P&L
D-14	Foreign-currency documents print "Rs"
D-15	Reprint ignores the invoice-time tax snapshot
D-16	App-wide table CSS leaks into the print template
D-17	Retry after commit returns an empty record ID
D-18	Open redirect in `/auth/callback?next=`
D-19	Branding path traversal
D-20	CSV formula injection
D-21	Audit records the editable display name
D-22	Back after sign-out shows workspace content

What held up

- Ledger integrity: every posted journal balanced and source-unique, apart from the two S1 cases above. Receivable, inventory and commission controls reconciled after every trading sequence.
- Commission engine: every commission case passed. That covers accrual, release on cleared cash, bounce reversal, recovery through payroll and plan date overlaps.
- Tenancy through the app: the API projection scopes every role and branch correctly. The tenant-isolation gap (D-07) applies only to direct database reads with a user's session.
- Idempotency, revision conflicts and atomic commits (snapshot plus relational projection) behaved as designed under parallel load.

P1 (release-critical) cases: 395 of 411 passed, 12 failed and 4 are manual.

2. Scope and approach

2.1 Test pack

The pack has 20 feature areas of 100 cases each. Every case has a stable ID, a priority (P1 blocks release to P4 cosmetic), a type, an action with data, and an expected result quoting the product's real messages. FBR e-invoicing, real bank feeds and real customs lookups are out of scope.

TYPE	MEANING	CASES
F	Functional	883
N	Negative / validation	211
E	Edge / boundary	184
S	Security / permission	214
U	UI / UX	194
D	Data integrity / accounting	292
X	Performance / limits	22

2.2 How cases were executed

LAYER	WHAT IT EXERCISES	CASES
Engine and domain	The ERP engine in memory: commands, role and module gates, validation, postings, reports, invariants. Also covers route handlers with faked Supabase, server-rendered components, static source and CSS checks, and the real database migrations in PGlite (RLS, storage policies, atomic commit).	1632
Browser	Playwright in Chromium at 1440×900 and 390×844 (touch and mobile emulation). Runs against a development server and a production build, each with its own disposable QA company. Includes axe-core accessibility scans, print emulation and A4 PDF pagination.	311
Both layers	Engine core proven in memory, then the user-facing part confirmed in the browser.	0
Manual only	Needs real hardware, Safari or Firefox, Lighthouse, OWASP ZAP, email delivery or a second real account.	57

2.3 Rules applied to every case

- Each case is one test whose title starts with its ID, so results trace back to the pack one to one.
- A failing test means the product is wrong. Failures were re-run on their own before being counted, to rule out flaky timing or lost sessions.
- Where a case asked to "verify" a design choice, the actual behaviour was recorded (section 5), not marked as a failure.
- Where a case guessed a harmless detail wrong, such as an account code or exact wording, the case was corrected to match the product (section 5).
- Testing used only disposable QA companies. Customer data, the demo company's records and the design partner's company were never touched.

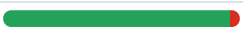
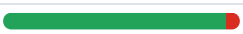
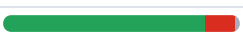
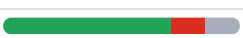
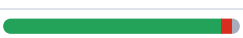
2.4 Environment

Engine pass	Vitest, Node 22, PGlite with the project's migrations
Browser pass	Playwright with Chrome, Next.js 16 development and production builds, hosted Supabase (QA tenants only)
Data	Synthetic fixtures built with the product's own engine

3. Results

Pass Fail Manual

3.1 By feature area

AREA	CASES	PASS	FAIL	MANUAL	PASS RATE*	DISTRIBUTION
Sign-in, password reset, sessions (AUTH)	100	75	7	18	91.5%	
Onboarding wizard and provisioning (ONB)	100	93	6	1	93.9%	
Settings: company & documents, features, access, audit (SET)	100	97	3	0	97.0%	
Products, opening stock, CSV import/export (PRD)	100	94	6	0	94.0%	
Customers and suppliers, walk-in names, credit limits (PTY)	100	96	3	1	97.0%	
Inventory: stock, transfers, corrections, valuation, warehouses (INV)	100	98	2	0	98.0%	
Sales documents and the invoice editor (SAL)	100	98	1	1	99.0%	
Purchasing, foreign currency, supplier payments (PUR)	100	93	7	0	93.0%	
Imports and landed cost (IMP)	100	94	6	0	94.0%	
Collections: receipts, cheques (PDC), follow-ups (COL)	100	96	4	0	96.0%	
Cash and bank (CSH)	100	96	3	1	97.0%	
Commissions (COM)	100	100	0	0	100.0%	
Payroll (PAY)	100	94	6	0	94.0%	
Accounting: journals, reversals, periods, invariants (ACC)	100	97	3	0	97.0%	
Reports and CSV export (RPT)	100	92	7	1	92.9%	
Printing, PDF and document view (PRN)	100	85	13	2	86.7%	
Shell: navigation, search, notifications, workspace switcher, Overview (NAV)	100	87	10	3	89.7%	
Security, tenancy, roles, API (SEC)	100	81	8	11	91.0%	
Responsive, mobile, accessibility, motion (UIX)	100	71	14	15	83.5%	
Data integrity, concurrency, failure recovery, migrations (DAT)	100	92	5	3	94.8%	
Total	**2000**	**1829**	**114**	**57**	**94.1%**	

* Pass rate is over executed cases (pass + fail).

3.2 By priority

PRIORITY	CASES	PASS	FAIL	MANUAL
P1	411	395	12	4
P2	1569	1416	101	52
P3	20	18	1	1
P4	0	0	0	0

3.3 By type

TYPE	CASES	PASS	FAIL	MANUAL
Functional	883	822	45	16
Negative / validation	211	203	7	1
Edge / boundary	184	167	12	5
Security / permission	214	189	11	14
UI / UX	194	149	29	16
Data integrity / accounting	292	282	6	4
Performance / limits	22	17	4	1

3.4 Failing cases

Each failing case and the defect it belongs to. One defect can fail several cases.

ID	P	CASE	EXPECTED	DEFECT
AUTH-019	P1	Sign out from the sidebar profile button	Returns to login; back button does not show workspace content	S2 **D-22** Back after sign-out shows workspace content

ID	P	CASE	EXPECTED	DEFECT
AUTH-032	P2	Forgot password with empty email	Validation; no request	S4 **D-50** Forgot password with empty email sends a request
AUTH-049	P2	Sign out in one tab, act in another tab	Other tab's next command gets 401 and the UI shows the login	S3 **D-34** 401 in another tab shows an error, not the login
AUTH-061	P2	Sign in while offline	Clear network error; retry works when online	S4 **D-49** Offline errors read "Failed to fetch"
AUTH-065	P2	Cookies	HttpOnly/secure session cookies; no token in localStorage	S3 **D-31** Session cookies readable by JavaScript
AUTH-083	P3	Loading state on first paint before hydration	Buttons disabled until hydrated; no lost clicks	S4 **D-51** Login submit enabled before hydration
AUTH-084	P2	Refresh on `/account/reset` after a successful change	Shows expired/complete state, not the form	S4 **D-52** Refreshing `/account/reset` after success shows the form again
ONB-061	P2	Reduced motion	Lines appear without stagger; ring slow	S4 **D-53** Reduced motion stops the provisioning ring
ONB-067	P2	Card on 390px phone	Fills screen; step titles collapse to numbers except current; sticky footer buttons; no horizontal scroll	S3 **D-35** Phone top bar is wider than the screen
ONB-069	P2	Documents step on phone	Preview below the form; scrollable inside card	S3 **D-35** Phone top bar is wider than the screen
ONB-077	P2	`set_modules` with unknown key `crypto`	Validation error	S4 **D-54** `set_modules` silently drops unknown keys
ONB-088	P2	Owner finishes setup while a second member is on the card in another browser	Second member's next action shows the workspace	S3 **D-48** Stale setup card in a second session
ONB-093	P2	Logo upload while offline	Clear error; can retry	S4 **D-49** Offline errors read "Failed to fetch"
SET-069	P2	Branding preview typography/colours match print	Yes	S4 **D-69** Branding preview font differs from print
SET-075	P2	Settings → Features counts "N turned on"	Matches	S4 **D-70** Features tab has no "N turned on" count
SET-085	P2	Default tax absent on new company	Tax placeholder 0 on lines	S3 **D-27** Omitted `taxRate` defaults to 18% on a company with no tax rate
PRD-014	P2	Price `abc` via API	Validation error	S3 **D-23** Raw "[DecimalError] Invalid argument" shown
PRD-016	P2	Price with 6 decimals `12.345678`	Stored at 4 dp, rounding defined (half-up)	S1 **D-05** Amount, quantity and FX precision is not enforced at input (the 4-dp gap)
PRD-031	P1	Opening qty 50, cost empty	Product saved; error "Product saved, but the opening stock was not booked: Amount must be greater than zero. …"; product exists with 0 stock	S3 **D-23** Raw "[DecimalError] Invalid argument" shown

ID	P	CASE	EXPECTED	DEFECT
PRD-035	P2	Opening qty `0.00001`	Rounds to 0 at 4 dp → treated as no stock or refused; no negative value	S1 **D-05** Amount, quantity and FX precision is not enforced at input (the 4-dp gap)
PRD-076	P2	20 MB CSV	Refused or streamed; no browser freeze	S3 **D-37** 20 MB CSV freezes the tab before being refused
PRD-095	P2	Name with only spaces	Refused (trim → <2)	S3 **D-29** Whitespace-only names are stored empty
PTY-030	P2	Salesperson creates customer	Refused via API; invoice editor "Yes, save customer" fails with message, "No" path still works	S3 **D-28** Salesperson's "No, don't save" fails on a fresh company
PTY-098	P2	10,000 customers	Suggestion datalist still usable (or capped with search)	S3 **D-36** Large data sets render slowly
PTY-099	P2	Supplier statement	Bills, returns, payments with running balance	S1 **D-01** Supplier statement adds payments to the balance
INV-063	P2	Set quantity dialog shows current qty and value	Yes	S4 **D-68** Set-quantity dialog lacks the current value
INV-086	P2	2,000 products × 3 warehouses	Stock tab renders < 2 s	S3 **D-36** Large data sets render slowly
SAL-013	P2	Qty 0.00001	Rounds to 0 → refused	S1 **D-05** Amount, quantity and FX precision is not enforced at input (the 4-dp gap)
PUR-007	P1	Currency USD, rate empty	"Amount must be greater than zero." (fxRate pos)	S3 **D-23** Raw "[DecimalError] Invalid argument" shown
PUR-020	P2	Bill rate differs from GRN rate	Difference posted to FX gain/loss (verify account)	S1 **D-03** Bill rate different from GRN rate leaves a GRNI residue
PUR-032	P1	Allocations ≠ amount	"Allocations must equal the receipt amount."	S1 **D-02** Supplier payment amount need not equal its allocations; the difference is booked as an FX gain/loss
PUR-044	P2	Supplier statement	PO not shown; GRN/bill/return/payment with running balance	S1 **D-01** Supplier statement adds payments to the balance
PUR-070	P2	Bill overdue in attention queue	Yes when past due	S3 **D-42** Overdue supplier bills not in the attention queue
PUR-085	P2	"PKR actually paid" only for non-PKR bills	Yes	S1 **D-02** Supplier payment amount need not equal its allocations; the difference is booked as an FX gain/loss
PUR-095	P2	Rate 5 decimals	Rounded to 4	S1 **D-05** Amount, quantity and FX precision is not enforced at input (the 4-dp gap)
IMP-037	P2	Customs lookup (simulated) shows GD lookup result	No "simulated" banner on the page any more; lookup still returns stub data	S3 **D-41** No customs/GD lookup (simulated)
IMP-054	P2	Add cost dialog placeholders	Amount empty with placeholder	S2 **D-11** Imports screen is read-only

ID	P	CASE	EXPECTED	DEFECT
IMP-055	P2	Classification help text	Explains each	S2 **D-11** Imports screen is read-only
IMP-056	P2	Search finds shipment by number/container	Yes	S3 **D-39** Search does not cover journals, cheques or shipments
IMP-082	P2	Shipment list sorting newest first	Yes	S3 **D-40** Missing list filters and sort
IMP-083	P2	Filter by status	Works	S3 **D-40** Missing list filters and sort
COL-053	P2	Follow-up on supplier via API	"Customer not found." (parties lookup) or refused	S4 **D-55** Follow-up accepted on a supplier
COL-075	P2	Receipts list filters by method/status	Yes	S3 **D-40** Missing list filters and sort
COL-076	P2	Search by cheque number	Found	S3 **D-39** Search does not cover journals, cheques or shipments
COL-084	P2	5,000 receipts	List/aging < 2 s	S3 **D-36** Large data sets render slowly
CSH-050	P2	Negative balance styled red	Yes	S4 **D-65** Negative bank balance not red
CSH-051	P2	Dialogs on phone	Bottom sheets	S4 **D-63** Phone sheets keep 20 px bottom corners
CSH-095	P2	Card hover lift only with mouse	Yes	S4 **D-64** Sticky hover shadow after a tap
PAY-025	P2	Reconcile = recreate run (cancel draft and create)	Documented path works	S2 **D-10** A stale draft payroll cannot be fixed
PAY-032	P2	Attendance 31 in file	Row error	S1 **D-06** Payroll attendance import accepts 31 days and overpays
PAY-062	P2	Settings → Payroll visible only with Payroll on	Yes	S2 **D-09** Payroll cannot run on a real company
PAY-063	P2	Lines table columns readable on phone	Scroll	S3 **D-44** No per-employee payroll lines view
PAY-064	P2	Status badges	Colours	S4 **D-66** Payroll status badges share a colour
PAY-094	P2	Error wording for prototype rules	No "prototype" banner in UI (removed); message text "Payroll calculated using prototype rules." still returned → flag for copy change	S4 **D-57** Internal wording visible
ACC-017	P1	Reverse an opening-balance journal	Same	S1 **D-04** An opening-stock journal can be reversed and breaks the inventory control
ACC-053	P2	Journal search by memo/number	Works	S3 **D-39** Search does not cover journals, cheques or shipments

ID	P	CASE	EXPECTED	DEFECT
ACC-091	P2	Debit/credit columns aligned, tabular numerals	Yes	S4 **D-67** Trial balance columns left-aligned
RPT-004	P2	Aged receivables as-of date in the past	Buckets recomputed for that date; receipts after date excluded	S2 **D-12** Aged balances for a past date include later receipts
RPT-011	P2	Statement print	Branded; company header; period	S3 **D-43** Report printing is unusable
RPT-012	P2	Supplier statement	Bills, returns, payments	S1 **D-01** Supplier statement adds payments to the balance
RPT-044	P2	Branch-limited Accountant: P&L	Not available or marked company-wide (journals not sent)	S2 **D-13** Branch-limited Accountant sees an empty P&L
RPT-048	P2	Print a report	Clean print CSS	S3 **D-43** Report printing is unusable
RPT-053	P2	Statement with bounced cheque	Receipt and reversal lines	S3 **D-25** Bounced cheque missing from the statement
RPT-072	P2	Sales analysis by salesperson with splits	Each gets share	S3 **D-26** Sales analysis ignores commission splits
PRN-016	P2	Qty 1,234,567	Thousands separators	S4 **D-74** Printed quantities lack thousands separators
PRN-020	P2	Draft invoice print	"DRAFT" status visible	S3 **D-24** Draft and cancelled documents print without a status
PRN-021	P2	Cancelled invoice print	"CANCELLED" visible	S3 **D-24** Draft and cancelled documents print without a status
PRN-026	P2	Purchase order print	Currency USD and rate; totals in both	S2 **D-14** Foreign-currency documents print "Rs"
PRN-039	P2	Print preview A4 portrait	Fits one page for 10 lines; page breaks for 60 lines without splitting a row	S2 **D-16** App-wide table CSS leaks into the print template
PRN-040	P2	Print preview Letter	Fits	S2 **D-16** App-wide table CSS leaks into the print template
PRN-044	P2	100-line invoice	Paginates; totals on last page	S2 **D-16** App-wide table CSS leaks into the print template
PRN-045	P2	Very long product name (120 chars)	Wraps within column	S2 **D-16** App-wide table CSS leaks into the print template
PRN-046	P2	Notes 2,000 chars	Wraps; not clipped	S2 **D-16** App-wide table CSS leaks into the print template
PRN-056	P2	Document chain shown in details (QUO → SO → DN → INV)	Links work	S4 **D-76** Document-chain rows are not links
PRN-069	P2	Tax snapshot shows registered status at invoice time	Yes	S2 **D-15** Reprint ignores the invoice-time tax snapshot

ID	P	CASE	EXPECTED	DEFECT
PRN-072	P2	Rate column formatting `Rs 2,400` / `$10.00`	Correct	S2 **D-14** Foreign-currency documents print "Rs"
PRN-091	P2	Address with newlines in header	Line breaks	S4 **D-75** Company address newlines lost in the print header
NAV-015	P1	Search "INV-2026"	Documents listed; Enter opens the first	S3 **D-38** Search dialog keyboard and record opening
NAV-026	P2	Arrow keys move selection, Escape closes	Yes	S3 **D-38** Search dialog keyboard and record opening
NAV-027	P2	Result click navigates to module and opens record	Yes	S3 **D-38** Search dialog keyboard and record opening
NAV-028	P2	Search with 20k records	< 100 ms per keystroke	S3 **D-36** Large data sets render slowly
NAV-037	P2	Metric cards stagger in	Yes; reduced motion no stagger	S4 **D-56** Metric cards never stagger
NAV-043	P2	No "Design partner", "Revision", "prototype" copy anywhere	Confirmed	S4 **D-57** Internal wording visible
NAV-054	P2	Escape closes drawer	Yes	S4 **D-59** Escape does not close the phone drawer
NAV-076	P2	Attention item "Draft sales documents to submit" opens Sales filtered to drafts	Yes	S4 **D-60** "Draft sales documents" link does not preset the filter
NAV-080	P2	Search dialog on phone	Full-width; keyboard 16px input	S4 **D-61** Phone search input is 15.2 px (iOS zooms on focus)
NAV-082	P2	Long company name in switcher	Truncated with ellipsis	S4 **D-58** Long company name wraps in the switcher
SEC-030	P1	Salesperson page load	Only own sales docs, customers, own commissions; employees salaries zero; no journals/audit/payroll	S1 **D-07** RLS lets any member read every document in the tenant
SEC-046	P2	Content-Type text/plain	400	S3 **D-32** Workspace POST accepts any content type
SEC-053	P2	CSV injection `=cmd()` in export	Prefixed/quoted on export (assert)	S2 **D-20** CSV formula injection
SEC-054	P2	Path traversal in branding upload	Ignored	S2 **D-19** Branding path traversal
SEC-066	P2	Audit records actor id, type, time, not payload secrets	Yes	S2 **D-21** Audit records the editable display name
SEC-068	P2	500 on engine exception returns generic message	And logs server-side	S3 **D-30** Unexpected errors return 400 with raw text and are not logged
SEC-089	P2	Open redirect in `next` param	Blocked	S2 **D-18** Open redirect in `/auth/callback?next=`
SEC-100	P1	End to end: eight roles each run a scripted set of allowed/denied commands	Matrix matches	S1 **D-08** Every non-Auditor role can change company settings, branding and modules

ID	P	CASE	EXPECTED	DEFECT
UIX-001	P1	Every module at 390×844	No horizontal page scroll (`scrollWidth ≤ innerWidth`)	S3 **D-35** Phone top bar is wider than the screen
UIX-003	P2	768×1024 tablet	Sidebar visible or drawer; readable	S4 **D-62** Tablet Overview overflows (799 px at 768)
UIX-005	P2	320px width	Usable; wraps	S3 **D-35** Phone top bar is wider than the screen
UIX-024	P2	Contrast: badges (green/amber/red/violet/cyan on pastels) ≥ 4.5:1	Each passes	S3 **D-47** Colour contrast failures
UIX-027	P2	Contrast: setup footer white on tinted scrim	Passes	S3 **D-47** Colour contrast failures
UIX-029	P1	Focus visible everywhere (violet ring)	Yes	S4 **D-72** Default (blue) focus ring on scroll regions
UIX-030	P2	Focus trapped in dialogs; returns to trigger on close	Yes	S3 **D-45** Focus is lost when a dialog closes
UIX-053	P2	Metric stagger 40ms	Yes	S4 **D-56** Metric cards never stagger
UIX-059	P2	Empty states have a next action	Yes	S4 **D-73** Empty states offer no next action
UIX-062	P2	Single radius scale 6/10/14/20	No stray radii	S4 **D-71** Stray border radii (3/4/5/8/9 px)
UIX-083	P2	Long names truncate with ellipsis where fixed width	Yes	S4 **D-58** Long company name wraps in the switcher
UIX-089	P2	Toast/notice readable on phone	Yes	S3 **D-46** Notice off-screen on phone
UIX-092	P2	axe-core no serious violations on every module	Yes	S3 **D-47** Colour contrast failures
UIX-099	P2	Setup card full-screen on phone	Yes	S3 **D-35** Phone top bar is wider than the screen
DAT-004	P1	Quantity precision 4 dp	0.00005 rounds consistently	S1 **D-05** Amount, quantity and FX precision is not enforced at input (the 4-dp gap)
DAT-011	P1	Retry with network error after server commit	Client retry returns original result	S2 **D-17** Retry after commit returns an empty record ID
DAT-058	P2	Browser closed mid-command	Server completes; reload shows result	S3 **D-33** Command dropped if the browser closes mid-request
DAT-060	P2	DB unreachable	500 with generic message; retry later works	S3 **D-30** Unexpected errors return 400 with raw text and are not logged
DAT-075	P2	Whitespace-only inputs	Rejected where required	S3 **D-29** Whitespace-only names are stored empty

4. Defect register

Each entry is one root cause. It lists the affected cases, what happens and what should happen, where in the code, and a suggested direction for the fix. Entries are ordered by severity; within a severity, money, stock and security come before user experience.

S1: release blockers

D-01 · S1 · Supplier statement adds payments to the balance

Cases	PUR-044, PTY-099, RPT-012.
What happens	a supplier payment increases the statement balance. In PUR-044 the statement shows 417,200 owed when 137,200 is owed.
Expected	bills raise the balance; payments and returns lower it.
Where	`src/modules/erp/reports.ts:61` `partyStatement`. The receipt amount is always negated and then multiplied by `sign` (−1 for suppliers).
Fix	sign each event by party type before applying `sign` (a supplier payment should be `+amount`), or drop the double flip.

D-02 · S1 · Supplier payment amount need not equal its allocations; the difference is booked as an FX gain/loss

Cases	PUR-032, PUR-085.
What happens	paying 5,000 against a PKR bill allocated at 4,000 is accepted, and 1,000 is posted to 5400 as an exchange loss. The "PKR actually paid" field is shown for every supplier, including all-PKR ones.
Expected	refuse with "Allocations must equal the receipt amount." unless an allocated bill is in a foreign currency.
Where	`engine.ts:136` (`create_receipt` checks equality for customers only), `engine.ts:106` `clearReceipt` (variance to 5400), `src/components/document-dialogs.tsx:152`.
Fix	require equality whenever every allocated bill is PKR, and show the field only for non-PKR bills.

D-03 · S1 · Bill rate different from GRN rate leaves a GRNI residue

Cases	PUR-020.
What happens	the GRN credits 2010 at the GRN rate, but the bill debits 2010 at the bill rate. 2010 keeps a 5,000 debit and nothing posts to 5400.
Expected	2010 cleared at the GRN value; the rate difference posted to FX (5400).
Where	`engine.ts:93`, the `supplier_invoice` posting when `parentId` is set.
Fix	debit 2010 with the GRN's booked value and post the difference to 5400.

D-04 · S1 · An opening-stock journal can be reversed and breaks the inventory control

Cases	ACC-017.
What happens	`reverse_journal` reverses the Dr 1200 / Cr 3000 journal from `bulk_stock_increase` (purpose `opening_stock`). Stock moves stay, so 1200 no longer equals the stock valuation.
Expected	refused with the "Use the source document's correction workflow…" message.
Where	`engine.ts:178`. The guard only catches document IDs, sourceIds containing ":", or memos starting "Opening"; this journal's memo is the user's reason.
Fix	tag stock-backed journals (sourceId or a flag) and refuse to reverse any journal that has stock moves.

D-05 · S1 · Amount, quantity and FX precision is not enforced at input (the 4-dp gap)

Cases	SAL-013, DAT-004, PRD-035, PRD-016, PUR-095.
What happens	values are stored as typed (price 12.345678, rate 280.12345, quantity 0.00001). Stock moves are rounded to 4 dp, so a line of 0.00005 moves 0.0001 (DAT-004). A quantity of 0.00001 is accepted (SAL-013). Opening quantity 0.00001 creates a zero-quantity move that carries value (PRD-035).
Expected	quantise to the defined precision (half-up), then refuse any quantity that rounds to 0.
Where	`engine.ts:20–22` (the `decimal`, `pos`, `nonneg` schemas never quantise) and `engine.ts:51` `move()` (`money(quantity)`).
Fix	quantise in the schemas (amount/quantity 4 dp; FX rate at a defined precision) and re-check `> 0` after rounding.

Why S1: invoiced and stocked quantities disagree. The [documented] DAT-005 case (FX stored as entered) must be decided together with this fix.

D-06 · S1 · Payroll attendance import accepts 31 days and overpays

Cases	PAY-032.
What happens	the attendance import caps at 31, but `save_employee` caps at 30. Basic pay is salary × attendance ÷ 30, so 31 days pays 31/30 of salary.
Expected	31 is a row error.
Where	`engine.ts:204` (`import_records` attendance uses `.max(31)`).
Fix	share one attendance schema (`max(30)`) between the import and `save_employee`.

D-07 · S1 · RLS lets any member read every document in the tenant

Cases	SEC-030.
What happens	a Salesperson querying PostgREST directly with the publishable key reads `goods_receipt`, purchase documents and other people's sales. The API projection is scoped correctly; the database is not.
Expected	role-scoped RLS matching `scopeWorkspace`.
Where	`supabase/migrations/202610120001_transactional_projection.sql:40–53`. The role-scoped loop omits `documents` (and likely `parties`, `receipts`, `stock_moves`), which keep the plain tenant-member policy.
Fix	add role-aware policies for the remaining projection tables, with RLS tests per role on the direct path.

D-08 · S1 · Every non-Auditor role can change company settings, branding and modules

Cases	SEC-100.
What happens	Salesperson, Recovery Officer, Warehouse Manager and Payroll Administrator can run `save_company`, `save_branding` and `set_modules` on the server.
Expected	Owner/Accountant only once setup is complete.
Where	`engine.ts:23`, the `allow` map (`save_company: ["Accountant","Sales Manager","Salesperson",…]`), which was widened for the setup wizard.
Fix	allow setup-time saves only while `setupComplete` is false; after that, restrict to Owner/Accountant.

S2

ID	TITLE	CASES	WHAT HAPPENS → EXPECTED	WHERE	FIX DIRECTION
D-09	Payroll cannot run on a real company	PAY-062	Engine demands "Settings → Payroll", which does not exist; no UI calls `save_settings` → a Payroll settings section	`engine.ts:124` `payrollConfigured`; `settings-view.tsx`	Add a Settings → Payroll form (EOBI, tax, numbering) shown while Payroll is on
D-10	A stale draft payroll cannot be fixed	PAY-025	After a commission changes, review is refused, there is no cancel command, and the month cannot be recreated → a cancel or recalculate path	`engine.ts:139–140`	Add `cancel_payroll` (draft only) or a recalculate action
D-11	Imports screen is read-only	IMP-054, IMP-055	No control to create a shipment or add or classify a cost (engine only) → add-cost dialog with classification help	`workbench.tsx:183` (imports view)	Build new-shipment and add-cost dialogs over the existing commands
D-12	Aged balances for a past date include later receipts	RPT-004	As of 30 Sep, an invoice paid on 10 Oct shows 0 → 1,000 in 1–30 days	`reports.ts:28` (current `outstanding()`)	Rebuild the open amount from allocations dated ≤ `asOf`
D-13	Branch-limited Accountant sees an empty P&L	RPT-044	Journals are withheld, so the P&L shows 0 revenue as if real → hide the ledger reports or mark them company-wide	`workbench.tsx:140` `ledgerView` ignores `branchIds`	Add `&& !actor.branchIds.length`
D-14	Foreign-currency documents print "Rs"	PRN-026, PRN-072	A USD PO prints "Rs 100" with no PKR equivalent → "$10.00" plus PKR totals	`money.ts:15` `formatMoney`; `document-preview.tsx:61–68`	Pass the currency to the formatter; add a PKR total line
D-15	Reprint ignores the invoice-time tax snapshot	PRN-069	Prints the party's current NTN/registration → the `taxSnapshot` values	`documents.ts:70`	Read `doc.taxSnapshot` for the tax ID and status
D-16	App-wide table CSS leaks into the print template	PRN-039, PRN-040, PRN-044, PRN-045, PRN-046	Totals table is 100% wide, so notes shrink to 0 px, "Balance due" is clipped on Letter and on long A4 invoices, long names don't wrap, and a 10-line invoice takes 2 pages → printed totals in full	`globals.css` `table / td` rules reach `.doc`	Scope the app table rules away from `.doc`; give totals a fixed width
D-17	Retry after commit returns an empty record ID	DAT-011	Replay returns `recordId ""` with "already completed", so the editor skips the follow-up submit → the original result	`engine.ts:127`	Store the result with the idempotency key and return it on replay

ID	TITLE	CASES	WHAT HAPPENS → EXPECTED	WHERE	FIX DIRECTION
D-18	Open redirect in `/auth/callback?next=`	SEC-089	`/\evil.com` passes the `startsWith('/')` check and lands on evil.com → stays on the same origin	`auth/callback/route.ts:10`	Resolve the URL and compare origins, or allow-list paths
D-19	Branding path traversal	SEC-054	`<tenant>/../<other>/logo.png` passes the prefix check, and paths are signed with the service role → refused	`engine.ts:186`	Accept only exact `` `${tenant}/(logo\ `` (png\ letterhead). jpg)`
D-20	CSV formula injection	SEC-053	`toCsv` writes `=cmd()`, `+`, `@` and `-` cells unchanged → neutralised	`master-data.ts:183`	Prefix risky leading characters with `'`
D-21	Audit records the editable display name	SEC-066	`actor: actor.name` → user ID plus role	`engine.ts:218`	Store `actor.userId`; show the name at read time
D-22	Back after sign-out shows workspace content	AUTH-019	Browser history cache restores `.app-shell` → login	`next.config.ts` headers; no `no-store` on app pages	Send `Cache-Control: no-store` on authenticated pages

Why: D-14, D-15 and D-16 are S2, not S1, because the ledger is correct; only the customer-facing document is wrong, with no in-app way to print it correctly. D-19 is S2, not S1, because exposure is limited to another tenant's logo or letterhead, and only if storage resolves `..`. D-12 and D-13 are S2 because the ledger and today's ageing are right; only the historical or branch view is wrong.

S3

ID	TITLE	CASES	WHAT HAPPENS → EXPECTED	WHERE	FIX DIRECTION
D-23	Raw "[DecimalError] Invalid argument" shown	PRD-014, PRD-031, PUR-007 (also SAL-009 [documented])	Zod 4 runs the chained `pos` / `nonneg` refines after the `decimal` refine fails, and `D("")` throws → "Enter a valid number…" or "Amount must be greater than zero."	`engine.ts:20-22`	Use `{abort:true}` / `superRefine`, or guard `D()` in the later refines
D-24	Draft and cancelled documents print without a status	PRN-020, PRN-021	No DRAFT/CANCELLED mark → a visible status banner	`document-preview.tsx` (ignores `view.status`)	Show a watermark or banner when status ≠ submitted
D-25	Bounced cheque missing from the statement	RPT-053	Only cleared receipts are listed → receipt plus reversal lines	`reports.ts:57`	List bounced receipts with a matching reversal
D-26	Sales analysis ignores commission splits	RPT-072	Split lines fall into "Unassigned" → 60/40 shares	`reports.ts:100` `keyOf`	Apportion line net by `splits`
D-27	Omitted `taxRate` defaults to 18% on a company with no tax rate	SET-085	API line without a rate → 18% instead of 0 (UI sends a rate)	`engine.ts:57` `lineInput`	Default from `settings.taxRate` (blank = 0); see the decisions section
D-28	Salesperson's "No, don't save" fails on a fresh company	PTY-030	Creating the shared walk-in account needs `save_party`, which is refused → the No path works	`document-editor.tsx:127`	Provision walk-in accounts at setup, or create them server-side inside `create_document`
D-29	Whitespace-only names are stored empty	PRD-095, DAT-075	`min(2)` runs before trim, so " " saves as "" → refused	`engine.ts:180-181`	`z.string().trim().min(2)`
D-30	Unexpected errors return 400 with raw text and are not logged	SEC-068, DAT-060	TypeError text, or "Unexpected request failure.", returned as 400 → 500, generic message, server log	`api/workspace/route.ts:14-22`	Map only known domain errors to 4xx; log and 500 the rest
D-31	Session cookies readable by JavaScript	AUTH-065	`@supabase/ssr` default `httpOnly:false` → HttpOnly	`src/lib/supabase/*` cookie options	Server-only session cookies, or an accepted risk with a CSP
D-32	Workspace POST accepts any content type	SEC-046	`text/plain` → 200 instead of 400	`api/workspace/route.ts:39-41`	Require `application/json`

ID	TITLE	CASES	WHAT HAPPENS → EXPECTED	WHERE	FIX DIRECTION
D-33	Command dropped if the browser closes mid-request	DAT-058	Session and snapshot are read before the body, so an aborted body loses the command; invoice stays draft → server completes it	`api/workspace/route.ts:37–39`	Read the body first (low priority: nothing is corrupted)
D-34	401 in another tab shows an error, not the login	AUTH-049	`command()` shows the message → redirect to login on 401	`workbench.tsx:146`	On 401, go to login
D-35	Phone top bar is wider than the screen	UIX-001, UIX-005, UIX-099, ONB-067, ONB-069	`.topbar-right` pushes the page to 392–431 px at 390 px (also behind the setup card; the Documents-step preview adds width) → no horizontal scroll	`globals.css` topbar rules	Let the top bar shrink: hide the date/breadcrumb and let the search collapse
D-36	Large data sets render slowly	INV-086, COL-084, NAV-028, PTY-098	Stock tab 7.6 s, Collections 6.2 s, search 300–976 ms per keystroke, 10k-customer datalist 1.5 s → under 2 s / 100 ms	Whole workspace rendered client-side; no pagination or virtualisation	Paginate or virtualise lists; cap datalist and search results
D-37	20 MB CSV freezes the tab before being refused	PRD-076	3 s main-thread block → refuse early	`csv-import-dialog.tsx`	Check size or row count before parsing; parse in a worker
D-38	Search dialog keyboard and record opening	NAV-015, NAV-026, NAV-027	Enter and arrow keys do nothing; a click opens the module, not the record → select and open	`workbench.tsx` search dialog	Add an active index and open the record dialog on select
D-39	Search does not cover journals, cheques or shipments	ACC-053, COL-076, IMP-056	Not findable by JV number, cheque number or container → found	Global search covers documents, parties and items only	Extend the search index
D-40	Missing list filters and sort	COL-075, IMP-082, IMP-083	No receipt method/status filter, no shipment status filter, shipments oldest first → filters, newest first	`workbench.tsx` Collections and Imports cards	Add the filter bar used on Sales; sort by date descending
D-41	No customs/GD lookup (simulated)	IMP-037	Not present → lookup returns stub data	Imports view	Add the simulated lookup or remove the case
D-42	Overdue supplier bills not in the attention queue	PUR-070	No payable entry → listed when past due	`workbench.tsx:57 attentionItems`	Add an overdue-bills queue
D-43	Report printing is unusable	RPT-011, RPT-048	Sidebar and grey background print, and the statement omits the customer when the parameter row is hidden → clean print with header and period	`globals.css` print rules; `reports-view.tsx`	Add `@media print` for reports and a printed title line
D-44	No per-employee payroll lines view	PAY-063	Runs show totals only → a per-employee lines table that scrolls on phone	Payroll view	Add a run-detail table
D-45	Focus is lost when a dialog closes	UIX-030	Focus drops to `<body>` → returns to the trigger	Dialog wrappers in `document-dialogs.tsx` / `workbench.tsx`	Keep a mounted trigger, or restore focus manually
D-46	Notice off-screen on phone	UIX-089	Notice renders above the heading out of view → visible after the action	`workbench.tsx` notice placement	Use a toast or sticky notice
D-47	Colour contrast failures	UIX-024, UIX-027, UIX-092	Danger badge 4.15:1, setup footer about 1.9:1, `.doc-title-meta th`, unfocusable scroll regions → 4.5:1 and axe clean	`globals.css` tokens; `.table-scroll`	Darken the danger token, make the footer scrim opaque, add `tabindex` to scroll regions
D-48	Stale setup card in a second session	ONB-088	"Setup is already complete." stays on the card → open the workspace	`setup-wizard.tsx`	On that error, reload the workspace

Why: D-23 is S3 even though PRD-031 and PUR-007 are P1: nothing wrong is saved, only the message is unreadable. D-35 is S3 because the app is still usable by scrolling sideways.

S4

ID	TITLE	CASES	WHERE / FIX DIRECTION
D-49	Offline errors read "Failed to fetch"	AUTH-061, ONB-093	Login and branding upload `catch`: map `TypeError` to a "No internet connection" message
D-50	Forgot password with empty email sends a request	AUTH-032	`login-screen.tsx` `reset()`: check the email first
D-51	Login submit enabled before hydration	AUTH-083	`login-screen.tsx`: render disabled until mounted
D-52	Refreshing `/account/reset` after success shows the form again	AUTH-084	`reset-password.tsx`: show a completed or expired state
D-53	Reduced motion stops the provisioning ring	ONB-061	`globals.css:533` global `!important` rule beats `.provision-ring`; exempt the ring
D-54	`set_modules` silently drops unknown keys	ONB-077	`engine.ts:187`: refuse unknown keys
D-55	Follow-up accepted on a supplier	COL-053	`engine.ts:193`: require `type === "customer"`
D-56	Metric cards never stagger	NAV-037, UIX-053	`workbench.tsx:218` `Metrics`: set `style={{'--i':i}}`
D-57	Internal wording visible	NAV-043, PAY-094	"Revision" label in `settings-view.tsx:34`; "prototype rules" message at `engine.ts:139`
D-58	Long company name wraps in the switcher	NAV-082, UIX-083	Sidebar switcher: add `nowrap` + `text-overflow: ellipsis`
D-59	Escape does not close the phone drawer	NAV-054	Add a keydown handler on `.sidebar.mobile-open`
D-60	"Draft sales documents" link does not preset the filter	NAV-076	Pass `status=draft` when navigating to Sales
D-61	Phone search input is 15.2 px (iOS zooms on focus)	NAV-080	`.search-dialog .search-input` overrides the 16 px phone rule; raise its specificity
D-62	Tablet Overview overflows (799 px at 768)	UIX-003	Wrap the metric grid to two columns below about 900 px
D-63	Phone sheets keep 20 px bottom corners	CSH-051 (and PRN-062 [documented])	`globals.css:454` radius beats the `:636` phone rule; reorder or raise specificity
D-64	Sticky hover shadow after a tap	CSH-095	Gate the card hover in `@media (hover:hover)`
D-65	Negative bank balance not red	CSH-050	`cash-bank-view.tsx`: add a negative class
D-66	Payroll status badges share a colour	PAY-064	Four statuses use three colours; add a distinct tone
D-67	Trial balance columns left-aligned	ACC-091	Add the `num` class to the debit/credit/net cells
D-68	Set-quantity dialog lacks the current value	INV-063	`inventory-view.tsx`: show the value next to the quantity
D-69	Branding preview font differs from print	SET-069	Preview inherits Geist; apply the print font stack to the `.doc` preview
D-70	Features tab has no "N turned on" count	SET-075	Add the count to the Features header
D-71	Stray border radii (3/4/5/8/9 px)	UIX-062	Replace legacy radii with the 6/10/14/20 tokens
D-72	Default (blue) focus ring on scroll regions	UIX-029	Add the violet `:focus-visible` to `.table-scroll` and `.branding-preview-frame`
D-73	Empty states offer no next action	UIX-059	Add a primary action to the "No records yet." variants
D-74	Printed quantities lack thousands separators	PRN-016	`documents.ts:45`: format the quantity
D-75	Company address newlines lost in the print header	PRN-091	`.doc-identity span { white-space: pre-line }`
D-76	Document-chain rows are not links	PRN-056	`document-details.tsx`: make each row open its document

5. Behaviour to decide and case corrections

5.1 Not defects, but worth deciding

- **Tax default 18% (SAL-023, and SET-085 = D-27).** `lineInput` defaults an omitted `taxRate` to 18. Decide whether the default comes from Settings, and add a Settings field for it, since `save_settings` has no UI either (see D-09).
- **Sign-out ends every session (AUTH-019/049/090, ONB-073).** `auth.signOut()` uses global scope. Decide between global and local scope (`scope:'local'`).
- **Login pre-fills `owner@sahulaterp.com` (AUTH-001).** This is fine for the demo, but it reveals an account email on production.
- **Reset demo (SET-048, SEC-058).** The route always rebuilds the demo tenant, even when it is pressed from a non-demo workspace. Hide the button outside the demo company, or confirm which company will be reset.
- **Idempotent replay returns `recordId ""` (SEC-039).** This is recorded as accepted behaviour, but it causes D-17.
- **FX rate stored as entered (DAT-005)** conflicts with the PUR-095 expectation (D-05). Pick the FX precision before fixing D-05.
- **Cash & bank reconciliation compares the account total with itself (CSH-034)**, so it can never show a variance. This is a placeholder that looks like a control.
- **Security posture:** SameSite=Lax cookies with no Origin check and `?tenant=` accepted (SEC-048); no CSP (SEC-050); no request rate limiting beyond the password throttle (SEC-060).
- **Ledger gaps:** 2400 statutory payable is never settled in-app (PAY-081/082); bank transfers have no balance check (CSH-013); future dates are accepted everywhere (INV-017, DAT-063); a landed cost has no currency (IMP-045).
- **Snapshots:** branding is not snapshotted on documents (PRN-068), and party names are not snapshotted (PTY-086). Compare with the CLAUDE.md requirement for transaction snapshots.
- **Navigation:** no URL routing for modules or reports (RPT-090, NAV-060), so Back leaves the app.
- **Perf watch:** DAT-051 (10k documents, under 4 s) passed in the final run but measured 4.27 s in the stopped DAT shard. Treat it as borderline.

5.2 Behaviour recorded where a case asked to verify (227)

ID	CASE	ACTUAL BEHAVIOUR
AUTH-001	Open `/` signed out	email field is pre-filled with owner@sahulaterp.com
AUTH-002	Sign in with valid email and password	signs in at /?tenant=QA so it lands on that company; a bare / would open the first non-demo membership
AUTH-006	Empty password	also minLength=8
AUTH-010	200-character password	no account has a 200-char password; the field and the request carry all 200 characters and Supabase answers with the generic error
AUTH-011	SQL-ish email `' OR 1=1--@x.com`	the browser email validation blocks it before any request
AUTH-012	Script tag in email field	blocked by email validation, nothing rendered
AUTH-019	Sign out from the sidebar profile button	sign-out ends every session of the account
AUTH-022	Sign in with `?tenant=<own tenant>` in the URL	login keeps pathname+search via window.location.assign
AUTH-049	Sign out in one tab, act in another tab	sign-out ends every session of the account
AUTH-068	`member:create` for an existing email	verified from script source; not run against hosted
AUTH-073	Error message wording for wrong password	shows Supabase's own message "Invalid login credentials" verbatim
AUTH-076	Email 254 chars	passes browser validation, reaches Supabase and gets the generic error
AUTH-078	Emoji as password	no account has an emoji password; the field and request keep the emoji and Supabase answers with the generic error
AUTH-088	Sign in with an email that has a plus tag `a+qa@x.com`	no a+qa account exists; the address is accepted by the form and sent unchanged, Supabase answers generically
AUTH-090	Sign out while a command is in flight	sign-out ends every session of the account
ONB-005	Name filled, address empty	the wizard refuses client-side with "Enter the company name and address."; the API only refuses at complete_setup
ONB-012	Invalid email `abc`	input type=email blocks in the browser; API refuses "Invalid email address"
ONB-073	Sign out from the wizard	sign-out ends every session of the account
ONB-075	Steps saved via API out of order (`save_setup_step` 4 before 2)	any step 1-4 is accepted in any order
ONB-088	Owner finishes setup while a second member is on the card in another browser	simulated with the same owner in a second browser context; no second member account

ID	CASE	ACTUAL BEHAVIOUR
SET-029	Password field pasted with trailing newline	no trimming; refused as "Password incorrect."
SET-048	Reset demo on a non-demo company via API	the route only ever resets the demo tenant; non-members get 403
SET-072	Accent colour white `#ffffff`	no contrast validation; white headings are a design limitation
SET-079	Hide module while that module's view is open	an open tab moves to Overview on its next successful command
SET-083	Address empty in Settings	the form blocks it (textarea required); the API save_company accepts a blank address after setup
PRD-015	Price `1e3` typed	browser number input sends 1000; the API accepts "1e3" and stores it as typed, valued as 1000
PRD-019	Unit `1000 pcs` (user puts quantity in unit)	limitation; hint "pcs, box, kg…"
PRD-021	Category blank	new-product form defaults to "General"; a cleared category saves blank and a category plan pays its default rate
PRD-059	Export Unicode	plain UTF-8 without BOM, text/csv;charset=utf-8
PRD-070	501+ rows	limit is 2,000 rows per file; 2,001 refused by the API and by the dialog "Import at most 2,000 rows at a time."
PRD-074	Semicolon-delimited file	the parser auto-detects ";" so columns map correctly rather than failing
PRD-076	20 MB CSV	the whole file is parsed on the main thread, then refused with "Import at most 2,000 rows at a time."
PTY-022	Email invalid	
PTY-046	Customer statement for Walk-in customer	
PTY-073	Count in Settings → Features (customers)	counts all
PTY-085	Credit limit exceeded on sales order	the check applies to sales invoices only
PTY-086	Customer renamed after invoices	no name snapshot on documents
PTY-087	Walk-in customer renamed by user	lookup is by name
INV-015	Set quantity 999,999,999	the resulting value 1,499,999,923,500 exceeds one trillion; the below-one-trillion rule applies to inputs only
INV-017	Date in the future	accepted (no future-date guard)
INV-032	Salesperson opens Inventory	Inventory is not in the Salesperson sidebar; the stock data is still sent (for selling) and the view has no edit/transfer buttons
INV-054	Archive a warehouse with stock	there is no warehouse archive; set_archived accepts only party/item
INV-066	Unit cost 0.00001	for 15+ units the value rounds up to 0.0002 and is accepted
INV-076	Transfer dialog warehouse selects exclude archived	warehouses cannot be archived, so there is nothing to exclude
INV-077	Transfer multiple lines	a transfer moves one product; each is guarded on its own and an over-quantity transfer changes nothing
SAL-009	Product chosen, qty empty	engine throws raw "[DecimalError
SAL-023	Tax 0 and blank	engine defaults taxRate to 18 when the field is omitted entirely
SAL-032	Salesperson submits an invoice	role gate refuses first with "Your role does not permit this action."
SAL-061	Due date before date	accepted, no ordering check
SAL-080	Escape closes editor with unsaved changes	closes immediately, no confirmation, typed data discarded
SAL-090	Delivery note print	quantities AND rates/amounts/totals are shown
SAL-093	Submit a document whose product was archived after drafting	create refuses archived products but submit of an existing draft still succeeds
PUR-004	New supplier name → "Save supplier?" Yes	Importer class, 30-day default terms
PUR-039	Payment by cheque (outgoing)	recorded as status received with no journal; follows the same deposit/clear state machine, posting on clear
IMP-023	All weights 0, basis weight	allocation refused "Allocation needs positive weights."; no fallback
IMP-034	Shipment status flow in_transit → customs → received → costed	engine has no in_transit/customs commands; new shipments start "received", adding a cost sets "received", allocation sets "costed"

ID	CASE	ACTUAL BEHAVIOUR
IMP-041	Deposit later refunded	no refund command; a manual journal Dr bank / Cr 1400 clears it
IMP-045	Cost in USD?	a cost has no currency; on a USD GRN the amount is posted as PKR as entered
IMP-049	Two shipments for the same GRN	allowed; each allocates its own costs
IMP-050	Delete cost before allocation	not supported; no delete command exists
IMP-051	Edit cost after allocation	no edit command; allocated costs are immutable
IMP-059	Allocation then supplier return	return removes stock at the bill line cost (excl. landed cost); landed cost stays on remaining units; payable reduced by bill price only
IMP-064	Turn Imports on later	nothing is provisioned; deposit/expense accounts are part of the fixed chart
IMP-085	Recoverable tax appears in tax summary/reports	no separate tax summary; shown as 1300 Recoverable tax on the balance sheet and trial balance
IMP-092	Turn off Inventory with Imports on	engine refuses unless Imports is also turned off ("Imports & costing cannot stay on without Inventory."); dropping both hides both
IMP-097	Export shipments CSV (if present)	there is no shipment CSV export
COL-026	Bounce from cleared	recovery is recorded only via payroll when commission was already paid (see COL-071)
COL-035	Journal on cheque receipt	no cheques-in-hand account
COL-046	Salesperson records receipt	refused, "Your role does not permit this action."
COL-059	Statement shows cheque only when cleared	received/deposited cheques are not shown as pending lines
COL-060	Receipt date before invoice date	allowed; aging unaffected, no crash
COL-061	Receipt in future date	allowed when the month is not locked
COL-069	Collections hidden module	with Collections off no customer receipt can be recorded
COL-079	Deposit without account id	API accepts it; clearing then posts to 1000 "Bank & cash (legacy)"
COL-080	Deposit into cash account type	allowed by the engine; UI lists only bank accounts
COL-087	Follow-up on archived customer	allowed
COL-091	Clear cheque dated earlier than deposit	allowed; journal and statement use the clear date
COL-094	Receipt row actions by state	received → Deposit/Bounced, deposited → Cleared/Bounced; cleared and bounced cheques leave "Cheques in hand" and keep only Print, so a cleared cheque cannot be bounced from the UI
CSH-009	Add accounts until limit	cap is 99 accounts, codes 1001–1099
CSH-013	Transfer more than balance	no balance check; source goes negative
CSH-025	Activity filter by date	Cash & bank activity has no date filter; Reports → General ledger filters an account by date
CSH-034	Cash/bank control reconciliation	the reconciliation row compares the account total with itself, so it can never show a variance
CSH-055	Date defaults today (Asia/Karachi)	no engine default
CSH-063	Account with 10,000 lines	no pagination; card shows latest 50 movements
CSH-064	Export activity CSV (if present)	not present in Cash & bank; use Reports → General ledger export
CSH-065	Search finds account by name	not present; search covers documents, parties and items only
CSH-066	Account details text shows on card	bank cards show details; cash cards show "Cash"
CSH-078	Memo truncation with tooltip	the memo wraps in full; no truncation or tooltip
CSH-085	Account select order	creation order everywhere, not bank-first or alphabetical
CSH-087	Account names with Urdu	id falls back to "record" slug; code assigned normally
COM-011	Category plan: Lighting = 2.5	unlisted categories fall back to the plan rate
COM-014	Tiered plan 0 = 2, 500000 = 3	marginal tiers on each line's own net base, not cumulative across invoices
COM-015	Target and multiplier	once the salesperson's month-to-date base incl. this line reaches the target, that line's commission is multiplied
COM-052	Invoice with inactive employee via API	allowed; commission accrues

ID	CASE	ACTUAL BEHAVIOUR
COM-057	Export employees	export columns are the same for all roles, values come from the projection
COM-061	Commission with Collections hidden	refused, so cash releases always have a collections path
COM-062	Commissions hidden then invoice submitted	accrual is not gated by the module
COM-066	Journal on payroll pay	2200 is cleared at payroll approval; payment settles 2300
COM-069	Rate 0	a zero entitlement row is created and no journal is posted
COM-072	Tier threshold equal to base	the amount up to the threshold earns the lower tier; only the excess earns the next tier
COM-080	Sales analysis by salesperson includes commission	not present; rows carry invoices, quantity, net, returns only
COM-084	Plan form shows only fields for its type	every type shows all fields — category, class, tier rates, target and multiplier
COM-087	Search finds employee	not present; search covers documents, parties and items only
COM-094	Employee deactivated with unpaid commission	payroll skips inactive employees, so it is not paid until reactivated
PAY-010	Gross = salary + allowance (attendance-prorated per rule)	basic = salary × attendance ÷ 30; allowance is not prorated
PAY-011	Attendance 0	basic 0, allowance still paid; with no allowance the employee EOBI makes deductions exceed earnings and the whole run is refused
PAY-026	Pay posts: Dr salary expense / Dr employer EOBI / Cr payroll payable / Cr statutory payable / commission payable cleared	expense posts at approval, payment credits legacy 1000
PAY-033	Payroll print/payslip (if present)	not present; printable kinds are sales, purchase and receipt documents only
PAY-041	Turn Payroll on later	no account provisioning needed; chart is fixed
PAY-050	Run for a future month	allowed
PAY-053	Reconciliation: payroll payable	no payroll row in the Control reconciliation
PAY-054	Cancel/delete draft run	not offered — no command exists
PAY-076	Export payroll CSV (if present)	not present in the Payroll view
PAY-077	Search finds payroll run	not present; search covers documents, parties and items only
PAY-080	P&L shows salary and employer EOBI	employer EOBI is inside 5200 Salary expense, no separate line
PAY-081	Balance sheet shows payables until settled	2400 is never settled in-app
PAY-082	Settle payroll payable via Cash & bank expense/transfer	payroll pay settles 2300 automatically; 2400 needs a manual journal because record_expense only allows 5100/5200/5300
PAY-083	Attendance 30 default	import and the form default to 30; the API requires attendance
PAY-085	Reports: payroll summary (if present)	not present; payroll appears only through P&L / ledger accounts
PAY-089	Line commission tooltip lists invoices	not present — runs show no line or commission detail and no tooltip
ACC-010	Lines with zero on both sides	the command still reports "Balanced journal posted."
ACC-012	Memo empty	an empty string is allowed; omitting memo fails validation
ACC-048	Manual journal to control accounts (1100/2000/1200)	allowed; reconciliation then shows a variance
ACC-055	Journal print/export	not present in the Accounting view; Reports offers General ledger / Trial balance CSV
ACC-056	Rounding: 3 lines of 33.3333 vs 100	manual journals are not auto-allocated
ACC-087	`commit_workspace` rejects unbalanced?	the database rejects it ("not balanced") and keeps the old revision
RPT-014	Sales analysis by salesperson	net and returns per person present; COGS, margin and commission not present
RPT-016	Sales analysis by item	margin not present
RPT-020	Aged payables total = payable control	aging excludes unbilled landed-cost charges; aging + landed-cost payable = control
RPT-024	Profit & loss for a month	no separate gross-profit line
RPT-037	Export empty report	the Export button is disabled when the table has no rows
RPT-055	Sales analysis margin after landed cost	not present; COGS incl. landed cost is visible in P&L 5000
RPT-061	Balance sheet includes cheques in hand (if asset)	an uncleared cheque posts nothing, so it is not an asset until cleared
RPT-063	Trial balance excludes zero accounts?	no option to show them

ID	CASE	ACTUAL BEHAVIOUR
RPT-065	Reconciliation after company reset	verified on a fresh company; the reset script is hosted
RPT-066	Report parameters persist while switching tabs	shared parameters persist across reports; leaving Reports resets them; no crash
RPT-070	Customer statement for customer with no activity	opening and closing 0 with no lines; the table shows an "Opening balance" row rather than "No activity"
RPT-078	Negative numbers formatting	"Rs -1,234.5" with a leading hyphen-minus
RPT-079	Totals row sticky/bold	bold with a heavy top border; not sticky
RPT-087	Direct API export endpoint (if any) auth	not present; exports are generated in the browser
RPT-090	Deep link to a report (if routed)	reports are not routed — /reports and ?report= open the app at Overview
RPT-092	Stock valuation excludes zero-stock items?	always omitted, no toggle
RPT-093	Aged payables for USD bills	no currency column
RPT-097	Report refresh after a new command in another tab	the open tab does not refresh by itself
RPT-099	Reports accessible screen-reader table headers	header cells are <th> in <thead> without an explicit scope attribute
PRN-010	Terms line	walk-in invoices show the shared walk-in account terms, here "0 days"
PRN-014	Unit "1000 pcs"	
PRN-023	Quotation print	Due row still printed from dueDate
PRN-025	Delivery note print	prices and amounts are printed too
PRN-028	Supplier bill print	Due is the date entered on the bill, not recomputed from supplier terms
PRN-032	Customer statement print	printed from Reports → statement via browser print; no /print/statement route
PRN-037	Logo URL signed and short-lived	600 s expiry; expired-link failure needs hosted storage
PRN-050	Emoji in notes	glyph rendering needs a browser
PRN-058	Details: returns list	a return appears as its credit note in "Document chain", the line shows "1 returned" and totals show "Returns / credits"
PRN-062	Details dialog on phone	the sheet keeps 20px radius on its bottom corners too — globals.css `.dialog-content{border-radius:20px}` (colour layer) overrides the phone sheet rule `…0 0`
PRN-063	Print URL with wrong tenant param but your own doc	a non-member tenant falls back to the first membership
PRN-064	Print URL without tenant param	memberships[0
PRN-067	Print in a company still in setup	print is allowed; there are no documents so any id is not found
PRN-068	Branding changed after invoice	branding is not snapshotted
PRN-070	Amount in words (if present)	not present on the template
PRN-074	Tax column `0%` vs blank	
PRN-075	Save as PDF filename suggestion	Chromium names the PDF after the tab title, which stays "SahulatERP · Your business, connected"; PrintActions' "Sales invoice <number>" title is not applied
PRN-079	Back from print returns to app state	Print opens a new tab; closing it leaves the app tab exactly as it was
PRN-080	Table header repeats on each printed page	repetition itself needs a browser
PRN-083	Print via keyboard Ctrl/Cmd+P	headless Chromium cannot show the native dialog; the page leaves the shortcut to the browser (keydown not prevented) and the browser prints the same @media print output the button uses
PRN-084	Print a document while its module is hidden	allowed; document kept
PRN-085	Statement print for walk-in customer	all walk-in documents are grouped under the one shared account
PRN-090	Invoice number barcode/QR (if present)	not present
PRN-093	Letterhead + NTN	NTN is omitted with the letterhead, consistently
PRN-100	End to end: branded invoice with logo, NTN, footer, walk-in name, partial payment and return prints correctly on A4	A4 layout needs a browser
NAV-008	Workspace switcher lists memberships with demo tag	current company carries a check icon labelled "Current company"; there is no separate highlight colour

ID	CASE	ACTUAL BEHAVIOUR
NAV-009	Switch company	target navigation intercepted and verified; the other company is not opened per the isolation rule
NAV-010	Switcher with one company	the owner account belongs to 6 companies, so a one-company menu cannot be produced without creating a user; verified the trigger shows the current company and it appears exactly once in the menu
NAV-022	Search empty string	placeholder "Invoice number, customer, SKU…" and the description; the results area stays empty, also for spaces only
NAV-023	Search 2 chars	results appear from the first character with no "type more" message; "AI" returns the first 8 documents (their type text "sales invoice" matches), so the Ali Traders customer is pushed past the 8-result cap
NAV-030	Bell menu lists items grouped; click goes to module	one row per queue labelled "{count} · {Module}" rather than grouped headings; queues are commission awaiting approval, unallocated shipment costs, open customer invoices, cheques due, follow-ups due, draft sales
NAV-057	Tabbed modules (Sales: Documents/Customers; Inventory: Stock/Products/Warehouses)	tabs expose role=tablist/tab and aria-selected; no role=tabpanel/aria-controls and no arrow-key movement between tabs
NAV-060	Browser back/forward	module switches add no history entries; Back leaves the app for the previous page and Forward reloads the same company on Overview
NAV-063	Preview as role chip and Exit	non-Owner members could not be signed in here; demo vs non-demo verified
NAV-069	Dialog overlay click	the document editor also closes on an overlay click, discarding typed lines
NAV-083	Many companies (10) in switcher	with 10 entries the menu fits within 1440×900 and 1366×768 without scrolling; the menu has no max-height, so it does not scroll itself
NAV-090	Breadcrumb "Workspace" not a link	
NAV-097	Offline then command	the notice reads the browser text "Failed to fetch"
SEC-005	RLS: query `workspace_snapshots` with anon key and user JWT for other tenant	an Owner/Accountant/Auditor can read their own tenant's snapshot; a salesperson none
SEC-010	Service-role key absent from client bundle	no build output scanned, source only
SEC-011	`NEXT_PUBLIC_*` contains only URL and publishable key	.env.example also has NEXT_PUBLIC_APP_URL
SEC-039	Same key, different payload	treated as a replay — unchanged workspace, recordId "", "This operation was already completed."
SEC-044	Payload 5 MB	no 413 at app level; field limits reject it
SEC-047	GET /api/workspace	signed-in GET returns the role-scoped workspace
SEC-048	CSRF: cross-site POST with cookies	session cookies are SameSite=Lax (@supabase/ssr default); the route has no Origin check and also accepts ?tenant= in the URL
SEC-050	Security headers (X-Frame-Options/CSP)	X-Content-Type-Options, Referrer-Policy, X-Frame-Options DENY, Permissions-Policy on all routes; no CSP; HSTS left to the host
SEC-051	XSS via every free-text field (name, notes, memo, footer, reason)	list/dialog rendering needs a browser
SEC-058	`/api/demo-reset` on non-demo	route always targets the demo tenant
SEC-060	Rate limiting on commands (200/min)	none in the app; only the 5-per-15-min password throttle exists
SEC-080	Enumerate ids by guessing (uuid)	
SEC-091	Unicode normalisation attack on names (duplicate check)	names are compared with trim + toLowerCase only; NFC "Café" and NFD "Café" are both accepted
SEC-095	Log review: no PII in logs	only "Empty workspace command body" (method/length/type) and database rejection messages are logged
UIX-016	Hover styles not stuck after tap	colour-only :hover rules on nav and buttons are not gated
UIX-031	Skip to content link (if present)	not present
UIX-047	Animations ≤ 320ms UI, ease-out	spinners, skeleton pulse, provisioning ring and the 800 ms setup reveal are longer by design
UIX-049	No layout-animating properties (width/height) except accordions	only .setup-progress animates width
UIX-054	Hover lift only pointer:fine	the nav-icon hover scale is not pointer-gated

ID	CASE	ACTUAL BEHAVIOUR
UIX-057	Money format `Rs 1,234` consistent	formatter checked; every screen needs a browser pass
UIX-061	No raw hex in components (tokens only)	only the branding accent swatch palette in branding-form.tsx
UIX-065	Gradient only on brand mark, welcome strip border, LIVE, final setup button, provisioning ring/progress	also the welcome-strip icon tile
UIX-079	Number inputs show numeric keyboard	document editor and record forms checked in source only
UIX-080	Select elements styled consistently	filter-bar selects are a compact 12.64 px / 41 px size; form selects are 14 px / 42 px; radius, border, background, colour and font are identical everywhere
UIX-084	Urdu text rendering	Geist has no Arabic script; Chromium falls back to a system Arabic font
DAT-005	FX precision 4 dp	fxRate 280.12345 is stored as entered and used at full precision; posted PKR amounts round half-up to 4 dp
DAT-013	10 concurrent invoice creations	requests are serialised by the revision check
DAT-026	Snapshot with unknown extra keys	preserved, except the retired fbr and bankLines keys which are dropped
DAT-027	Snapshot schema_version bump path	schemaVersion stays 1; additive fields are defaulted by decodeWorkspace; no version switch exists
DAT-032	Audit includes failed commands?	no
DAT-039	Numbering width 4 then 10,000th doc	a plain string sort then puts …-10000 before …-9999
DAT-042	Item rename	
DAT-048	Demo reset restores exact seed (hash)	content is identical; ids and audit timestamps are regenerated, so a raw hash differs
DAT-049	Snapshot size growth 10k documents	extrapolated from 200 invoices
DAT-050	Command latency with 10k docs	engine only, excluding network/DB
DAT-051	Page load with 10k docs	9,9xx extra documents are drafts cloned from an engine-built draft (no postings, so the ledger stays balanced); a 10k-document company cannot be built command-by-command in test time
DAT-057	Walk-in account creation then document fails	the walk-in account remains; acceptable
DAT-063	Clock skew client vs server	future dates are accepted
DAT-074	Unicode normalisation of names	not normalised; NFC and NFD forms are distinct
DAT-081	Snapshot JSON stable key order (diffability)	not normalised; insertion order from the engine
DAT-088	Playwright suites independent of order	re-runs this pack's UI file and then its print file, each alone in a child Playwright run (the reverse of the default 16 → 19 order), and compares every outcome with its tag: defect-tagged tests fail, manual-tagged tests skip, the rest pass
DAT-089	Playwright global warm-up prevents compile 500s	the server under test is a production build, so there is no on-demand compile to provoke; the warm-up is wired in the config and every route it touches answers below 500
DAT-091	No retry on 4xx	a 4xx whose body is not JSON is retried once
DAT-094	Stale tab commits from old revision	the server re-reads the latest snapshot on every POST, so the command applies on top without a conflict

5.3 Case corrections (69)

These cases guessed a harmless detail wrong, such as an account code, a number format or exact wording. The test asserts what the product actually does, and the pack should be updated to match.

ID	CASE	WHAT IS ACTUALLY TRUE
ONB-050	Provisioned accounts	the chart is a fixed list (accountNames) present for every company regardless of modules; nothing is provisioned per module
ONB-051	Provisioned number series	series are derived from existing documents on first use, not provisioned records
ONB-068	Card on iPad portrait	the Documents step stacks the preview under the form below 1000 px, so iPad portrait shows one column
ONB-090	Address with 2,000 characters	the address limit is 500 characters (form maxLength 500, API max 500)
ONB-092	Footer containing only spaces	stored as typed (not trimmed); prints an empty-looking footer paragraph
SET-024	Try to turn off Sales via API but keep Commissions	message lists modules in catalogue order: "Collections & PDCs, Commissions cannot stay on without Sales."

ID	CASE	WHAT IS ACTUALLY TRUE
SET-058	Settings reachable for every role	only Owner, Accountant and Auditor see Settings; other roles have no Settings entry
SET-068	Branding preview uses latest posted invoice if any, else sample	it uses the first submitted invoice in the list, not the latest
SET-086	Numbering prefix in settings	settings.numbering ("DP" for a new company) is stored but not used; documents use fixed kind prefixes like INV-YYYY-0001
PRD-002	Empty SKU	API error is the min-length validation (route shows "Check the highlighted information."); "SKU is required" is the CSV import message
PTY-023	Phone `0300–1234567`	party phone is not printed on documents (print shows name, city, email, NTN)
PTY-060	Import duplicate name in file	message is "X appears more than once in the file" without the "customer" prefix
PTY-093	`save_party` for another tenant's id	the id is treated as a new record in this tenant; no other tenant's data is read or changed
INV-008	Set quantity without reason	set_stock_quantity returns the min-length validation (form field is required); "Adjustment reason is required." is the stock_adjust message
INV-046	Overview attention: low stock queue	the Overview attention list has no low-stock entry; low stock shows only as Inventory badges/filter
INV-060	Movement history order	the report is chronological; Item details shows the newest 20 movements first with no running quantity
INV-072	Stock tab totals row	the Stock tab has no totals row; the total is in Reports → Stock valuation
INV-074	Archived product with stock in Stock tab	shown, but without an "archived" mark
SAL-099	Invoice for a customer in another branch (branch-limited member)	message is "This warehouse is outside your permitted branches."
PUR-018	GRN submit journal	credit goes to 2010 Goods received, not billed
PUR-031	Allocation exceeds bill outstanding	message is "Invalid invoice allocation."
PUR-034	Allocate to a bill of another supplier	message is "Invalid invoice allocation."
PUR-035	Allocate to a draft bill	message is "Invalid invoice allocation."
PUR-044	Supplier statement	GRNs are not listed either
PUR-051	PO print	totals are in the document currency only; PKR equivalent appears only via the exchange-rate line
PUR-061	Input tax on bill with tax 18	input tax account is 1300 Recoverable tax
PUR-076	Bill without GRN (direct bill)	a direct bill DOES move stock in and debits 1200
IMP-066	Amount 999,999,999,999	accepted (below one trillion); 1,000,000,000,000 is refused
IMP-069	Origin/port blank	defaults apply only when the fields are omitted; blank strings are stored blank
IMP-073	Sales analysis margin after allocation	Sales analysis has no margin column; the overview margin (summary) reflects landed cost
IMP-086	Expense classification appears in P&L under clearing/import expenses	posts to 5300 "Operating expense", not a separate clearing/import expense line
COL-005	Allocation exceeds invoice outstanding	message is "Invalid invoice allocation."
COL-006	Allocation to another customer's invoice	message is "Invalid invoice allocation."
COL-007	Allocation to a draft invoice	message is "Invalid invoice allocation."
COL-044	Receipt numbering RCT-? per year	REC-0001, REC-0002… one global series without a year, shared with supplier payments
CSH-046	Amount 999,999,999,999	accepted (limit is below one trillion); 1,000,000,000,000 is refused
CSH-048	Memo with newline	surrounding whitespace is trimmed; an inner newline is kept as typed
CSH-094	Mobile card tap opens activity	summary cards are not tappable; the account name in the Accounts list opens the activity
COM-028	Invariant earned ≥ released ≥ paid	paid may exceed released after a bounce until payroll recovers it
COM-055	Import duplicate employee	message is "Hamza Khan appears more than once in the file" without the word "employee"
COM-081	Entitlement badges earned/released/approved	states read unreleased / pending (released, awaiting approval) / approved / recoverable

ID	CASE	WHAT IS ACTUALLY TRUE
COM-091	Approve then invoice bounce	approved resets to false when the release changes
PAY-043	Employee with only commission	net = commission − income tax − employee EOBI
PAY-048	Numbering per year	PAY-YYYY-MM (one run per month), not PAY-2026-0001
ACC-008	Negative amount	rejected at validation with "Amount cannot be negative."
ACC-014	Reverse a manual journal	memo "Reversal · JV-00001"; message is the generic "Changes saved."
ACC-033	Journal numbering sequential	JV-00001 style, not J-2026-0001
ACC-054	Filter by date range	the date range lives in Reports → General ledger; Accounting lists the latest 8
ACC-090	Journal detail on phone	no separate journal detail screen; the Recent posted journals table is the journal view
RPT-038	Date pickers default to current month	range defaults to 1 January of the current year → today (year to date); as-at defaults to today
RPT-046	Report link from Overview "Needs attention"	rows open the module queue, not a report
RPT-073	Sales analysis "no salesperson" bucket	labelled "Unassigned"
RPT-076	Report headings and currency label PKR	amounts are shown with "Rs", not "PKR"
PRN-002	Logo uploaded	logo box is 22mm (≈83px) square with object-fit: contain
PRN-043	Fonts: Geist loads in print; fallback OK offline	the print template uses Arial/Helvetica with "Noto Naskh Arabic" fallback, not Geist
PRN-053	Logo transparent PNG on accent background	the logo sits on the white document header, not on an accent background
PRN-073	Discount column shows `10%` only when non-zero	"0%" is printed for zero discounts as well, not only when non-zero
PRN-078	Print page title = document number	title is "<Kind> <number>", e.g. "Sales invoice INV-…", set client-side by PrintActions
PRN-081	Page margins 15–20 mm	@page margins are 14mm sides/top and 16mm bottom, A4
NAV-019	Search employee	search covers documents, customers, suppliers and products only, as its description says; employees are never returned for any role, so salaries cannot leak
NAV-031	Attention: draft sales to submit, cheques due, low stock, follow-ups due, unallocated shipments, released commissions, overdue invoices/bills	queues are commission awaiting approval, unallocated shipment costs, open customer invoices, cheques due, follow-ups due and draft sales; there is no low-stock or overdue supplier-bill queue
NAV-038	Sales & collections chart last 6 months	weekly totals up to 12 weeks ending at the latest activity, not 6 months; bars violet (accent) and cyan (accent-2) with legend
NAV-061	Reload keeps company (via cookie) but returns to Overview	the company is kept by the ?tenant= query that stays in the URL, not by a cookie
NAV-064	Footer "SAHULAT ERP · Made for the way you do business · PKR · Asia/Karachi"	reads "SAHULAT ERP Made for the way you do business." and "PKR · Asia/Karachi" without the extra dots
SEC-002	POST with missing `x-tenant-id`	403 TENANT_REQUIRED, not 400
SEC-041	Revision conflict: two clients commit from same revision	the client does not auto-retry; it shows "Another change was saved first. Refresh and try again."
DAT-003	Allocation of 100 across 3 lines	33.3333 / 33.3333 / 33.3334 — the last line takes the remainder
DAT-014	Concurrent receipt and return on same invoice	a receipt above the outstanding is refused; a return after full payment is allowed and leaves a customer credit (negative outstanding) by design
DAT-090	Client retries once on ≥500 with same key	postCommand retries once with the same body only when the response is not JSON or a 5xx without a message; a 5xx with a message is not retried

6. Manual-only cases (57)

These cases cannot be automated in this environment. They are the checklist for a hands-on session on real devices before the first customer goes live.

ID	P	CASE	WHY MANUAL
AUTH-031	P1	"Forgot your password?" with a valid email	reset email delivery with link to /auth/callback?next=/account/reset (redirectTo verified in source)
AUTH-034	P2	Forgot password twice quickly	two quick reset requests: hosted Supabase rate limit surfaced cleanly
AUTH-036	P1	Set new password of 8 chars	Set a new 8-char password through the emailed reset link, sign in with it, old one fails
AUTH-039	P2	New password 72+ chars	72+ char new password: hosted GoTrue limit, never silently truncated
AUTH-040	P2	New password equal to old password	new password equal to old: hosted Supabase policy
AUTH-042	P2	Reuse the reset link after success	Reusing the emailed reset link after success shows the expired state
AUTH-043	P2	Reset link opened in a different browser	reset link opened in a different browser
AUTH-050	P2	Session expiry after the refresh token lapses	session expiry after refresh-token lapse returns to login
AUTH-051	P2	Two accounts in two browsers simultaneously	Two accounts in two browsers have independent sessions
AUTH-059	P2	Password manager autofill	password-manager autofill fills both fields
AUTH-066	P1	Reset-password flow for a user, then sign in as another user in same browser	Reset-password flow for one user then sign-in as another in the same browser: no leakage
AUTH-067	P2	Member created by `npm run member:create` with MEMBER_DEMO=1	member:create with MEMBER_DEMO=1 against hosted: sees own and demo company
AUTH-070	P2	`company:reset` for a member	company:reset against hosted: next sign-in shows setup card
AUTH-079	P2	20 failed logins in a minute	20 failed logins/minute: hosted Supabase rate limit surfaced
AUTH-085	P2	Reset page with another user's recovery token	reset page with another user's recovery token changes only that user
AUTH-086	P3	Browser with cookies disabled	cookies disabled: clear failure, no loop
AUTH-097	P2	Sign in, change password elsewhere, continue	password changed elsewhere: session continues until refresh fails
AUTH-100	P2	Full round trip: sign up by script → sign in → setup → sign out → forgot → reset → sign in	full round trip script sign-up → sign in → setup → sign out → forgot → reset → sign in
ONB-097	P2	Company reset by script, user already signed in	company reset by script while signed in; next navigation shows the setup card (needs hosted Supabase)
PTY-090	P2	Suggestions on iOS Safari	Suggestions on iOS Safari: datalist works or free text still saves
SAL-066	P2	Two users create invoices simultaneously	two users create invoices at once: needs real Supabase commits (code review: api/workspace/route.ts:59 returns 409 "Another change was saved first. Refresh and try again." with no
CSH-086	P2	Balance after company reset	balances after company reset are 0 — reset runs the hosted company:reset script; fresh-company zero state is covered by CSH-002
RPT-098	P2	CSV opens in Excel with correct columns	exported CSV opens in Excel with correct columns
PRN-054	P2	Print on phone (iOS Share → Print)	iPhone Share → Print layout fits without clipping
PRN-099	P2	Print page in Safari	print page in Safari matches Chrome
NAV-070	P2	First load TTI < 3 s on 4G	first load TTI < 3 s on 4G (Lighthouse)
NAV-075	P2	Hover motion only with mouse	touch device: no stuck hover after tap
NAV-095	P2	Idle for 1 hour then act	idle 1 hour then act: works or clean re-login
SEC-056	P2	Logo signed URL reuse after expiry	logo signed URL reused after its 600 s expiry fails (hosted Storage)
SEC-061	P2	Session fixation: sign in doesn't reuse pre-login session id	session fixation: sign-in issues a new session (hosted Supabase Auth)
SEC-062	P2	Logout invalidates refresh token	logout invalidates the refresh token (hosted Supabase Auth)
SEC-063	P2	Password reset token single-use	password reset token is single-use (hosted Supabase Auth)
SEC-069	P2	Dependency audit `npm audit`	npm audit: no high/critical (needs the npm registry)

ID	P	CASE	WHY MANUAL
SEC-085	P2	Vercel env has no service key exposed as NEXT_PUBLIC	Vercel env exposes no service key as NEXT_PUBLIC_*
SEC-086	P2	HTTPS only; HSTS	HTTPS only with HSTS on the deployed host
SEC-087	P2	Cookie secure flags in prod	production cookies carry Secure
SEC-096	P2	Backup/restore of snapshot table	backup/restore of the production snapshot table, then rebuild the projection
SEC-097	P2	Penetration smoke with OWASP ZAP baseline	OWASP ZAP baseline scan: no high alerts
SEC-098	P2	Auth brute force	auth brute force hits hosted Supabase limits
UIX-009	P2	Sheet content scrolls with overscroll contained	sheet content scrolls with overscroll contained (no rubber-band) on iOS
UIX-011	P2	Keyboard open on phone with sheet	phone keyboard open with sheet keeps actions reachable
UIX-019	P2	Landscape phone	landscape phone usable
UIX-043	P2	Screen reader reads setup steps and progress	screen reader announces setup steps and progress
UIX-072	P2	Windows high-contrast mode	Windows high-contrast mode keeps borders visible
UIX-073	P2	Safari desktop rendering	Safari desktop matches Chrome
UIX-074	P2	Firefox rendering	Firefox matches Chrome
UIX-075	P2	iOS Safari bottom bar overlap	iOS Safari bottom bar does not cover sticky actions
UIX-076	P2	Android Chrome keyboard resize	Android Chrome keyboard resizes the layout
UIX-077	P2	Datalist on iOS	datalist on iOS offers suggestions or free text
UIX-078	P2	Date input on iOS	date input on iOS opens the native picker
UIX-090	P2	Lighthouse accessibility ≥ 95 on login, overview, sales	Lighthouse accessibility ≥ 95 on login, overview, sales
UIX-091	P2	Lighthouse performance ≥ 80 mobile	Lighthouse mobile performance ≥ 80
UIX-094	P2	Voice control names match visible labels	voice control names match visible labels
UIX-100	P1	Full tour on iPhone hardware	full tour on iPhone hardware
DAT-023	P2	Migration dry-run on hosted copy	migration dry-run on a hosted copy
DAT-062	P2	Supabase auth slow	slow hosted Supabase auth: UI waits, no duplicate
DAT-098	P2	Vercel deploy rollback	Vercel deploy rollback: previous build reads current snapshots

7. How to reproduce

STEP	COMMAND	RESULT
Engine pass (all 20 areas)	`npm run test:qa`	Writes `reports/qa-run/vitest.json`. Open defects are failing tests tagged [defect]. This suite is kept out of `npm test`.
Browser pass	`npm run test:qa:browser`	Writes `reports/qa-run/playwright.json`. Parallel runs set `QA_TENANT_ID`, `QA_BASE_URL` and `QA_PW_JSON`.
Merge results	`npm run qa:report`	Writes `RESULTS.md` and `results.json`.
This PDF	`npx tsx scripts/qa-pdf.ts`	Writes `reports/qa-run/SahulatERP-SQA-Report.pdf`.
Reset a QA company	`npm run qa:tenant`	Touches only the disposable QA tenants.

Artefacts

- `docs/qa/README.md`: master test plan, entry and exit criteria, and invariants.
- `docs/qa/cases/`: the 2,000 cases.
- `tests/qa/`: engine-pass tests; `tests/qa-browser/`: browser-pass specs.
- `reports/qa-run/DEFECTS.md`: defect register; `RESULTS.md`: per-case results.

Appendix A. Every case and its result

Layer: E = engine pass, B = browser pass, E+B = both. Case text is as written in the pack; see section 5 for corrections.

Sign-in, password reset, sessions (AUTH): 75 pass · 7 fail · 18 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
AUTH-001	P1	F	Open `/` signed out	"Welcome back" login screen, email field, password field, "Sign in securely", "Forgot your password?"	E	Pass
AUTH-002	P1	F	Sign in with valid email and password	Lands on the member's own company (Overview or setup card); no error	B	Pass
AUTH-003	P1	N	Wrong password	Error under the form; stays on login; password field kept focusable; no page reload	B	Pass
AUTH-004	P1	N	Unknown email	Same generic error as wrong password (no account enumeration)	B	Pass
AUTH-005	P2	N	Empty email, click sign in	Browser/HTML validation blocks submit; no request sent	E	Pass
AUTH-006	P2	N	Empty password	Blocked client-side	E	Pass
AUTH-007	P2	E	Email with surrounding spaces `user@x.com`	Trimmed and accepted	B	Pass
AUTH-008	P2	E	Email in different case `USER@X.COM`	Accepted (Supabase emails are case-insensitive)	B	Pass
AUTH-009	P3	E	Password with leading/trailing spaces	Not trimmed; must match exactly what was set	B	Pass
AUTH-010	P2	E	200-character password	Accepted if it matches; no truncation	B	Pass
AUTH-011	P2	N	SQL-ish email `' OR 1=1--@x.com`	Generic failure; no 500	B	Pass
AUTH-012	P2	N	Script tag in email field	Rendered as text in any error; no XSS	B	Pass
AUTH-013	P2	E	Unicode email local part	Rejected by email validation or generic failure; no crash	B	Pass
AUTH-014	P2	F	Toggle password visibility (eye icon)	Characters shown/hidden; cursor position kept	B	Pass
AUTH-015	P3	U	Tab order	Email → password → eye → sign in → forgot link	E	Pass
AUTH-016	P2	F	Press Enter in password field	Submits the form	B	Pass
AUTH-017	P2	F	Double-click "Sign in securely"	One request; button shows spinner and is disabled while busy	B	Pass
AUTH-018	P1	F	Sign in, reload the page	Still signed in (cookie session)	B	Pass
AUTH-019	P1	F	Sign out from the sidebar profile button	Returns to login; back button does not show workspace content	B	Fail
AUTH-020	P1	S	After sign-out, open `/api/workspace` directly	401	E	Pass
AUTH-021	P1	S	After sign-out, POST a command with a stale cookie	401 and nothing changes	E	Pass
AUTH-022	P2	F	Sign in with `?tenant=<own tenant>` in the URL	Lands on that tenant, query preserved through the redirect	E	Pass
AUTH-023	P1	S	Sign in with `?tenant=<tenant you are not a member of>`	Falls back to your own company; no data from the other tenant	E	Pass
AUTH-024	P2	E	`?tenant=not-a-uuid`	Ignored; own company shown	E	Pass
AUTH-025	P2	E	`?tenant=` (empty)	Ignored	E	Pass
AUTH-026	P2	F	Member of a demo company and an own company, no `?tenant`	Lands on the own (non-demo) company	E	Pass
AUTH-027	P2	F	Member of two non-demo companies, no `?tenant`	Lands on the first by membership order; switcher lists both	E	Pass
AUTH-028	P1	F	Account with no memberships	"No workspace assigned" screen with Sign out	E	Pass
AUTH-029	P2	F	Member whose membership is inactive	Treated as no membership for that company	E	Pass
AUTH-030	P2	F	Member whose tenant snapshot is missing	"Workspace unavailable" screen, no stack trace	E	Pass
AUTH-031	P1	F	"Forgot your password?" with a valid email	Confirmation text shown; reset email arrives with link to `/auth/callback?next=/account/reset`	—	Manual
AUTH-032	P2	N	Forgot password with empty email	Validation; no request	E	Fail
AUTH-033	P2	S	Forgot password with unknown email	Same confirmation text (no enumeration)	B	Pass
AUTH-034	P2	E	Forgot password twice quickly	Second request accepted or rate-limited with a clear message; no 500	—	Manual
AUTH-035	P1	F	Open reset link	Reset page "Choose a password of at least 8 characters"	E	Pass
AUTH-036	P1	F	Set new password of 8 chars	Saved; can sign in with it; old password fails	—	Manual
AUTH-037	P1	N	New password of 7 chars	Refused with length message	B	Pass
AUTH-038	P2	N	New password empty	Blocked	B	Pass
AUTH-039	P2	E	New password 72+ chars	Accepted or refused with a clear message; never silently truncated	—	Manual
AUTH-040	P2	E	New password equal to old password	Accepted (no history policy) or clear refusal	—	Manual
AUTH-041	P1	N	Open reset page without a valid session	"Expired" state with link back to login	B	Pass
AUTH-042	P2	N	Reuse the reset link after success	Expired state	—	Manual
AUTH-043	P2	E	Reset link opened in a different browser	Works (token in link) or expired state; never a blank page	—	Manual
AUTH-044	P2	F	`/auth/callback` without `code`	Redirects to login without error	E	Pass
AUTH-045	P2	N	`/auth/callback?code=garbage`	Redirects to login; no 500	E	Pass
AUTH-046	P2	S	`/auth/callback?next=https://evil.com`	Open redirect refused; stays on the app	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
AUTH-047	P2	S	`/auth/callback?next=//evil.com`	Same	E	Pass
AUTH-048	P3	F	`/auth/signout` via GET	Method not allowed or no-op; POST only signs out	E	Pass
AUTH-049	P2	F	Sign out in one tab, act in another tab	Other tab's next command gets 401 and the UI shows the login	B	Fail
AUTH-050	P2	F	Session expiry after the refresh token lapses	Next navigation returns to login; no infinite spinner	—	Manual
AUTH-051	P2	F	Two accounts in two browsers simultaneously	Independent sessions; each sees only its memberships	—	Manual
AUTH-052	P2	F	Same account in two tabs	Both work; commands from both apply	B	Pass
AUTH-053	P3	U	Login page at 390px wide	Single column, inputs 16px font, no zoom on focus, no horizontal scroll	B	Pass
AUTH-054	P3	U	Login right-hand panel at <900px	Hidden; form fills the width	E	Pass
AUTH-055	P3	U	Brand mark and "SahulatERP" wordmark	Gradient mark, violet "ERP"	E	Pass
AUTH-056	P3	U	Benefit list on the right	Three plain lines, no fake statistics	E	Pass
AUTH-057	P3	U	No "Working prototype" or "simulated" copy anywhere on login	Confirmed absent	E	Pass
AUTH-058	P3	U	Reduced-motion OS setting	No slide animations on login	B	Pass
AUTH-059	P2	F	Password manager autofill	Both fields filled; sign in works	—	Manual
AUTH-060	P2	F	Browser back after signing in	Does not return to the login form with the app signed in	B	Pass
AUTH-061	P2	E	Sign in while offline	Clear network error; retry works when online	B	Fail
AUTH-062	P2	E	Sign in while Supabase auth is slow (>10s)	Spinner persists; no duplicate submit; eventually resolves	B	Pass
AUTH-063	P2	S	Inspect network: password sent only over HTTPS to Supabase	No password in URL or logs	B	Pass
AUTH-064	P2	S	Response headers of `/`	No service-role key or secrets in HTML/JS	E	Pass
AUTH-065	P2	S	Cookies	HttpOnly/secure session cookies; no token in localStorage	E	Fail
AUTH-066	P1	S	Reset-password flow for a user, then sign in as another user in same browser	No cross-user leakage	—	Manual
AUTH-067	P2	F	Member created by `npm run member:create` with MEMBER_DEMO=1	Can sign in; sees own company and demo company	—	Manual
AUTH-068	P2	F	`member:create` for an existing email	Password updated; no second company created	E	Pass
AUTH-069	P2	N	`member:create` without MEMBER_PASSWORD	Script fails with a clear message	E	Pass
AUTH-070	P2	F	`company:reset` for a member	Next sign-in shows setup card; account unchanged	—	Manual
AUTH-071	P2	N	`company:reset` for an email with no account	Script error "No account for …"	E	Pass
AUTH-072	P2	N	`company:reset` for a member with two own companies	Script refuses ("Expected exactly one own company")	E	Pass
AUTH-073	P3	U	Error message wording for wrong password	Plain English, no Supabase internals	B	Pass
AUTH-074	P3	U	Focus ring on inputs	Violet ring, visible with keyboard	B	Pass
AUTH-075	P3	U	Dark-mode OS setting	App stays light (light-only product); theme-color matches	E	Pass
AUTH-076	P2	E	Email 254 chars	Accepted by validation or clear error	B	Pass
AUTH-077	P2	E	Paste email with newline	Newline stripped or rejected; no crash	B	Pass
AUTH-078	P3	E	Emoji as password	Works if it was set that way	B	Pass
AUTH-079	P2	S	20 failed logins in a minute	Rate limit or lockout message from Supabase surfaced cleanly	—	Manual
AUTH-080	P2	F	Auditor signs in to a company still in setup	"… is being set up" screen with Sign out, never the wizard	E	Pass
AUTH-081	P2	F	Non-owner role signs in to a company in setup	Setup card shown (any member can finish setup)	E	Pass
AUTH-082	P2	F	Sign-in landing when `?tenant` is a demo company you belong to	Demo company opens; sidebar shows "Demo workspace"	E	Pass
AUTH-083	P3	U	Loading state on first paint before hydration	Buttons disabled until hydrated; no lost clicks	E	Fail
AUTH-084	P2	F	Refresh on `/account/reset` after a successful change	Shows expired/complete state, not the form	B	Fail
AUTH-085	P2	S	Reset page with another user's recovery token	Only that token's user is changed	—	Manual
AUTH-086	P3	E	Browser with cookies disabled	Login fails with a clear message; no loop	—	Manual
AUTH-087	P3	E	Browser with JavaScript disabled	Static login renders; submit does nothing harmful	E	Pass
AUTH-088	P2	F	Sign in with an email that has a plus tag `a+qa@x.com`	Works	B	Pass
AUTH-089	P3	U	Forgot-password confirmation copy	Says to check email; does not reveal whether the account exists	E	Pass
AUTH-090	P2	F	Sign out while a command is in flight	Command may complete server-side; UI returns to login without error	B	Pass
AUTH-091	P2	S	`/api/branding/logo` while signed out	401	E	Pass
AUTH-092	P2	S	`/print/sales_invoice/<id>` while signed out	Login or 401, never the document	E	Pass
AUTH-093	P2	F	Deep link `/print/...` then sign in	After sign-in the user reaches the app (returning to the deep link is acceptable)	B	Pass
AUTH-094	P3	U	Page title on login	"SahulatERP · Your business, connected"	E	Pass
AUTH-095	P3	U	Favicon present	Yes	E	Pass
AUTH-096	P2	E	Very slow network (3G throttle)	Login usable; no duplicate submissions	B	Pass
AUTH-097	P2	F	Sign in, change password elsewhere, continue	Existing session continues until refresh fails, then login	—	Manual
AUTH-098	P2	S	Member removed from a company while signed in	Next load falls back to remaining companies or "No workspace assigned"	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
AUTH-099	P3	U	Screen reader on login	Inputs labelled "Email address", "Password"; button text read	E	Pass
AUTH-100	P2	F	Full round trip: sign up by script → sign in → setup → sign out → forgot → reset → sign in	All steps succeed end to end	—	Manual

Onboarding wizard and provisioning (ONB): 93 pass · 6 fail · 1 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
ONB-001	P1	F	Sign in to a fresh company	Centred setup card over blurred dashboard; "Step 1 of 4"; stepper shows Company highlighted	E	Pass
ONB-002	P1	F	Company name + address, Continue	Step 2 Documents; progress bar grows to 50%	E	Pass
ONB-003	P1	N	Continue with empty name	"Enter the company name and address."	E	Pass
ONB-004	P1	N	Name of 1 character	Same refusal	E	Pass
ONB-005	P1	N	Name filled, address empty	Same refusal	E	Pass
ONB-006	P2	E	Name of exactly 2 characters	Accepted	E	Pass
ONB-007	P2	E	Name 120+ characters	Accepted or trimmed with clear limit; invoice preview does not overflow	E	Pass
ONB-008	P2	E	Name with emoji and Urdu script	Saved and printed correctly	E	Pass
ONB-009	P2	E	Name with leading/trailing spaces	Trimmed	E	Pass
ONB-010	P2	E	Address with multiple lines	Preserved on documents	E	Pass
ONB-011	P2	F	Email prefilled with the signed-in email	Yes, editable	E	Pass
ONB-012	P2	N	Invalid email `abc`	Browser validation or saved as text; printed as typed	E	Pass
ONB-013	P2	F	Phone `+92 300 1234567`	Saved; shown on invoice header	E	Pass
ONB-014	P2	F	NTN optional field left blank	Allowed; "NTN / STRN" line absent on print	E	Pass
ONB-015	P2	F	Close the tab after step 1, sign in again	Resumes at step 2 with company saved	E	Pass
ONB-016	P2	F	Back from step 2 to step 1	Fields show saved values; slide-left transition	B	Pass
ONB-017	P1	F	Step 2: upload PNG logo under 1 MB	Preview updates; "Replace logo" button appears	E	Pass
ONB-018	P1	F	Upload JPG logo	Accepted	E	Pass
ONB-019	P2	N	Upload a 3 MB PNG as logo	Refused with size message	E	Pass
ONB-020	P2	N	Upload a PDF as logo	Refused by type	E	Pass
ONB-021	P2	N	Upload SVG as logo	Refused (no SVG by product decision)	E	Pass
ONB-022	P2	N	Upload a file renamed `.png` that is actually text	Refused or broken-image handled; no crash	E	Pass
ONB-023	P2	F	Upload letterhead (≤2 MB)	Replaces whole header in preview	E	Pass
ONB-024	P2	N	Letterhead 5 MB	Refused	E	Pass
ONB-025	P2	F	Pick each accent colour swatch	Preview heading/table colour changes immediately	E	Pass
ONB-026	P2	F	Custom colour input	Hex saved; preview updates	E	Pass
ONB-027	P2	N	Custom colour `zzzzzz`	Rejected or ignored; no broken CSS	E	Pass
ONB-028	P2	F	Footer text with bank details	Appears at the bottom of the preview	E	Pass
ONB-029	P2	E	Footer 500 chars	Wraps; no overflow	E	Pass
ONB-030	P2	E	Footer with `<script>`	Rendered as text	E	Pass
ONB-031	P2	F	Continue on step 2 with nothing changed	Allowed (standard template)	E	Pass
ONB-032	P2	F	Replace logo twice	Latest shown; old file removed from storage	E	Pass
ONB-033	P2	S	Another tenant's member requests this logo URL	401	E	Pass
ONB-034	P1	F	Step 3: presets "All features", "Importer-distributor", "Just the basics" shown as pills	Yes, with descriptions on desktop	E	Pass
ONB-035	P1	F	Click "All features"	Every module on, including Payroll	E	Pass
ONB-036	P1	F	Click "Importer-distributor"	8 modules on (no Payroll)	E	Pass
ONB-037	P1	F	Click "Just the basics"	Sales, Purchasing, Inventory, Cash & bank, Reports	E	Pass
ONB-038	P1	F	Basics, then turn on Commissions	Collections turns on automatically (dependency)	E	Pass
ONB-039	P1	F	Turn on Imports & costing	Purchasing and Inventory turn on	E	Pass
ONB-040	P1	F	Turn off Sales while Commissions on	Commissions and Collections turn off; warning text shown on cards	E	Pass
ONB-041	P1	N	Turn everything off, Continue	"Turn on at least one feature."	E	Pass
ONB-042	P2	F	Active preset pill highlights when selection matches exactly	Yes; deselects when any switch changes	E	Pass
ONB-043	P2	U	Each module tile has coloured icon, benefit line, description, "Will set up" line	Yes	E	Pass
ONB-044	P2	U	Selected tile has violet outline and lifted shadow	Yes	B	Pass
ONB-045	P2	F	Keyboard: Tab to switch, Space toggles	Works; `role=switch` with `aria-label "<Module> module"`	E	Pass
ONB-046	P1	F	Step 4 Review shows company, documents, features with Edit buttons	Yes; "N turned on" count correct	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
ONB-047	P2	F	Edit from Review → step 1, change name, Continue through	Review reflects change	B	Pass
ONB-048	P1	F	"Create my workspace"	Provisioning screen: ring, "Setting up {company}", staggered module lines; stays ≥2.6s	B	Pass
ONB-049	P1	F	After provisioning	Dashboard unblurs; Overview visible; modules in sidebar match selection	E	Pass
ONB-050	P1	D	Provisioned accounts	Chart has receivable/revenue/output-tax (Sales), payable/input-tax (Purchasing), inventory/COGS (Inventory), bank & cash (Cash & bank) per "Sets up" text	E	Pass
ONB-051	P1	D	Provisioned number series	QUO/SO/DN/INV/CN, PO/GRN/BILL/PR present as enabled	E	Pass
ONB-052	P1	D	Inventory on with no warehouse	A first branch and warehouse created ("Main store")	E	Pass
ONB-053	P1	D	Cash & bank on	A main bank account and cash-in-hand created	E	Pass
ONB-054	P2	D	Commissions on	A draft "Standard plan" exists	E	Pass
ONB-055	P1	N	Call `complete_setup` twice	Second: "Setup is already complete."	E	Pass
ONB-056	P1	N	`complete_setup` via API with no modules	"Turn on at least one feature."	E	Pass
ONB-057	P1	N	`complete_setup` via API with blank company	"Enter the company name and address first."	E	Pass
ONB-058	P2	F	Provisioning fails (simulate network drop)	Card returns to Review with error; no partial module state visible	E	Pass
ONB-059	P2	F	Press "Create my workspace" twice quickly	One provisioning; button disabled while busy	E	Pass
ONB-060	P2	U	Provisioning lines equal selected modules + "Opening your books" + "Preparing your workspace"	Yes	B	Pass
ONB-061	P2	U	Reduced motion	Lines appear without stagger; ring slow	B	Fail
ONB-062	P1	S	Payroll off: POST `create_payroll`	400 "Payroll is turned off for this company. Turn it on in Settings → Features."	E	Pass
ONB-063	P1	S	Imports off: `create_shipment`	Refused with the module-off message	E	Pass
ONB-064	P2	F	Sidebar after basics	No Commissions, Imports, Payroll, Collections entries	E	Pass
ONB-065	P2	F	Header action buttons respect modules	No "Record receipt" when Collections off	E	Pass
ONB-066	P2	F	Search excludes records of off modules	Yes	B	Pass
ONB-067	P2	U	Card on 390px phone	Fills screen; step titles collapse to numbers except current; sticky footer buttons; no horizontal scroll	B	Fail
ONB-068	P2	U	Card on iPad portrait	Centred with scrim; both columns on Documents step	B	Pass
ONB-069	P2	U	Documents step on phone	Preview below the form; scrollable inside card	B	Fail
ONB-070	P2	U	Backdrop	Dashboard visible but blurred; not clickable; not focusable with Tab (`inert`)	B	Pass
ONB-071	P2	U	Footer under card	"Setting up {company} as {name}", switch-company links, Sign out	E	Pass
ONB-072	P2	F	Switch to another company from the footer	Navigates with `?tenant`; wizard state of the first is kept	B	Pass
ONB-073	P2	F	Sign out from the wizard	Login screen	B	Pass
ONB-074	P2	F	Escape key on the card	Does nothing (setup cannot be dismissed)	B	Pass
ONB-075	P2	E	Steps saved via API out of order (`save_setup_step` 4 before 2)	Accepted or refused consistently; wizard opens at the saved step without crashing	E	Pass
ONB-076	P2	E	`save_setup_step` with step 9	Validation error	E	Pass
ONB-077	P2	E	`set_modules` with unknown key `crypto`	Validation error	E	Fail
ONB-078	P2	E	`set_modules` with `["commissions"]` only	Server adds sales and collections (requirements)	E	Pass
ONB-079	P2	F	Step 1 while busy (Continue pressed)	Inputs disabled; spinner on button	E	Pass
ONB-080	P2	F	Hydration: click Continue immediately after load	Click not lost (button disabled until hydrated)	E	Pass
ONB-081	P2	F	Reload during step 3 after toggling but before Continue	Toggles reset to last saved set	E	Pass
ONB-082	P2	F	Company name change after setup (Settings)	Not blocked by wizard state	E	Pass
ONB-083	P2	U	Progress bar widths 25/50/75/100%	Yes, animated	E	Pass
ONB-084	P2	U	Stepper done steps show check marks	Yes	E	Pass
ONB-085	P2	U	Step transition forward slides from right, back from left	Yes	B	Pass
ONB-086	P2	D	Audit history after setup	Entries for save_company, save_branding, set_modules, save_setup_step, complete_setup	E	Pass
ONB-087	P2	D	Revision increments once per step save	Yes	E	Pass
ONB-088	P2	F	Owner finishes setup while a second member is on the card in another browser	Second member's next action shows the workspace	B	Fail
ONB-089	P2	F	Auditor opens a company mid-setup	"… is being set up" screen	E	Pass
ONB-090	P2	E	Address with 2,000 characters	Accepted up to limit; preview truncates gracefully	E	Pass
ONB-091	P2	E	Tax id with letters and dashes `ABC-123-X`	Saved as typed	E	Pass
ONB-092	P2	E	Footer containing only spaces	Treated as empty	E	Pass
ONB-093	P2	F	Logo upload while offline	Clear error; can retry	B	Fail
ONB-094	P2	S	Upload path traversal filename `../../x.png`	Stored under tenant path; filename ignored	E	Pass
ONB-095	P2	F	Branding saved on step 2 appears in Settings → Company & documents	Yes	E	Pass
ONB-096	P2	F	Invoice printed after setup	Logo, accent colour, footer, NTN all as entered	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
ONB-097	P2	F	Company reset by script, user already signed in	Next navigation shows setup card again	—	Manual
ONB-098	P2	U	Demo company never shows setup card	Correct (demo seeded complete)	E	Pass
ONB-099	P2	X	Setup with All features on a slow connection	Provisioning completes; UI never shows a half-provisioned sidebar	B	Pass
ONB-100	P1	F	End to end: fresh tenant → all four steps → Create → first invoice printed	All pass with no console errors	E	Pass

Settings: company & documents, features, access, audit (SET): 97 pass · 3 fail · 0 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
SET-001	P1	F	Open Settings	Tabs: Company & documents, Features, Access, Audit history	E	Pass
SET-002	P1	F	Change company name, Save changes	"Company details and document branding saved."; sidebar switcher updates	E	Pass
SET-003	P1	N	Clear company name, Save	Refused (name ≥ 2)	E	Pass
SET-004	P2	F	Save button disabled until something changes	Yes	E	Pass
SET-005	P2	F	Change NTN, Save, print invoice	"NTN / STRN: …" printed	E	Pass
SET-006	P2	F	Change accent colour, Save	Preview updates live; print uses new colour	E	Pass
SET-007	P2	F	Replace logo in Settings	New logo in preview and print	E	Pass
SET-008	P2	F	Remove logo	Preview falls back to company initial	E	Pass
SET-009	P2	F	Upload letterhead, then remove	Header returns to logo + name	E	Pass
SET-010	P2	N	Logo > 1 MB	Refused	E	Pass
SET-011	P2	E	Footer emptied	Printed documents show no footer line	E	Pass
SET-012	P2	F	Edit, then switch tab without saving	Changes lost without crash (acceptable) or kept; no partial save	B	Pass
SET-013	P2	F	Auditor opens Company & documents	Read-only; no Save button	E	Pass
SET-014	P2	F	Salesperson saves company name	Allowed (all roles may edit company/branding per access list)	E	Pass
SET-015	P1	F	Features tab lists all modules with switches and record counts	Yes	E	Pass
SET-016	P1	F	Turn on an off module (e.g. Payroll)	No password; switch on; sidebar gains Payroll immediately	E	Pass
SET-017	P1	F	Turn off a module	Dialog "Hide {Module}?" with record counts, password field	E	Pass
SET-018	P1	N	Wrong password	"Password incorrect." stays open	E	Pass
SET-019	P1	F	Correct password	Module hidden; sidebar entry gone; records kept	E	Pass
SET-020	P1	S	`set_modules` via API removing a module without `confirmPassword`	401	E	Pass
SET-021	P1	S	`set_modules` via API with wrong password	401	E	Pass
SET-022	P1	F	Turn module back on	Records and settings exactly as before	E	Pass
SET-023	P1	F	Turn off Sales with Commissions on	Dialog lists Collections and Commissions as also hidden	E	Pass
SET-024	P1	N	Try to turn off Sales via API but keep Commissions	"Commissions, Collections & PDCs cannot stay on without Sales."	E	Pass
SET-025	P2	F	Any role (not only Owner) can confirm with their own password	Yes by product decision	E	Pass
SET-026	P2	N	Auditor tries to toggle	Switches disabled	E	Pass
SET-027	P2	F	Cancel the hide dialog	Switch stays on	B	Pass
SET-028	P2	F	Press Escape on the hide dialog	Closes, no change	B	Pass
SET-029	P2	E	Password field pasted with trailing newline	Trimmed or refused cleanly	E	Pass
SET-030	P2	S	Password sent only in the command body over HTTPS	Not in URL, not logged in audit	E	Pass
SET-031	P2	F	Hidden module's drafts	"Drafts inside a hidden module cannot be submitted until it is turned back on."	E	Pass
SET-032	P1	S	Submit a sales draft via API while Sales is hidden	Refused with module-off message	E	Pass
SET-033	P2	F	Hidden module's records in search	Not shown	B	Pass
SET-034	P2	F	Hidden module's accounts in Accounting	Still present (books are never hidden)	E	Pass
SET-035	P2	F	Reports when Collections hidden	Receivables still report (accounting data)	E	Pass
SET-036	P2	F	Hide Reports	Sidebar Reports gone; Overview still works	E	Pass
SET-037	P2	F	Hide Cash & bank with balances	Allowed (hidden, not deleted); Accounting balances unchanged	E	Pass
SET-038	P2	F	Record counts in the dialog	Match the actual lists (documents, customers, products…)	E	Pass
SET-039	P2	F	Presets not shown in Settings	Correct (`showPresets=false`)	E	Pass
SET-040	P2	F	Tiles in Settings use the same colours as sidebar	Yes	B	Pass
SET-041	P2	U	Dialog backdrop blurred and tinted	Yes	B	Pass
SET-042	P2	U	Dialog on phone	Bottom sheet; password input 16px	B	Pass
SET-043	P2	F	Access tab: company name, tax id, revision	Correct	B	Pass
SET-044	P2	F	"View as role"	Enabled only for Owner in a demo company	B	Pass
SET-045	P2	F	View as Salesperson	Sidebar reduces; "Previewing as Salesperson" chip; Exit preview	B	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
SET-046	P1	S	While previewing as Salesperson, POST an owner-only command	Allowed server-side (real role) — preview is visual only, and the UI says so	E	Pass
SET-047	P2	F	Reset demo (demo company, Owner)	Confirmation dialog; data restored to seed	E	Pass
SET-048	P2	S	Reset demo on a non-demo company via API	Refused	E	Pass
SET-049	P2	S	Reset demo as Accountant	Button absent; API refused	E	Pass
SET-050	P2	F	Audit history tab	Chronological commands with actor, type, time	E	Pass
SET-051	P2	F	Audit entry for a hide-module action	Present without the password	E	Pass
SET-052	P2	F	Audit after 200 commands	Paginated or scrollable; no freeze	E	Pass
SET-053	P2	E	Company name with emoji	Saved; printed	E	Pass
SET-054	P2	E	Phone with letters	Saved as typed (free text)	E	Pass
SET-055	P2	E	Email invalid	Saved as typed or validated; consistent with wizard	E	Pass
SET-056	P2	F	Save company while another member saves branding	Both persist (separate commands)	E	Pass
SET-057	P2	D	Revision increments per save	Yes	E	Pass
SET-058	P2	F	Settings reachable for every role	Yes (view)	E	Pass
SET-059	P2	F	Keyboard navigate tabs	Arrow keys/Tab reach tabs; `role=tab`, `aria-selected`	E	Pass
SET-060	P2	U	Active tab violet underline	Yes	B	Pass
SET-061	P2	F	Deep behaviour: hide Inventory with stock on hand	Allowed; stock valuation unchanged in Accounting	E	Pass
SET-062	P2	F	Hide Purchasing with open supplier bills	Allowed; payables unchanged	E	Pass
SET-063	P2	F	Hide Collections with pending cheques	Allowed; cheques can't be moved until re-enabled	E	Pass
SET-064	P2	N	Toggle quickly on/off/on	Single final state; no duplicate dialogs	E	Pass
SET-065	P2	F	Hide module, sign out, sign in	Still hidden	E	Pass
SET-066	P2	F	Hide module in tab A, tab B shows it	Tab B's next command returns new workspace; sidebar updates	E	Pass
SET-067	P2	S	`set_modules` as Auditor via API	"Your role does not permit this action."	E	Pass
SET-068	P2	F	Branding preview uses latest posted invoice if any, else sample	Yes	E	Pass
SET-069	P2	U	Branding preview typography/colours match print	Yes	B	Fail
SET-070	P2	F	Save branding with same values	No error; revision may increment	E	Pass
SET-071	P2	E	Accent colour black `#000000`	Allowed; print readable	E	Pass
SET-072	P2	E	Accent colour white `#ffffff`	Allowed but heading invisible — note as design limitation or validation	E	Pass
SET-073	P2	S	Branding upload path uses tenant id folder	Yes; cross-tenant read 401	E	Pass
SET-074	P2	S	Branding upload as Auditor	Refused	E	Pass
SET-075	P2	F	Settings → Features counts "N turned on"	Matches	B	Fail
SET-076	P2	U	Long module descriptions wrap in tiles	Yes	B	Pass
SET-077	P2	U	Dependency hint "Uses Sales" shown in violet	Yes	E	Pass
SET-078	P2	U	"Turning this off also turns off …" shown in amber	Yes	E	Pass
SET-079	P2	F	Hide module while that module's view is open	View closes to Overview; no blank page	B	Pass
SET-080	P2	F	Settings on a company still in setup	Not reachable (setup card)	B	Pass
SET-081	P2	F	Company & documents save as Recovery Officer	Allowed	E	Pass
SET-082	P2	E	NTN 50 characters	Accepted; prints	E	Pass
SET-083	P2	E	Address empty in Settings	Refused (address required)	E	Pass
SET-084	P2	F	Default sales-tax rate (if present in settings)	Applied as prefilled tax on sales lines	E	Pass
SET-085	P2	F	Default tax absent on new company	Tax placeholder 0 on lines	E	Fail
SET-086	P2	F	Numbering prefix in settings	Documents use it (`MC-…` for "My Company" seeded prefix)	E	Pass
SET-087	P2	S	Direct `save_company` with 10 KB address	Validation max length error	E	Pass
SET-088	P2	F	Reset demo twice in a row	Second works; seed identical	E	Pass
SET-089	P2	F	"Your workspace is ready." notice shown after setup only	Not shown after later saves	E	Pass
SET-090	P2	U	Error notice from a failed save is red	Yes (noticeError)	B	Pass
SET-091	P2	U	Success notice is neutral/green	Yes	B	Pass
SET-092	P2	F	Notice clears when the next command starts	Yes	B	Pass
SET-093	P2	F	Hide module dialog lists "N records" per type	Zero-count types omitted	E	Pass
SET-094	P2	F	Hide Reports requires password too	Yes (all hides)	E	Pass
SET-095	P2	F	Turning a module on provisions what it needs if missing	E.g. Cash & bank on later creates accounts once, never duplicates	E	Pass
SET-096	P2	D	Turn Cash & bank off and on three times	Still exactly one main bank and one cash account	E	Pass
SET-097	P2	D	Turn Commissions off/on	Plans and entitlements unchanged	E	Pass
SET-098	P2	F	Audit history shows role previews?	Not logged (client-only)	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
SET-099	P2	X	Features tab with 50k documents	Counts computed in < 2s	E	Pass
SET-100	P1	F	End to end: hide Commissions with password, verify API refusal, re-enable, approve an entitlement	All pass	E	Pass

Products, opening stock, CSV import/export (PRD): 94 pass · 6 fail · 0 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PRD-001	P1	F	Inventory → Products → New product with SKU, name, price	"{name} saved."; appears in list	E	Pass
PRD-002	P1	N	Empty SKU	HTML required blocks; API: "SKU is required"	E	Pass
PRD-003	P1	N	Empty name	Blocked; API: name min 2	E	Pass
PRD-004	P1	N	Name of 1 char	Refused	E	Pass
PRD-005	P1	N	Duplicate SKU, different case `led-24` vs `LED-24`	"SKU LED-24 already exists."	E	Pass
PRD-006	P2	E	SKU with surrounding spaces	Trimmed before uniqueness check	E	Pass
PRD-007	P2	E	SKU 50 chars	Accepted; 51 refused	E	Pass
PRD-008	P2	E	SKU with slash, hash, Urdu	Accepted; shows in lists and print	E	Pass
PRD-009	P1	N	Price left empty	HTML required blocks submit (price is required)	E	Pass
PRD-010	P2	E	Price `0`	Accepted (free item)	E	Pass
PRD-011	P1	N	Price `-5`	Blocked by min=0; API "Amount cannot be negative."	E	Pass
PRD-012	P2	E	Price `0.0001`	Accepted; shows Rs 0 in list (4 dp stored)	E	Pass
PRD-013	P2	E	Price `999999999999` (1 trillion − 1)	Accepted; `1000000000000` refused "Enter a valid number below one trillion"	E	Pass
PRD-014	P2	N	Price `abc` via API	Validation error	E	Fail
PRD-015	P2	N	Price `1e3` typed	Browser number input accepts → 1000	E	Pass
PRD-016	P2	E	Price with 6 decimals `12.345678`	Stored at 4 dp, rounding defined (half-up)	E	Fail
PRD-017	P2	F	Cost left empty	Saved as 0 (placeholder shown)	E	Pass
PRD-018	P2	F	Unit left blank	Saved blank; invoice shows quantity only	E	Pass
PRD-019	P2	E	Unit `1000 pcs` (user puts quantity in unit)	Saved as text; prints "199 1000 pcs" — documented limitation; hint text says "pcs, box, kg…"	E	Pass
PRD-020	P2	E	Unit 20 chars; 21 refused	Yes	E	Pass
PRD-021	P2	F	Category blank	Saved as "General" placeholder? No: saved blank — verify list shows blank, commission by category treats as uncategorised	E	Pass
PRD-022	P2	F	Reorder level 0	Badge logic: "Available" when qty>0	E	Pass
PRD-023	P2	F	Reorder level 10, stock 5	"Low stock" badge; Overview attention item	E	Pass
PRD-024	P2	N	Reorder level `-1`	Blocked min 0; API int min 0	E	Pass
PRD-025	P2	N	Reorder level `2.5`	HTML step 1 blocks; API int refused	E	Pass
PRD-026	P2	F	Weight/volume default placeholders 1	Saved 1 when left empty	E	Pass
PRD-027	P2	E	Weight 0	Accepted; landed-cost by weight excludes it (verify IMP)	E	Pass
PRD-028	P2	F	HS code free text	Saved	E	Pass
PRD-029	P1	F	Opening stock: qty 50, cost 1500	Product saved AND stock 50 @ 1,500 = Rs 75,000 in Stock tab; notice "Increased stock for 1 product"	E	Pass
PRD-030	P1	D	Opening stock journal	Dr Inventory 75,000 / Cr Opening balance equity (3000)	E	Pass
PRD-031	P1	N	Opening qty 50, cost empty	Product saved; error "Product saved, but the opening stock was not booked: Amount must be greater than zero. …"; product exists with 0 stock	E	Fail
PRD-032	P2	N	Opening qty 0, cost 500	No stock movement; product saved silently	E	Pass
PRD-033	P2	N	Opening qty `-3`	Blocked by min 0	E	Pass
PRD-034	P2	E	Opening qty `0.5` kg	Accepted (step 0.0001)	E	Pass
PRD-035	P2	E	Opening qty `0.00001`	Rounds to 0 at 4 dp → treated as no stock or refused; no negative value	E	Fail
PRD-036	P2	F	Two warehouses: warehouse select shown	Yes; one warehouse: hidden	E	Pass
PRD-037	P2	F	Opening stock into second warehouse	Stock shows in that warehouse only	E	Pass
PRD-038	P2	N	Opening stock while the current month is locked	Product saved; stock refused "Accounting period YYYY-MM is locked."	E	Pass
PRD-039	P2	S	Warehouse Manager limited to branch A chooses warehouse in branch B	"This warehouse is outside your permitted branches."	E	Pass
PRD-040	P2	F	Edit existing product	No Opening stock section	E	Pass
PRD-041	P2	F	Edit shows stored price `2400` not `2400.0000`	Yes	E	Pass
PRD-042	P2	F	Edit SKU to another product's SKU	"SKU … already exists."	E	Pass
PRD-043	P2	F	Edit price	Future invoices prefill new price; posted documents unchanged	E	Pass
PRD-044	P2	F	Edit archived product keeps archived flag	Yes	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PRD-045	P1	F	Archive product	Moves to Archived filter; hidden from invoice product list	E	Pass
PRD-046	P1	N	Create invoice line with archived product via API	"{name} is archived. Restore it first."	E	Pass
PRD-047	P2	F	Restore product	Back in active list	E	Pass
PRD-048	P2	F	Archive product with stock on hand	Allowed; stock valuation still counts it	E	Pass
PRD-049	P2	S	Salesperson tries to archive	Button absent; API "Your role does not permit this action."	E	Pass
PRD-050	P2	F	Search products by SKU fragment	Case-insensitive match	E	Pass
PRD-051	P2	F	Search by name with Urdu	Works	E	Pass
PRD-052	P2	F	Status filter Active/Archived/All	Counts correct	E	Pass
PRD-053	P2	F	Clear filters link	Resets	B	Pass
PRD-054	P2	U	Table on phone	Horizontal scroll hint; first column bold	B	Pass
PRD-055	P1	F	Export CSV	File `items-YYYY-MM-DD.csv`, headers from spec, one row per visible product	E	Pass
PRD-056	P2	F	Export with filters applied	Only visible rows	E	Pass
PRD-057	P2	F	Export when list empty	Button disabled	E	Pass
PRD-058	P2	E	Export product with comma and quotes in name	CSV properly quoted	E	Pass
PRD-059	P2	E	Export Unicode	UTF-8 BOM or plain UTF-8 opens in Excel correctly	E	Pass
PRD-060	P1	F	Import CSV: download template	Headers `sku,name,category,unit,price,cost,…`; example row	E	Pass
PRD-061	P1	F	Import 5 valid rows	Preview shows mapping; Import → 5 products	E	Pass
PRD-062	P1	F	Header auto-mapping "Item Code" → sku	Recognised	E	Pass
PRD-063	P1	N	Row with missing SKU	"Nothing was imported. Fix these rows…" with row number; zero imported	E	Pass
PRD-064	P1	N	Two rows with same SKU in file	"SKU X appears more than once in the file"; nothing imported	E	Pass
PRD-065	P1	N	Row SKU already exists (mode create)	"SKU X already exists"; nothing imported	E	Pass
PRD-066	P2	F	Mode update: existing SKU rows	Updated in place; new ones created	E	Pass
PRD-067	P2	N	Price `abc` in a row	Row error; nothing imported	E	Pass
PRD-068	P2	N	Negative price row	Row error	E	Pass
PRD-069	P2	E	500 rows	Imported; list responsive	E	Pass
PRD-070	P2	E	501+ rows	Clear limit message or chunked; no timeout	E	Pass
PRD-071	P2	E	Empty file	"No rows" message	E	Pass
PRD-072	P2	E	File with only headers	Same	E	Pass
PRD-073	P2	E	Windows line endings	Parsed	E	Pass
PRD-074	P2	E	Semicolon-delimited file	Not parsed as columns → clear mapping failure, not silent garbage	E	Pass
PRD-075	P2	E	Excel `.xlsx` chosen	Refused by type	E	Pass
PRD-076	P2	E	20 MB CSV	Refused or streamed; no browser freeze	B	Fail
PRD-077	P2	E	Quoted fields with embedded commas and newlines	Parsed correctly	E	Pass
PRD-078	P2	E	UTF-8 BOM at start	First header still mapped	E	Pass
PRD-079	P2	E	Required column unmapped	Import button disabled with hint listing required columns	E	Pass
PRD-080	P2	F	Error list shows at most 20 rows then "…and N more"	Yes	E	Pass
PRD-081	P2	S	Import as Salesperson	Button absent; API refused	E	Pass
PRD-082	P2	D	Import never creates stock	Correct; opening stock is set separately	E	Pass
PRD-083	P2	D	Import is atomic	One bad row → zero rows created (verified by count)	E	Pass
PRD-084	P2	D	Import idempotency: retry same key	No duplicates	E	Pass
PRD-085	P2	F	View product (Stock tab → View)	Item details: stock by warehouse, movement history	E	Pass
PRD-086	P2	F	Item details → Set quantity (count correction)	See INV cases; from product context works	E	Pass
PRD-087	P2	U	Required asterisks on SKU, name, price	Yes	E	Pass
PRD-088	P2	U	Placeholders grey, typed values dark	Yes	B	Pass
PRD-089	P2	U	"Opening stock" section separated by a rule	Yes	E	Pass
PRD-090	P2	U	Dialog on phone is a bottom sheet with sticky actions	Yes	B	Pass
PRD-091	P2	F	Cancel after typing	Nothing saved	B	Pass
PRD-092	P2	F	Escape closes	Nothing saved	B	Pass
PRD-093	P2	F	Submit with Enter in SKU field	Saves	B	Pass
PRD-094	P2	E	Product name 120 chars; 121 refused	Yes	E	Pass
PRD-095	P2	E	Name with only spaces	Refused (trim → <2)	E	Fail
PRD-096	P2	F	Same name, different SKU	Allowed	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PRD-097	P2	F	Product count in Settings → Features hide dialog	Matches	E	Pass
PRD-098	P2	X	5,000 products	List search < 300 ms; invoice product select usable	E	Pass
PRD-099	P2	D	Price shown in list formatted `Rs 2,400`	Yes	E	Pass
PRD-100	P1	F	End to end: import 10 → set opening stock on two → sell one → export	Export reflects data; stock correct	E	Pass

Customers and suppliers, walk-in names, credit limits (PTY): 96 pass · 3 fail · 1 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PTY-001	P1	F	Sales → Customers → New customer, name only	Saved with defaults	E	Pass
PTY-002	P1	N	Name empty / 1 char	Refused	E	Pass
PTY-003	P1	N	Duplicate customer name (case-insensitive)	"A customer named X already exists."	E	Pass
PTY-004	P2	F	Same name as a supplier	Allowed (types are separate)	E	Pass
PTY-005	P2	E	Name with trailing spaces	Trimmed; duplicate check on trimmed	E	Pass
PTY-006	P2	E	Name 120 chars	Accepted; longer refused	E	Pass
PTY-007	P2	F	Terms 0 days	Due date = invoice date	E	Pass
PTY-008	P2	F	Terms 365	Due date a year later	E	Pass
PTY-009	P2	N	Terms -1	Refused	E	Pass
PTY-010	P2	N	Terms 2.5	Refused (integer)	E	Pass
PTY-011	P1	F	Credit limit 0	No credit check on invoices	E	Pass
PTY-012	P1	F	Credit limit 100,000; outstanding 90,000; new invoice 20,000	"Customer credit limit exceeded. An owner must enter an override reason."	E	Pass
PTY-013	P1	F	Same with Owner override reason	Submits; reason stored on invoice	E	Pass
PTY-014	P1	S	Accountant supplies override reason	Still refused (owner only)	E	Pass
PTY-015	P2	E	Limit exactly equal to exposure + invoice	Allowed (not exceeded)	E	Pass
PTY-016	P2	E	Limit exceeded by 0.0001	Refused	E	Pass
PTY-017	P2	F	Credit check counts only submitted invoices minus paid/returned	Verified with a return	E	Pass
PTY-018	P2	F	Credit check ignores drafts	Yes	E	Pass
PTY-019	P2	F	Registered = true with tax id	Tax snapshot on invoice marks registered	E	Pass
PTY-020	P2	F	Registered = true, tax id empty	Allowed or hinted; print shows blank	E	Pass
PTY-021	P2	F	Customer class Retailer/Wholesaler/…	Used by class-based commission plans	E	Pass
PTY-022	P2	E	Email invalid	Saved as text (no strict validation) or validated consistently	E	Pass
PTY-023	P2	E	Phone `0300-1234567`	Saved; printed	E	Pass
PTY-024	P2	F	Edit customer type via API customer→supplier	"A party's type cannot be changed."	E	Pass
PTY-025	P1	F	Archive customer	Hidden from invoice suggestions	E	Pass
PTY-026	P1	N	Create document for archived customer via API	"{name} is archived. Restore it first."	E	Pass
PTY-027	P2	F	Archive customer with outstanding balance	Allowed; receivables still include them (verify report)	E	Pass
PTY-028	P2	F	Restore	Back in suggestions	E	Pass
PTY-029	P2	S	Recovery Officer creates customer	Allowed? Access list: save_party = Accountant, Sales Manager, Warehouse Manager (+Owner). Recovery Officer refused	E	Pass
PTY-030	P2	S	Salesperson creates customer	Refused via API; invoice editor "Yes, save customer" fails with message, "No" path still works	E	Fail
PTY-031	P1	F	Invoice editor: type existing name exactly	Matched; hint shows city/phone/terms; no prompt	B	Pass
PTY-032	P1	F	Type existing name in different case	Matched	B	Pass
PTY-033	P1	F	Type a new name, Submit	"Save customer?" dialog with No/Yes	B	Pass
PTY-034	P1	F	Yes, save customer	Customer created with city/phone; invoice booked to it	E	Pass
PTY-035	P1	F	No, don't save	Invoice booked to "Walk-in customer"; typed name shown in list and on print; no customer created	E	Pass
PTY-036	P1	F	First "No" ever	"Walk-in customer" account created automatically	E	Pass
PTY-037	P1	F	Second "No"	Reuses the same walk-in account (count stays 1)	E	Pass
PTY-038	P2	F	Walk-in customer archived, then "No"	Error "Walk-in customer is archived. Restore it in Sales → Customers first."	E	Pass
PTY-039	P2	F	Type "Walk-in customer" exactly	Matches the shared account; no prompt; no display name	E	Pass
PTY-040	P2	F	"No" then submit fails (stock) and retry	No second prompt; same draft updated	E	Pass
PTY-041	P2	F	Change the name after answering	Prompt asked again for the new name	B	Pass
PTY-042	P2	F	Edit a walk-in draft	Name field shows typed name; saving keeps it	E	Pass
PTY-043	P2	F	Convert walk-in order → delivery → invoice	Typed name carried	E	Pass
PTY-044	P2	F	Return on walk-in invoice	Credit note shows typed name	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PTY-045	P2	F	Receipt for a walk-in invoice	Choose "Walk-in customer"; open invoices listed by number	E	Pass
PTY-046	P2	F	Customer statement for Walk-in customer	All walk-in invoices together (documented)	E	Pass
PTY-047	P2	F	Search "Rashid" after a walk-in sale	Invoice found by typed name	E	Pass
PTY-048	P2	E	Typed name equals an existing customer but with extra inner spaces	Not matched (exact trimmed compare) → prompt; acceptable	B	Pass
PTY-049	P2	E	Typed name 121 chars	Refused by API max 120 with message	E	Pass
PTY-050	P2	E	Typed name with `<b>`	Printed as text	E	Pass
PTY-051	P2	F	Supplier editor: same flow with "Save supplier?" and "Walk-in supplier"	Yes	E	Pass
PTY-052	P2	F	New supplier via PO "Yes"	Supplier created with class Importer, 30-day terms	E	Pass
PTY-053	P2	F	Cancel the Save dialog (X)	Back to editor; nothing saved	B	Pass
PTY-054	P2	U	Dialog copy mentions the typed name and walk-in account	Yes	B	Pass
PTY-055	P2	U	Hint under name: "New customer. You can choose whether to save them."	Yes	E	Pass
PTY-056	P2	F	City and phone optional fields appear only for new names	Yes	B	Pass
PTY-057	P2	F	Customer list search by city	Works	E	Pass
PTY-058	P1	F	Export customers CSV	Correct columns	E	Pass
PTY-059	P1	F	Import customers CSV valid	Created	E	Pass
PTY-060	P1	N	Import duplicate name in file	"customer X appears more than once in the file"	E	Pass
PTY-061	P1	N	Import existing name (create mode)	"customer X already exists"	E	Pass
PTY-062	P2	F	Import update mode	Updates by name	E	Pass
PTY-063	P2	N	Import credit limit `abc`	Row error	E	Pass
PTY-064	P2	N	Import terms `-5`	Row error	E	Pass
PTY-065	P2	E	Import registered `yes/true/1`	All accepted as true	E	Pass
PTY-066	P2	E	Import 500 customers	OK	E	Pass
PTY-067	P2	F	Suppliers list under Purchasing	Separate tab	E	Pass
PTY-068	P2	F	Supplier terms 60 → bill due date	+60 days	E	Pass
PTY-069	P2	F	Supplier credit limit field	Ignored for purchases (no supplier credit check) — verify no error	E	Pass
PTY-070	P2	F	Customer with 0-day terms on quotation	Due date same day; no validity logic	E	Pass
PTY-071	P2	F	Delete customer	No delete; archive only (verify no delete control)	E	Pass
PTY-072	P2	D	Archiving customer keeps historical invoices and statements	Yes	E	Pass
PTY-073	P2	F	Count in Settings → Features (customers)	Matches including archived? Verify spec: counts all	E	Pass
PTY-074	P2	U	Customers table phone column on phone width	Scrolls	B	Pass
PTY-075	P2	F	Receipt dialog lists only customers (not suppliers)	Yes	B	Pass
PTY-076	P2	F	Payment dialog lists only suppliers	Yes	B	Pass
PTY-077	P2	F	Follow-up dialog lists customers only	Yes	B	Pass
PTY-078	P2	S	Salesperson sees customers list	Allowed (view)	E	Pass
PTY-079	P2	S	Warehouse Manager archives customer	Allowed per access list	E	Pass
PTY-080	P2	E	Credit limit `1e12`	Refused (below one trillion rule)	E	Pass
PTY-081	P2	E	Credit limit `0.5`	Accepted	E	Pass
PTY-082	P2	F	Credit override reason 500 chars; 501 refused	Yes	E	Pass
PTY-083	P2	F	Override reason empty string with limit exceeded	Refused	E	Pass
PTY-084	P2	F	Credit limit exceeded on quotation	No check (only sales_invoice)	E	Pass
PTY-085	P2	F	Credit limit exceeded on sales order	Verify product rule: check applies to invoice only	E	Pass
PTY-086	P2	F	Customer renamed after invoices	Lists and statements show new name; printed PDFs regenerated show new name (no snapshot) — document	E	Pass
PTY-087	P2	F	Walk-in customer renamed by user	New "No" sales create a fresh "Walk-in customer" (name match) — document behaviour	E	Pass
PTY-088	P2	F	Two customers "Ali Traders" and "Ali Traders Ltd"	Typing "Ali Traders" matches the exact one	B	Pass
PTY-089	P2	U	Datalist suggestions show city · phone	Yes	B	Pass
PTY-090	P2	U	Suggestions on iOS Safari	Datalist works or free text still saves	—	Manual
PTY-091	P2	F	Paste a name with newline	Trimmed	E	Pass
PTY-092	P2	S	`save_party` with type `employee`	Validation error	E	Pass
PTY-093	P2	S	`save_party` for another tenant's id	Not found in this tenant	E	Pass
PTY-094	P2	D	Audit shows party creations from the invoice path	Yes (save_party command)	E	Pass
PTY-095	P2	F	Customer class free text `VIP`	Saved; commission class plan can reference	E	Pass
PTY-096	P2	E	Tax id with spaces	Saved as typed	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PTY-097	P2	F	Registered customer tax snapshot on invoice after customer changes to unregistered	Old invoice snapshot unchanged	E	Pass
PTY-098	P2	X	10,000 customers	Suggestion datalist still usable (or capped with search)	B	Fail
PTY-099	P2	F	Supplier statement	Bills, returns, payments with running balance	E	Fail
PTY-100	P1	F	End to end: new customer via invoice "Yes", limit 50,000, invoice 60,000 refused, override by owner, receipt clears	All pass; statement balances	E	Pass

Inventory: stock, transfers, corrections, valuation, warehouses (INV): 98 pass · 2 fail · 0 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
INV-001	P1	F	Inventory → Stock	Row per product: quantity, average cost, value, badge	E	Pass
INV-002	P1	F	Single warehouse	Tab says "Stock"; no warehouse column, filter or transfer button; no "Main branch" text anywhere	E	Pass
INV-003	P1	F	Add second warehouse	Warehouse column, filter, "Transfer stock" and per-warehouse breakdown appear	E	Pass
INV-004	P1	F	View product → movement history	Opening stock row with qty +50, value 75,000, type adjustment	E	Pass
INV-005	P1	F	Set quantity (count correction) 50 → 45, reason	Movement −5 @ avg cost; journal Dr 5300 shrinkage / Cr 1200	E	Pass
INV-006	P1	F	Set quantity 45 → 60 with unit cost 1,600	+15 @ 1,600; average cost recomputed = (45×1500+15×1600)/60	E	Pass
INV-007	P1	N	Set quantity to the same number	"Enter a different quantity to save a stock correction."	E	Pass
INV-008	P1	N	Set quantity without reason	"Adjustment reason is required."	E	Pass
INV-009	P1	N	Increase without unit cost	"Enter a unit cost above zero for added stock."	E	Pass
INV-010	P1	N	Set quantity to −1	Blocked; API nonneg	E	Pass
INV-011	P1	D	Stale expected quantity (someone else changed it)	"Stock changed since you opened this item. Current quantity in Main store: N"	E	Pass
INV-012	P2	F	Purpose opening_stock on increase	Offsets to 3000 equity	E	Pass
INV-013	P2	F	Purpose count_correction on increase	Offsets to 5300	E	Pass
INV-014	P2	E	Set quantity 0.0001	Accepted	E	Pass
INV-015	P2	E	Set quantity 999,999,999	Accepted; value below one trillion rule	E	Pass
INV-016	P2	N	Set quantity in a locked period	"Accounting period … is locked."	E	Pass
INV-017	P2	N	Date in the future	Accepted or refused consistently (document)	E	Pass
INV-018	P2	E	Date `2026-02-30`	Validation error (iso date)	E	Pass
INV-019	P1	F	Bulk stock increase: two products	Two movements, one journal with total	E	Pass
INV-020	P1	N	Bulk increase with duplicate product	"Each product can appear only once in a stock increase."	E	Pass
INV-021	P1	N	Bulk increase line qty 0	"Amount must be greater than zero."	E	Pass
INV-022	P2	N	Bulk increase value rounds to 0 (qty 0.0001 cost 0.0001)	"Each added stock line must have a positive value."	E	Pass
INV-023	P2	N	Bulk increase 501 lines	Validation max 500	E	Pass
INV-024	P2	N	Bulk increase reason empty	Validation	E	Pass
INV-025	P1	F	Transfer 10 units Main → Lahore	Main −10, Lahore +10 at same value; no P&L effect	E	Pass
INV-026	P1	N	Transfer same warehouse	"Choose two different warehouses."	E	Pass
INV-027	P1	N	Transfer 60 when 50 available	"Insufficient stock for LED Panel 24W in Main store. Requested: 60, available: 50."	E	Pass
INV-028	P2	N	Transfer 0	Refused	E	Pass
INV-029	P2	F	Transfer dialog closes on success	Yes (scoped auto-close)	E	Pass
INV-030	P2	F	Transfer in a locked period	Refused	E	Pass
INV-031	P2	S	Warehouse Manager of branch A transfers from branch B	"This warehouse is outside your permitted branches."	E	Pass
INV-032	P2	S	Salesperson opens Inventory	Allowed to view? Verify canView; transfer buttons absent	E	Pass
INV-033	P2	D	Average cost after transfer unchanged in both	Yes	E	Pass
INV-034	P1	D	Sale of 10 @ avg 1,500	COGS 15,000; stock 40; value 60,000	E	Pass
INV-035	P1	D	Purchase receipt 20 @ 1,800	Avg = (40×1500+20×1800)/60 = 1,600	E	Pass
INV-036	P1	D	Sale after that at avg 1,600	COGS uses 1,600	E	Pass
INV-037	P1	D	Return of 1 unit from an invoice	Stock +1 at the invoice line's cost; value restored	E	Pass
INV-038	P2	D	Stock never negative via any path	Direct invoice, delivery, transfer, correction all guarded	E	Pass
INV-039	P2	D	Rounding: 3 units @ 10.0001 then sell 1	Value never drifts beyond 0.0001; final stock 0 has value 0	E	Pass
INV-040	P2	D	Sell exactly all stock	Qty 0, value exactly 0.0000	E	Pass
INV-041	P2	F	Stock level filter Available / Low / Out	Correct sets	B	Pass
INV-042	P2	F	Category filter	Works	B	Pass
INV-043	P2	F	Search by SKU	Works	B	Pass
INV-044	P2	F	Badge "Out of stock" red at 0	Yes	E	Pass
INV-045	P2	F	"Low stock" when qty ≤ reorder level and > 0	Yes	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
INV-046	P2	F	Overview attention: low stock queue	Lists the items	E	Pass
INV-047	P1	F	Warehouses tab: add warehouse with branch	Created; unique name	E	Pass
INV-048	P1	N	Duplicate warehouse name	"Warehouse X already exists."	E	Pass
INV-049	P2	N	Warehouse name empty	Refused	E	Pass
INV-050	P1	F	Add branch	"Branch X already exists." on duplicate	E	Pass
INV-051	P2	F	Branch picker appears in header only with ≥2 branches	Yes	B	Pass
INV-052	P2	F	Branch picker filters documents/stock	Yes	B	Pass
INV-053	P2	S	Member with branchIds [A]	Sees only branch A data; picker locked	E	Pass
INV-054	P2	F	Archive a warehouse with stock	Refused or prevented (define) ; empty warehouse archive allowed	E	Pass
INV-055	P2	F	Default warehouse on documents	First warehouse when single	E	Pass
INV-056	P2	F	Document warehouse select hidden when single	Yes	B	Pass
INV-057	P2	F	Stock valuation report = Stock tab total	Equal to the paisa	E	Pass
INV-058	P2	D	Inventory control (1200) = valuation	Control reconciliation "reconciled"	E	Pass
INV-059	P2	F	Movement history types: adjustment, sale, purchase, transfer, return	Labelled; source document linked	E	Pass
INV-060	P2	F	Movement history order	Chronological; running quantity	E	Pass
INV-061	P2	U	Item details on phone	Sections stack; history scrolls	B	Pass
INV-062	P2	F	Set quantity dialog closes on success	Yes	B	Pass
INV-063	P2	F	Set quantity dialog shows current qty and value	Yes	B	Fail
INV-064	P2	E	Reason 2,000 chars; 2,001 refused	Yes	E	Pass
INV-065	P2	E	Unit cost with 4 decimals	Stored	E	Pass
INV-066	P2	E	Unit cost 0.00001	Rounds to 0 → "Enter a unit cost above zero"	E	Pass
INV-067	P2	D	Correction journal memo contains reason	Yes	E	Pass
INV-068	P2	D	Journal source id = movement id; re-posting same source	"This source has already been posted."	E	Pass
INV-069	P2	D	Idempotent retry of set_stock_quantity	Single movement	E	Pass
INV-070	P2	F	Stock in two warehouses, sell from Lahore only	Main unchanged	E	Pass
INV-071	P2	F	Delivery from warehouse with 0 of the item	Insufficient stock message names the warehouse	E	Pass
INV-072	P2	F	Stock tab totals row	Sum of values	E	Pass
INV-073	P2	F	Export stock CSV (report)	Matches tab	E	Pass
INV-074	P2	F	Archived product with stock in Stock tab	Shown (stock is real) with archived mark	E	Pass
INV-075	P2	F	Product with stock in only one of two warehouses	Breakdown shows 0 for the other	E	Pass
INV-076	P2	U	Transfer dialog warehouse selects exclude archived	Yes	E	Pass
INV-077	P2	F	Transfer multiple lines	Each guarded individually; all-or-nothing	E	Pass
INV-078	P2	N	Transfer with unknown item id	"Product not found."	E	Pass
INV-079	P2	N	Transfer with unknown warehouse	"Warehouse not found."	E	Pass
INV-080	P2	F	stock_adjust (free adjustment) non-zero	Posted	E	Pass
INV-081	P2	N	stock_adjust 0	"Enter a non-zero adjustment."	E	Pass
INV-082	P2	F	Count correction when average cost is 0 (no history)	Uses product standard cost; if 0 requires unit cost	E	Pass
INV-083	P2	D	Valuation after opening stock, purchase, sale, return, correction, transfer	Equals sum of movements' values	E	Pass
INV-084	P2	D	Stock movements report = history	Yes	E	Pass
INV-085	P2	F	Overview "Inventory value" metric	Equals valuation	E	Pass
INV-086	P2	X	2,000 products × 3 warehouses	Stock tab renders < 2 s	E	Fail
INV-087	P2	X	50,000 movements for one item	History paginates/scrolls; no freeze	B	Pass
INV-088	P2	F	Warehouse rename	Reflected in history and documents	E	Pass
INV-089	P2	F	Branch rename	Reflected in picker	E	Pass
INV-090	P2	S	Auditor views stock	Read-only; no buttons	E	Pass
INV-091	P2	S	`set_stock_quantity` as Salesperson via API	Refused	E	Pass
INV-092	P2	F	Hidden Inventory module	Stock view gone; sale still moves stock (engine)	E	Pass
INV-093	P2	F	Re-enable Inventory	View returns with all data	E	Pass
INV-094	P2	E	Warehouse name with emoji	Saved	E	Pass
INV-095	P2	E	50 warehouses	Selects remain usable	B	Pass
INV-096	P2	F	Transfer then cancel?	Transfers post immediately; no draft; reversing requires opposite transfer	E	Pass
INV-097	P2	D	Transfer journal	None (no accounting effect) or balanced zero — verify no unbalanced posting	E	Pass
INV-098	P2	U	Quantity formatting `50 pcs` with unit	Yes; blank unit → `50`	E	Pass
INV-099	P2	U	Average cost formatted `Rs 1,500`	Yes	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
INV-100	P1	F	End to end: opening stock → purchase → transfer → sale → return → correction; reconciliation reconciled	All pass	E	Pass

Sales documents and the invoice editor (SAL): 98 pass · 1 fail · 1 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
SAL-001	P1	F	Sales → New invoice: customer, product, qty 1, "Submit sales invoice"	Created and submitted in one click; row "Submitted"; stock −1; journal posted	E	Pass
SAL-002	P1	F	Same with "Save draft"	Row "Draft"; no stock or journal effect	E	Pass
SAL-003	P1	F	Submit from the list on a draft	Submitted	E	Pass
SAL-004	P1	F	Submit from document details	Submitted	E	Pass
SAL-005	P1	N	Submit with insufficient stock	Editor keeps draft; red "Saved as a draft but not submitted. Insufficient stock for … Requested: 9, available: 5."	E	Pass
SAL-006	P1	F	Fix qty and press Submit again	Same draft updated and submitted; exactly one row	E	Pass
SAL-007	P1	N	Empty customer	"Enter the customer's name."	E	Pass
SAL-008	P1	N	No product line	"Add at least one product line."	E	Pass
SAL-009	P1	N	Product chosen, qty empty	"Enter a quantity on line 1."	E	Pass
SAL-010	P1	N	Qty 0	Same	E	Pass
SAL-011	P2	N	Qty negative	Blocked by min; API "Amount must be greater than zero."	E	Pass
SAL-012	P2	E	Qty 0.0001	Accepted	E	Pass
SAL-013	P2	E	Qty 0.00001	Rounds to 0 → refused	E	Fail
SAL-014	P2	E	Qty 1,000,000 with stock	Accepted if stock	E	Pass
SAL-015	P1	F	Price prefilled from product	Yes; editable	B	Pass
SAL-016	P2	F	Product with price 0	Price field empty with placeholder 0.00	B	Pass
SAL-017	P2	E	Price empty on submit	Treated as 0; line total 0	E	Pass
SAL-018	P2	E	Price negative	Blocked/refused	E	Pass
SAL-019	P2	F	Discount 10%	Line total ×0.9	E	Pass
SAL-020	P2	E	Discount 100%	Line total 0	E	Pass
SAL-021	P2	N	Discount 101	Blocked max 100; API refused	E	Pass
SAL-022	P2	F	Tax 18	Tax = net×18%; totals update live	E	Pass
SAL-023	P2	E	Tax 0 and blank	Both 0	E	Pass
SAL-024	P2	N	Tax 150	Refused	E	Pass
SAL-025	P2	F	Default tax from settings prefilled on sales lines	Yes when configured	B	Pass
SAL-026	P1	N	Same product on two lines	"Use one line per product."	E	Pass
SAL-027	P2	F	Add line / remove line	Totals recalc; cannot remove last line? (verify: can, then "Add at least one")	E	Pass
SAL-028	P2	E	100 lines	Accepted; 101 refused	E	Pass
SAL-029	P2	F	Salesperson column for sales	Employee select; required for Salesperson role (own id)	E	Pass
SAL-030	P1	S	Salesperson role creates invoice with another salesperson on a line	"Salespeople can create their own sales only."	E	Pass
SAL-031	P1	S	Salesperson edits another's draft	"Salespeople can edit their own drafts only."	E	Pass
SAL-032	P1	S	Salesperson submits an invoice	"Sales documents require manager approval." (or allowed per role matrix — assert actual)	E	Pass
SAL-033	P2	S	Warehouse Manager creates invoice	"Warehouse users can create receipts and deliveries only."	E	Pass
SAL-034	P2	S	Recovery Officer creates invoice	Refused	E	Pass
SAL-035	P2	S	Accountant creates and submits invoice	Allowed	E	Pass
SAL-036	P1	F	Quotation → Submit → Convert → sales order	Lines copied; parentId set; chain shown	E	Pass
SAL-037	P1	N	Convert a draft quotation	"Submit the source document first."	E	Pass
SAL-038	P1	F	Sales order → Deliver part of a line	Delivery note draft with the entered quantities	E	Pass
SAL-039	P1	N	Deliver more than remaining	"Converted quantity exceeds the remaining source quantity."	E	Pass
SAL-040	P1	F	Deliver remaining later	Second delivery; order fully delivered; no more Deliver button	E	Pass
SAL-041	P1	F	Submit delivery	Stock out; COGS posted; no receivable yet	E	Pass
SAL-042	P1	F	Delivery → Invoice	Invoice draft from delivery; submitting does NOT move stock again	E	Pass
SAL-043	P1	F	Direct invoice (no delivery) submit	Moves stock	E	Pass
SAL-044	P2	N	Convert invoice	"This document cannot be converted."	E	Pass
SAL-045	P2	N	Convert with different warehouse on child via API	"Source party, warehouse and currency must match."	E	Pass
SAL-046	P2	N	Convert quotation directly to delivery via API	"Invalid document conversion."	E	Pass
SAL-047	P2	F	Cancel a draft with reason	Status cancelled; reason appended to notes; parent quantity freed	E	Pass
SAL-048	P1	N	Cancel a submitted invoice	"Only draft documents can be cancelled. Use a return or reversal for posted documents."	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
SAL-049	P1	N	Edit a submitted invoice	"Only draft documents can be edited…"	E	Pass
SAL-050	P1	N	Submit twice (double click / retry)	"Only draft documents can be submitted." or idempotent no-op; one journal	E	Pass
SAL-051	P1	F	Return 1 unit, reason damaged	Credit note CN-… submitted; stock +1; invoice returned amount; balance reduced	E	Pass
SAL-052	P1	N	Return more than unreturned	"Return exceeds the unreturned quantity."	E	Pass
SAL-053	P2	N	Return same line twice in one request	"Duplicate return line."	E	Pass
SAL-054	P2	F	Return all lines	Balance due 0; credit note total = invoice	E	Pass
SAL-055	P2	F	Return after partial payment	Overpaid → "Customer credit" shown negative balance	E	Pass
SAL-056	P2	N	Return on a draft invoice	"Select a submitted sales or supplier invoice."	E	Pass
SAL-057	P2	F	Credit note print	Shows typed/customer name, lines, reason	E	Pass
SAL-058	P2	D	Return journal	Reverses revenue, tax, receivable and COGS/stock proportionally	E	Pass
SAL-059	P2	F	Invoice date in the future	Accepted; due date computed	E	Pass
SAL-060	P2	N	Invoice in locked period	"Accounting period YYYY-MM is locked."	E	Pass
SAL-061	P2	N	Due date before date	Accepted or refused consistently (document)	E	Pass
SAL-062	P2	E	Date `2026-13-01`	Validation	E	Pass
SAL-063	P2	F	Notes 2,000 chars; 2,001 refused	Yes	E	Pass
SAL-064	P2	E	Notes with newlines	Printed with line breaks	E	Pass
SAL-065	P1	F	Numbering INV-2026-0001, 0002 … per year	Sequential; gaps only from cancellations? (cancelled keeps its number)	E	Pass
SAL-066	P2	D	Two users create invoices simultaneously	Distinct numbers; one revision conflict retried transparently	—	Manual
SAL-067	P2	F	Document list filters: type, status, customer, dates, search	Combine correctly; Clear filters	B	Pass
SAL-068	P2	F	Search by typed walk-in name	Found	E	Pass
SAL-069	P2	F	List ordering newest first	Yes	B	Pass
SAL-070	P2	F	View → details: chain, lines, totals, payments, returns	Correct	B	Pass
SAL-071	P2	F	Details Print button	Opens print view	B	Pass
SAL-072	P2	F	Edit a draft: change customer	Allowed; credit check on submit uses new customer	E	Pass
SAL-073	P2	F	Edit a draft: remove all lines and save	Refused	E	Pass
SAL-074	P2	U	Editor on phone	Bottom sheet; line table scrolls horizontally; sticky actions	B	Pass
SAL-075	P2	U	Three buttons: Cancel, Save draft (outline), Submit (violet)	Yes	B	Pass
SAL-076	P2	U	Spinner on the pressed button only	Yes	B	Pass
SAL-077	P2	U	Placeholders Qty / 0.00 / 0 / 0	Yes	B	Pass
SAL-078	P2	U	Unit hint under qty after product chosen	Yes	B	Pass
SAL-079	P2	U	Line total and Net/Tax/Total formatting	`Rs 4,500`	B	Pass
SAL-080	P2	F	Escape closes editor with unsaved changes	Closes (no confirm) — document	B	Pass
SAL-081	P2	F	Overview "Recent sales" shows last 5	Yes with walk-in names	B	Pass
SAL-082	P2	F	Overview metric "Total sales"	Sum of submitted invoices net of returns	E	Pass
SAL-083	P2	F	Credit limit path (see PTY-012/013)	Works from editor with owner reason field visible only for Owner + matched customer with limit	E	Pass
SAL-084	P2	D	Tax snapshot stored on invoice	Registered flag/tax id frozen	E	Pass
SAL-085	P2	D	Invoice `paid` updates on receipt	Yes; `outstanding` = total – paid – returned	E	Pass
SAL-086	P2	D	Journal for invoice: Dr 1100 total / Cr 4000 net / Cr 2100 tax; plus Dr 5000 / Cr 1200 cost on direct	Yes	E	Pass
SAL-087	P2	D	Journal memo and source = invoice id	Yes	E	Pass
SAL-088	P2	F	Quotation has no stock or ledger effect on submit	Correct	E	Pass
SAL-089	P2	F	Sales order submit no stock effect	Correct	E	Pass
SAL-090	P2	F	Delivery note print	Quantities, no prices? (verify design)	E	Pass
SAL-091	P2	F	Convert order with multiple products partially twice	Remaining tracked per line	E	Pass
SAL-092	P2	F	Cancel a delivery draft frees order quantity	Yes	E	Pass
SAL-093	P2	N	Submit a document whose product was archived after drafting	"{name} is archived. Restore it first." on submit? (engine checks at create; verify submit path)	E	Pass
SAL-094	P2	S	Salesperson sees only own documents in list	Yes (scopeWorkspace)	E	Pass
SAL-095	P2	S	Sales Manager sees all	Yes	E	Pass
SAL-096	P2	F	Commission splits via API totalling 90%	"Commission splits must total 100%."	E	Pass
SAL-097	P2	X	Invoice with 100 lines submit	< 3 s	E	Pass
SAL-098	P2	F	Hidden Sales module	Sales view gone; API refused	E	Pass
SAL-099	P2	F	Invoice for a customer in another branch (branch-limited member)	"This branch is outside your permitted branches."	E	Pass
SAL-100	P1	F	End to end: quotation → order → two deliveries → invoice → partial return → receipt; reconciliation reconciled	All pass	E	Pass

Purchasing, foreign currency, supplier payments (PUR): 93 pass · 7 fail · 0 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PUR-001	P1	F	Purchasing → New purchase order, supplier, product, qty, "Submit purchase order"	Created and submitted; no stock, no ledger	E	Pass
PUR-002	P1	F	Save draft	Draft row	E	Pass
PUR-003	P1	N	Supplier empty	"Enter the supplier's name."	E	Pass
PUR-004	P1	F	New supplier name → "Save supplier?" Yes	Supplier created (Importer, 60? verify default 30 days)	E	Pass
PUR-005	P1	F	New supplier name → No	Booked to "Walk-in supplier"; typed name shown	E	Pass
PUR-006	P1	F	Currency USD, rate 280, line 100 × $10	USD total 1,000; PKR equivalent 280,000 shown	E	Pass
PUR-007	P1	N	Currency USD, rate empty	"Amount must be greater than zero." (fxRate pos)	E	Fail
PUR-008	P1	N	Currency PKR, rate 280	"PKR exchange rate must be 1."	E	Pass
PUR-009	P2	E	Rate 0.0001	Accepted	E	Pass
PUR-010	P2	E	Rate 1e12	Refused (below one trillion)	E	Pass
PUR-011	P2	N	Currency EUR via API	Validation (PKR/USD/CNY only)	E	Pass
PUR-012	P2	F	CNY order	Works like USD	E	Pass
PUR-013	P2	F	Rate placeholder "e.g. 280" shown when empty	Yes	B	Pass
PUR-014	P1	F	Submitted PO → Receive → GRN with full quantities	GRN draft; submit → stock in at PKR cost (price × rate)	E	Pass
PUR-015	P1	F	Receive part (40 of 100)	GRN for 40; PO shows 60 remaining on next Receive	E	Pass
PUR-016	P1	N	Receive 61 when 60 remain	"Converted quantity exceeds the remaining source quantity."	E	Pass
PUR-017	P1	F	Receive remaining 60	PO fully received; Receive button gone	E	Pass
PUR-018	P1	F	GRN submit journal	Dr 1200 inventory / Cr GRNI or 2000 payable (verify account) at PKR value	E	Pass
PUR-019	P1	F	GRN → Invoice → supplier bill submit	Payable in USD at the bill's rate; PKR value posted	E	Pass
PUR-020	P2	F	Bill rate differs from GRN rate	Difference posted to FX gain/loss (verify account)	E	Fail
PUR-021	P2	N	Convert GRN with different currency via API	"Source party, warehouse and currency must match."	E	Pass
PUR-022	P2	N	Convert PO directly to bill via API	"Invalid document conversion."	E	Pass
PUR-023	P2	N	Convert a draft PO	"Submit the source document first."	E	Pass
PUR-024	P1	F	Supplier return on a bill (1 unit)	PR-… submitted; stock −1; payable reduced	E	Pass
PUR-025	P1	N	Return more than received	"Return exceeds the unreturned quantity."	E	Pass
PUR-026	P2	N	Return on a GRN (not bill)	"Select a submitted sales or supplier invoice."	E	Pass
PUR-027	P2	N	Return when the stock was already sold	"Insufficient stock for …"	E	Pass
PUR-028	P1	F	Supplier payments → Record payment USD 1,000 allocating to the bill	Bill paid; payable cleared	E	Pass
PUR-029	P1	F	"PKR actually paid" higher than rate value	Exchange loss posted automatically	E	Pass
PUR-030	P1	F	PKR paid lower	Exchange gain	E	Pass
PUR-031	P1	N	Allocation exceeds bill outstanding	"Allocation exceeds the invoice's outstanding balance."	E	Pass
PUR-032	P1	N	Allocations ≠ amount	"Allocations must equal the receipt amount."	E	Fail
PUR-033	P2	N	Same bill twice in allocations	"Duplicate allocation."	E	Pass
PUR-034	P2	N	Allocate to a bill of another supplier	"Receipt must reference submitted invoices for the same party."	E	Pass
PUR-035	P2	N	Allocate to a draft bill	Same	E	Pass
PUR-036	P1	S	Recovery Officer pays a supplier	"Recovery officers cannot pay suppliers."	E	Pass
PUR-037	P2	F	Payment method cash from Cash in hand	Cash account credited	E	Pass
PUR-038	P2	F	Payment method bank with account chosen	That bank credited	E	Pass
PUR-039	P2	F	Payment by cheque (outgoing)	Recorded; no cheque register state machine for outgoing? verify behaviour	E	Pass
PUR-040	P2	N	Payment amount 0	Refused	E	Pass
PUR-041	P2	N	Payment in locked period	Refused	E	Pass
PUR-042	P2	F	Partial payment	Bill outstanding reduced; aged payables updated	E	Pass
PUR-043	P2	F	Two partial payments	Sum; second cannot exceed remaining	E	Pass
PUR-044	P2	F	Supplier statement	PO not shown; GRN/bill/return/payment with running balance	E	Fail
PUR-045	P1	S	Salesperson opens Purchasing	Not visible; API "Your role cannot create purchase documents."	E	Pass
PUR-046	P1	S	Sales Manager submits a PO	"Your role can submit sales documents only."	E	Pass
PUR-047	P2	S	Warehouse Manager creates PO	"Warehouse users can create receipts and deliveries only."	E	Pass
PUR-048	P2	S	Warehouse Manager submits GRN	Allowed	E	Pass
PUR-049	P2	S	Warehouse Manager submits bill	"Your role can submit warehouse documents only."	E	Pass
PUR-050	P2	S	Accountant full chain	Allowed	E	Pass
PUR-051	P2	F	PO print	Shows currency and rate; totals in USD and PKR	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PUR-052	P2	F	Bill print	Supplier, due date by terms	E	Pass
PUR-053	P2	F	Due date = date + supplier terms	Yes	B	Pass
PUR-054	P2	F	Edit draft PO currency after lines	Totals recompute	E	Pass
PUR-055	P2	N	Submit PO in locked period	Refused	E	Pass
PUR-056	P2	F	Cancel draft GRN	Frees PO remaining	E	Pass
PUR-057	P2	N	Cancel submitted GRN	Refused; use return	E	Pass
PUR-058	P2	D	GRN with 3 lines, avg cost per product	Each product's avg updated independently	E	Pass
PUR-059	P2	D	Rounding: $0.3333 × 3 at 280	PKR value rounded at 4 dp; journal balanced	E	Pass
PUR-060	P2	D	Payable control (2000) = open bills	Reconciled	E	Pass
PUR-061	P2	D	Input tax on bill with tax 18	Dr 1500 input tax (verify code)	E	Pass
PUR-062	P2	F	Tax 0 on purchases	No tax line	E	Pass
PUR-063	P2	F	Discount on PO line	PKR value net of discount	E	Pass
PUR-064	P2	E	100 lines PO	OK	E	Pass
PUR-065	P2	E	Qty 0.5 kg	OK	E	Pass
PUR-066	P2	F	Numbering PO-2026-0001, GRN-, BILL-, PR-	Yes	E	Pass
PUR-067	P2	F	Purchasing list filters by type/status/supplier/date	Yes	B	Pass
PUR-068	P2	F	Search by walk-in supplier typed name	Found	E	Pass
PUR-069	P2	F	Supplier payments tab lists payments	Yes with status	B	Pass
PUR-070	P2	F	Bill overdue in attention queue	Yes when past due	B	Fail
PUR-071	P2	F	Receive dialog shows remaining per line	Yes	B	Pass
PUR-072	P2	F	Receive dialog line set to 0	Line skipped; at least one line >0 required	E	Pass
PUR-073	P2	N	Receive all lines 0	Refused	E	Pass
PUR-074	P2	F	GRN warehouse = PO warehouse	Enforced	E	Pass
PUR-075	P2	F	Warehouse select on PO hidden when single	Yes	B	Pass
PUR-076	P2	F	Bill without GRN (direct bill)	Allowed? verify; if allowed, stock not moved	E	Pass
PUR-077	P2	F	Return on walk-in supplier bill	Works; typed name carried	E	Pass
PUR-078	P2	F	Supplier archived after PO	GRN conversion refused "archived. Restore it first."	E	Pass
PUR-079	P2	F	Product archived after PO	Conversion refused	E	Pass
PUR-080	P2	D	Idempotent retry of submit GRN	One stock move	E	Pass
PUR-081	P2	D	Idempotent retry of payment	One journal	E	Pass
PUR-082	P2	F	Payment note 1,000 chars; 1,001 refused	Yes	E	Pass
PUR-083	P2	F	Cheque number on outgoing cheque 100 chars	Yes	E	Pass
PUR-084	P2	U	Payment dialog shows open bills with outstanding	Oldest first; amount spreads oldest-first on typing total	B	Pass
PUR-085	P2	U	"PKR actually paid" only for non-PKR bills	Yes	B	Fail
PUR-086	P2	U	Editor on phone	Usable	B	Pass
PUR-087	P2	S	Branch-limited Accountant submits GRN in other branch	Refused	E	Pass
PUR-088	P2	F	Hidden Purchasing	View gone; API refused	E	Pass
PUR-089	P2	F	Imports off: no shipment button on GRN	Correct	E	Pass
PUR-090	P2	F	Imports on: "Create shipment" available for submitted GRN/bill	Yes (see IMP)	E	Pass
PUR-091	P2	D	FX: bill at 280, payment at 285 for $1,000	Loss 5,000 posted; payable cleared exactly	E	Pass
PUR-092	P2	D	FX: payment at 275	Gain 5,000	E	Pass
PUR-093	P2	D	Partial FX payments at different rates	Each computes vs bill rate on allocated portion	E	Pass
PUR-094	P2	E	Rate with 4 decimals 280.1234	Stored	E	Pass
PUR-095	P2	E	Rate 5 decimals	Rounded to 4	E	Fail
PUR-096	P2	F	Supplier payment print (voucher)	"Payment voucher" title	E	Pass
PUR-097	P2	X	500 POs in list	Filters < 1 s	B	Pass
PUR-098	P2	F	Overview shows purchases?	Not required; verify no error	E	Pass
PUR-099	P2	F	Aged payables report after chain	Buckets correct	E	Pass
PUR-100	P1	F	End to end: USD PO → two GRNs → bill → return 1 → payment with FX loss; reconciliation reconciled	All pass	E	Pass

Imports and landed cost (IMP): 94 pass · 6 fail · 0 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
IMP-001	P1	F	Create shipment from a submitted GRN	SHP-2026-001; supplier, origin default Shenzhen, port Karachi; status received	E	Pass
IMP-002	P1	N	Create from a draft GRN	"Choose a submitted receipt or supplier bill."	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
IMP-003	P1	N	Create from a PO	Same refusal	E	Pass
IMP-004	P2	F	Create from a submitted bill	Allowed	E	Pass
IMP-005	P2	F	Container, GD number, ETA fields	Saved; shown in list	E	Pass
IMP-006	P2	E	ETA invalid date	Validation	E	Pass
IMP-007	P2	E	GD number 100 chars	Saved	E	Pass
IMP-008	P1	F	Add cost: Freight 50,000, inventory, by value	Listed unallocated	E	Pass
IMP-009	P1	F	Add cost: Customs duty 80,000, inventory, by quantity	Listed	E	Pass
IMP-010	P1	F	Add cost: Sales tax at import 30,000, recoverable_tax	Listed	E	Pass
IMP-011	P1	F	Add cost: Container deposit 20,000, deposit	Listed	E	Pass
IMP-012	P1	F	Add cost: Clearing agent fee 5,000, expense	Listed	E	Pass
IMP-013	P1	N	Cost amount 0	Refused	E	Pass
IMP-014	P2	N	Cost label empty	Refused	E	Pass
IMP-015	P2	N	Classification `other` via API	Validation	E	Pass
IMP-016	P2	N	Basis `price` via API	Validation	E	Pass
IMP-017	P1	F	Allocate	Inventory costs added to each line's stock value by basis; tax → 1500-type recoverable; deposit → asset; expense → 5xxx	E	Pass
IMP-018	P1	D	By value: lines 70%/30% of value get 70/30 of freight	Exact split; remainder paisa on last line	E	Pass
IMP-019	P1	D	By quantity	Split by units	E	Pass
IMP-020	P1	D	By weight using product weight	Split by qty×weight	E	Pass
IMP-021	P1	D	By volume	Split by qty×volume	E	Pass
IMP-022	P2	D	Weight 0 on one product, basis weight	That line gets 0; others share; no NaN	E	Pass
IMP-023	P2	D	All weights 0, basis weight	Allocation refused or falls back to quantity; no division by zero	E	Pass
IMP-024	P1	D	Average cost after allocation	(stock value + allocated) / qty	E	Pass
IMP-025	P1	D	Allocation when part of the stock already sold	Remaining inventory gets its share; sold share hits COGS (reconcile)	E	Pass
IMP-026	P1	D	Allocation when all stock sold	Entire inventory cost to COGS; stock value stays 0	E	Pass
IMP-027	P1	N	Allocate twice	"All shipment costs are already allocated."	E	Pass
IMP-028	P2	F	Add another cost after allocation, allocate again	Only the new cost allocated; status costed	E	Pass
IMP-029	P2	N	Allocate in locked period	Refused	E	Pass
IMP-030	P2	S	Warehouse Manager allocates	Allowed? access: allocate_shipment Accountant only → refused	E	Pass
IMP-031	P2	S	Warehouse Manager creates shipment	Allowed	E	Pass
IMP-032	P2	S	Salesperson sees Imports	Hidden	E	Pass
IMP-033	P2	S	Branch-limited user, GRN in other branch	Refused	E	Pass
IMP-034	P2	F	Shipment status flow in_transit → customs → received → costed	Displayed; costed after allocation	E	Pass
IMP-035	P2	F	Shipment list columns: shipment, supplier, container/GD, ETA, status, unallocated	Correct totals	B	Pass
IMP-036	P2	F	Unallocated total = sum of unallocated costs	Yes	E	Pass
IMP-037	P2	F	Customs lookup (simulated) shows GD lookup result	No "simulated" banner on the page any more; lookup still returns stub data	B	Fail
IMP-038	P2	D	Journal on allocate: Dr 1200 (inventory part) / Dr 1500 (tax) / Dr deposit asset / Dr expense / Cr 2000 or accrued charges	Balanced; one journal per allocation	E	Pass
IMP-039	P2	D	Reconciliation after allocation	Inventory control = valuation	E	Pass
IMP-040	P2	D	P&L after allocation	Expense cost in P&L; inventory cost not	E	Pass
IMP-041	P2	D	Deposit later refunded	Manual journal reversal path; documented	E	Pass
IMP-042	P2	E	Cost 0.0001	Accepted; allocation rounds to 0 on small lines without error	E	Pass
IMP-043	P2	E	50 costs on one shipment	Allocated in one journal	E	Pass
IMP-044	P2	E	GRN with 100 lines	Allocation computes < 2 s	E	Pass
IMP-045	P2	E	Cost in USD?	Costs are PKR only; UI labels PKR	E	Pass
IMP-046	P2	F	Shipment for walk-in supplier GRN	Works	E	Pass
IMP-047	P2	F	Shipment from bill when GRN also exists	Lines from the bill used	E	Pass
IMP-048	P2	N	Shipment from cancelled GRN	Refused	E	Pass
IMP-049	P2	F	Two shipments for the same GRN	Allowed? verify; each allocates its own costs	E	Pass
IMP-050	P2	F	Delete cost before allocation	If supported; else not offered	E	Pass
IMP-051	P2	F	Edit cost after allocation	Not allowed (immutable)	E	Pass
IMP-052	P2	F	Shipment number sequence per year	SHP-2026-001, 002	E	Pass
IMP-053	P2	U	Shipment table on phone	Scroll hint	B	Pass
IMP-054	P2	U	Add cost dialog placeholders	Amount empty with placeholder	B	Fail
IMP-055	P2	U	Classification help text	Explains each	B	Fail

ID	P	T	CASE	EXPECTED	LAYER	RESULT
IMP-056	P2	F	Search finds shipment by number/container	Yes	B	Fail
IMP-057	P2	F	Attention queue: shipments with unallocated costs	Listed	B	Pass
IMP-058	P2	D	Allocation then stock transfer	Value moves with units	E	Pass
IMP-059	P2	D	Allocation then supplier return	Return at current avg cost incl. landed; payable reduced by bill price only (document)	E	Pass
IMP-060	P2	D	Allocation then count correction	Uses new avg	E	Pass
IMP-061	P2	D	Idempotent allocate retry	One journal	E	Pass
IMP-062	P2	F	Audit entries for shipment commands	Present	E	Pass
IMP-063	P2	F	Hidden Imports module with existing shipments	Hidden; data kept; API refused	E	Pass
IMP-064	P2	F	Turn Imports on later	Deposits and expense accounts provisioned once	E	Pass
IMP-065	P2	E	Label with emoji	Saved	E	Pass
IMP-066	P2	E	Amount 999,999,999,999	Refused (trillion rule)	E	Pass
IMP-067	P2	E	Negative amount	Refused	E	Pass
IMP-068	P2	F	ETA in the past	Allowed	E	Pass
IMP-069	P2	F	Origin/port blank	Defaults applied	E	Pass
IMP-070	P2	F	Shipment detail shows per-line landed cost	Value per unit after allocation	E	Pass
IMP-071	P2	D	Landed cost per unit = (price×rate + allocated)/qty	Exact	E	Pass
IMP-072	P2	D	Sales after landed cost use new COGS	Margin reflects	E	Pass
IMP-073	P2	F	Sales analysis margin after allocation	Updated	E	Pass
IMP-074	P2	S	Auditor views shipments	Read-only	E	Pass
IMP-075	P2	S	`add_shipment_cost` as Payroll Administrator	Refused	E	Pass
IMP-076	P2	F	Shipment with no costs allocate	"All shipment costs are already allocated." (nothing pending)	E	Pass
IMP-077	P2	F	Reports: stock valuation shows landed value	Yes	E	Pass
IMP-078	P2	F	GRN print unaffected by allocation	Yes	E	Pass
IMP-079	P2	D	Allocation rounding across 3 lines of 1/3 each	Sum equals cost exactly	E	Pass
IMP-080	P2	D	Allocation with lines of quantity 0.5	Works	E	Pass
IMP-081	P2	F	Attention clears after allocation	Yes	E	Pass
IMP-082	P2	F	Shipment list sorting newest first	Yes	B	Fail
IMP-083	P2	F	Filter by status	Works	B	Fail
IMP-084	P2	E	200 shipments	List OK	B	Pass
IMP-085	P2	F	Recoverable tax appears in tax summary/reports	Yes	E	Pass
IMP-086	P2	F	Expense classification appears in P&L under clearing/import expenses	Yes	E	Pass
IMP-087	P2	D	Deposit classification on balance sheet as asset	Yes	E	Pass
IMP-088	P2	F	Shipment keyboard navigation	Buttons reachable	B	Pass
IMP-089	P2	U	Status badges colours	Pastel per status	B	Pass
IMP-090	P2	F	Shipment from GRN of a supplier later archived	Still viewable	E	Pass
IMP-091	P2	F	Shipment when Inventory hidden	Imports requires Inventory; cannot be on alone	E	Pass
IMP-092	P2	N	Turn off Inventory with Imports on	Imports turned off too (dependency)	E	Pass
IMP-093	P2	D	Allocation date defaults today; custom date in open period	Journal dated accordingly	E	Pass
IMP-094	P2	N	Allocation date in future locked?	Only locked months refused	E	Pass
IMP-095	P2	F	Container number with spaces	Saved as typed	E	Pass
IMP-096	P2	X	Allocation on GRN with 100 lines and 20 costs	< 3 s	E	Pass
IMP-097	P2	F	Export shipments CSV (if present)	Columns correct	B	Pass
IMP-098	P2	F	Attention item link opens Imports	Yes	B	Pass
IMP-099	P2	D	Trial balance after all import postings	Balanced	E	Pass
IMP-100	P1	F	End to end: USD GRN → shipment → 5 classified costs → allocate → sell → P&L and reconciliation correct	All pass	E	Pass

Collections: receipts, cheques (PDC), follow-ups (COL): 96 pass · 4 fail · 0 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
COL-001	P1	F	Record receipt, customer, cash 5,000 to Cash in hand, allocate to invoice	Receipt posted; invoice paid 5,000; cash +5,000	E	Pass
COL-002	P1	F	Typing total spreads across open invoices oldest first	Allocation auto-filled; editable	B	Pass
COL-003	P1	N	Amount 0	Refused	E	Pass
COL-004	P1	N	Allocations don't sum to amount	"Allocations must equal the receipt amount."	E	Pass
COL-005	P1	N	Allocation exceeds invoice outstanding	"Allocation exceeds the invoice's outstanding balance."	E	Pass
COL-006	P1	N	Allocation to another customer's invoice	"Receipt must reference submitted invoices for the same party."	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
COL-007	P1	N	Allocation to a draft invoice	Same	E	Pass
COL-008	P2	N	Duplicate invoice in allocations	"Duplicate allocation."	E	Pass
COL-009	P2	N	No allocations	Validation min 1	E	Pass
COL-010	P2	E	Amount 0.0001	Accepted	E	Pass
COL-011	P2	E	Exact outstanding	Invoice fully paid; balance 0	E	Pass
COL-012	P2	F	Overpayment attempt	Refused (exceeds outstanding)	E	Pass
COL-013	P1	F	Bank receipt to chosen bank account	Bank balance up	E	Pass
COL-014	P2	N	Bank receipt with archived account	"{name} is archived."	E	Pass
COL-015	P2	N	Account id of another tenant	Not found	E	Pass
COL-016	P1	F	Cheque receipt with number and due date next week	Status received; NOT counted in collections; invoice not paid yet (verify design: paid on clear)	E	Pass
COL-017	P1	N	Cheque without number	"Enter the cheque number."	E	Pass
COL-018	P2	E	Cheque number 100 chars; 101 refused	Yes	E	Pass
COL-019	P2	F	Cheque due date = today	Allowed	E	Pass
COL-020	P2	F	Post-dated 90 days	Allowed; shown in cheques in hand	E	Pass
COL-021	P1	F	Deposit cheque choosing bank account	Status deposited; account stored	E	Pass
COL-022	P1	F	Clear cheque	Status cleared; bank +amount; invoice paid; commission released	E	Pass
COL-023	P1	N	Clear from received (skip deposit)	"Invalid cheque transition."	E	Pass
COL-024	P1	N	Deposit twice	"Invalid cheque transition."	E	Pass
COL-025	P1	F	Bounce from deposited	Status bounced; nothing collected	E	Pass
COL-026	P1	F	Bounce from cleared	Bank reversed; invoice unpaid again; commission release reversed; recovery recorded	E	Pass
COL-027	P2	N	Bounce a bounced cheque	Invalid transition	E	Pass
COL-028	P2	N	update_cheque on a cash receipt	"This payment is not a cheque."	E	Pass
COL-029	P2	N	Clear in locked period	Refused	E	Pass
COL-030	P2	F	Clear date custom	Journal dated accordingly	E	Pass
COL-031	P2	F	Cheques in hand list: received/deposited with actions	Buttons per state	B	Pass
COL-032	P2	F	Attention: cheques due today/overdue	Listed	B	Pass
COL-033	P2	F	Overview "Collections" metric	Cleared customer receipts only	E	Pass
COL-034	P2	D	Journal on cash receipt: Dr cash / Cr 1100	Yes	E	Pass
COL-035	P2	D	Journal on cheque receipt	None until cleared (or Dr cheques-in-hand; verify design)	E	Pass
COL-036	P2	D	Journal on bounce after clear	Exact reversal with source linkage	E	Pass
COL-037	P2	D	Receivable control after receipts	Reconciled	E	Pass
COL-038	P2	D	Idempotent retry of create_receipt	One receipt	E	Pass
COL-039	P2	D	Idempotent retry of clear	One journal	E	Pass
COL-040	P1	F	Record receipt for walk-in customer	Invoices listed by number and typed name	E	Pass
COL-041	P2	F	Receipt note 1,000 chars	Saved	E	Pass
COL-042	P2	F	Receipt employeeId (collector)	Stored; optional	E	Pass
COL-043	P2	F	Receipt print	"Receipt" title, "Received from", allocations	E	Pass
COL-044	P2	F	Receipt numbering RCT-? per year	Sequential	E	Pass
COL-045	P1	S	Recovery Officer records customer receipt	Allowed	E	Pass
COL-046	P2	S	Salesperson records receipt	Allowed? access: create_receipt roles → verify and assert	E	Pass
COL-047	P2	S	Warehouse Manager records receipt	Refused	E	Pass
COL-048	P2	S	Branch-limited user clears cheque for other branch invoice	Refused	E	Pass
COL-049	P1	F	Log follow-up with note and next date	"Follow-up saved · next action YYYY-MM-DD."	E	Pass
COL-050	P1	N	Follow-up empty note	"Enter a follow-up note."	E	Pass
COL-051	P2	N	Follow-up invalid next date	Validation	E	Pass
COL-052	P2	F	Follow-up next date today	Attention shows due	E	Pass
COL-053	P2	F	Follow-up on supplier via API	"Customer not found." (parties lookup) or refused	E	Fail
COL-054	P2	F	Follow-up list per customer newest first	Yes	B	Pass
COL-055	P2	F	Follow-up dialog closes on success	Yes (follow_up auto-close)	B	Pass
COL-056	P2	S	Salesperson logs follow-up	Refused (Recovery Officer, Sales Manager, Owner)	E	Pass
COL-057	P2	F	Aged receivables after partial receipt	Buckets by due date; amounts net of paid	E	Pass
COL-058	P2	F	Customer statement running balance	Invoice +, receipt –, credit note –	E	Pass
COL-059	P2	F	Statement shows cheque only when cleared	Yes (or as pending line; assert design)	E	Pass
COL-060	P2	E	Receipt date before invoice date	Allowed? assert; no negative aging crash	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
COL-061	P2	E	Receipt in future date	Allowed or refused consistently	E	Pass
COL-062	P2	F	Receipt against invoice fully returned	Outstanding 0 → refused	E	Pass
COL-063	P2	F	Receipt against invoice partially returned	Up to remaining	E	Pass
COL-064	P2	F	Credit note then receipt larger than remaining	Refused	E	Pass
COL-065	P2	F	Receipt dialog on phone	Bottom sheet; allocations table scroll	B	Pass
COL-066	P2	U	Method radio/select shows cheque fields only for cheque	Yes	B	Pass
COL-067	P2	U	Account select lists bank and cash accounts, not archived	Yes	B	Pass
COL-068	P2	U	Amount placeholder	Empty with placeholder	B	Pass
COL-069	P2	F	Collections hidden module	Receipts still visible in Accounting; attention cheque items gone	E	Pass
COL-070	P2	F	Cheque register also visible in Cash & bank	Same data	B	Pass
COL-071	P2	D	Bounce after commission paid in payroll	Recovery recorded on commission; next payroll deducts	E	Pass
COL-072	P2	D	Clear after bounce?	Invalid (bounced is terminal)	E	Pass
COL-073	P2	D	Two cheques for the same invoice	Both allowed up to outstanding	E	Pass
COL-074	P2	D	Commission release proportional to cash	50% cash → 50% released	E	Pass
COL-075	P2	F	Receipts list filters by method/status	Yes	B	Fail
COL-076	P2	F	Search by cheque number	Found	B	Fail
COL-077	P2	E	Cheque number with spaces/dashes	Saved	E	Pass
COL-078	P2	E	Note with emoji	Saved	E	Pass
COL-079	P2	F	Deposit without account id	Allowed? API accepts optional; UI requires select → assert	E	Pass
COL-080	P2	F	Deposit into cash account type	Refused or allowed; assert design	E	Pass
COL-081	P2	S	Auditor views receipts	Read-only	E	Pass
COL-082	P2	F	Receipt against invoice in locked period but receipt dated open period	Allowed	E	Pass
COL-083	P2	F	Receipt allocation to 10 invoices	Each updated	E	Pass
COL-084	P2	X	5,000 receipts	List/aging < 2 s	B	Fail
COL-085	P2	F	Attention "Cheques due" count badge on bell	Matches	B	Pass
COL-086	P2	F	Follow-up actor name recorded	Signed-in member	E	Pass
COL-087	P2	F	Follow-up on archived customer	Allowed? assert	E	Pass
COL-088	P2	D	Receipt journal memo contains receipt number	Yes	E	Pass
COL-089	P2	D	Bank account code used in journal = account's ledger code	Yes	E	Pass
COL-090	P2	F	Receipt for supplier via Collections	Not offered (Purchasing → Supplier payments)	B	Pass
COL-091	P2	F	Clear cheque dated earlier than deposit	Allowed; order by date in statement	E	Pass
COL-092	P2	F	Bulk: 20 cheques cleared one by one	Each independent	E	Pass
COL-093	P2	U	Status badges: received amber, deposited info, cleared green, bounced red	Yes	B	Pass
COL-094	P2	U	Receipt row actions by state	Deposit/Cleared/Bounced only valid ones	B	Pass
COL-095	P2	F	Print cheque receipt before clearing	Shows status received	E	Pass
COL-096	P2	F	Recovery Officer sees only sales documents and receipts	Yes	E	Pass
COL-097	P2	F	Customer with no open invoices in receipt dialog	"No open invoices" state; cannot save	E	Pass
COL-098	P2	D	Outstanding = total − paid − returned always	Spot-check 5 invoices	E	Pass
COL-099	P2	D	Trial balance after bounce sequence	Balanced	E	Pass
COL-100	P1	F	End to end: invoice 10,000 → cash 4,000 → cheque 6,000 deposit → clear → bounce → re-receive cash; statement and commission correct	All pass	E	Pass

Cash and bank (CSH): 96 pass · 3 fail · 1 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
CSH-001	P1	F	Open Cash & bank	Cards per account with balance; total	E	Pass
CSH-002	P1	F	Fresh company	"Main bank account" and "Cash in hand" provisioned with 0	E	Pass
CSH-003	P1	F	Add bank account "HBL Current" with details	Created; new ledger code	E	Pass
CSH-004	P1	N	Duplicate name (case-insensitive)	"An account named HBL Current already exists."	E	Pass
CSH-005	P1	N	Name 1 char	Refused (min 2)	E	Pass
CSH-006	P2	E	Name 80 chars; 81 refused	Yes	E	Pass
CSH-007	P2	E	Details 200 chars; 201 refused	Yes	E	Pass
CSH-008	P2	N	Type `wallet` via API	Validation	E	Pass
CSH-009	P2	E	Add accounts until limit	"Too many bank accounts." at the cap (assert number)	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
CSH-010	P1	F	Transfer 10,000 bank → cash	Bank −10,000, cash +10,000; journal memo "Transfer A → B"	E	Pass
CSH-011	P1	N	Transfer same account	"Choose two different accounts."	E	Pass
CSH-012	P1	N	Transfer 0	Refused	E	Pass
CSH-013	P2	F	Transfer more than balance	Allowed (overdraft) — assert design; balance negative shown	E	Pass
CSH-014	P2	N	Transfer in locked period	Refused	E	Pass
CSH-015	P2	F	Transfer memo 200 chars; 201 refused	Yes	E	Pass
CSH-016	P2	F	Transfer to archived account	"{name} is archived."	E	Pass
CSH-017	P1	F	Record expense "Fuel" 2,000 from cash	Cash −2,000; expense account 5300 default	E	Pass
CSH-018	P2	F	Expense account choice 5100/5200/5300	Posted to chosen	E	Pass
CSH-019	P1	N	Expense memo 1 char	Refused (min 2)	E	Pass
CSH-020	P1	N	Expense amount 0	Refused	E	Pass
CSH-021	P2	N	Expense account `9999` via API	Validation	E	Pass
CSH-022	P2	N	Expense in locked period	Refused	E	Pass
CSH-023	P1	F	Account activity view	Lines with date, memo, in/out, running balance	E	Pass
CSH-024	P2	F	Running balance correct after mixed receipts, payments, transfers, expenses	Verified	E	Pass
CSH-025	P2	F	Activity filter by date	Works	E	Pass
CSH-026	P1	F	Archive an empty account	Archived; hidden from selects	E	Pass
CSH-027	P1	N	Archive account with balance	"{name} still has a balance. Transfer it out before archiving."	E	Pass
CSH-028	P2	F	Restore archived account	Back	E	Pass
CSH-029	P2	F	Rename account	Journals show new name? (memo frozen) — balances unchanged	E	Pass
CSH-030	P2	F	Change account type bank→cash	Allowed; code unchanged	E	Pass
CSH-031	P2	F	Cheque register tab	Cheques with states and actions	E	Pass
CSH-032	P2	F	Deposit from register	Same as Collections	E	Pass
CSH-033	P2	D	Balance = journal balance of the account's code	Equal	E	Pass
CSH-034	P2	D	Cash/bank control reconciliation	Reconciled	E	Pass
CSH-035	P2	D	Transfer journal balanced two lines	Yes	E	Pass
CSH-036	P2	D	Expense journal: Dr expense / Cr cash	Yes	E	Pass
CSH-037	P2	D	Idempotent transfer retry	One journal	E	Pass
CSH-038	P2	S	Salesperson opens Cash & bank	Hidden; API refused for transfer/expense	E	Pass
CSH-039	P2	S	Recovery Officer records expense	Refused (Accountant/Owner)	E	Pass
CSH-040	P2	S	Accountant archives account	Allowed	E	Pass
CSH-041	P2	S	Auditor	Read-only	E	Pass
CSH-042	P2	F	Cash & bank hidden	Receipts still choose accounts (bankAccounts always sent)	E	Pass
CSH-043	P2	F	Receipts dialog account list uses these accounts	Yes	E	Pass
CSH-044	P2	F	Supplier payment from new HBL account	Posted to its code	E	Pass
CSH-045	P2	E	Amount 0.0001	Accepted	E	Pass
CSH-046	P2	E	Amount 999,999,999,999	Refused	E	Pass
CSH-047	P2	E	Memo with emoji	Saved	E	Pass
CSH-048	P2	E	Memo with newline	Trimmed/one line	E	Pass
CSH-049	P2	U	Cards show type icon (bank/cash)	Yes	B	Pass
CSH-050	P2	U	Negative balance styled red	Yes	B	Fail
CSH-051	P2	U	Dialogs on phone	Bottom sheets	B	Fail
CSH-052	P2	F	Transfer dialog closes on success	Yes	B	Pass
CSH-053	P2	F	Expense dialog closes on success	Yes	B	Pass
CSH-054	P2	F	Notice on success shows amount with 2 dp	"Transferred 10000.00 from …"	E	Pass
CSH-055	P2	F	Date defaults today (Asia/Karachi)	Yes	E	Pass
CSH-056	P2	E	Date 2026-02-29 (not leap)	Validation	E	Pass
CSH-057	P2	F	Transfer between two bank accounts	Works	E	Pass
CSH-058	P2	F	Transfer between two cash accounts	Works	E	Pass
CSH-059	P2	F	Three transfers then balances	Sum zero across accounts	E	Pass
CSH-060	P2	F	Expense with 4-decimal amount	Stored 4 dp	E	Pass
CSH-061	P2	F	Overview attention: negative cash?	Not required; no crash	E	Pass
CSH-062	P2	F	Total balance card = sum	Yes	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
CSH-063	P2	F	Account with 10,000 lines	Activity paginates or scrolls smoothly	E	Pass
CSH-064	P2	F	Export activity CSV (if present)	Correct	E	Pass
CSH-065	P2	F	Search finds account by name	Yes	E	Pass
CSH-066	P2	F	Account details text shows on card	Yes	B	Pass
CSH-067	P2	S	`save_bank_account` with archived=true and balance via API	Refused	E	Pass
CSH-068	P2	S	`bank_transfer` with other tenant's account id	Not found	E	Pass
CSH-069	P2	F	Deleting accounts not possible	Correct	E	Pass
CSH-070	P2	D	Ledger code uniqueness across accounts	Each distinct	E	Pass
CSH-071	P2	D	Code survives rename	Yes	E	Pass
CSH-072	P2	D	Provisioning twice (module off/on)	No duplicate accounts	E	Pass
CSH-073	P2	F	Receipt into cash then expense same amount	Cash back to 0	E	Pass
CSH-074	P2	F	Receipt, then transfer, then payment chain	Balances correct	E	Pass
CSH-075	P2	F	Cheque cleared into HBL	HBL activity line	E	Pass
CSH-076	P2	F	Bounce after clear	Reversal line in activity	E	Pass
CSH-077	P2	U	Activity shows in/out columns with running balance	Yes	B	Pass
CSH-078	P2	U	Memo truncation with tooltip	Readable	B	Pass
CSH-079	P2	F	Period lock then view activity	Viewing allowed	E	Pass
CSH-080	P2	F	Owner reopens period, then expense	Allowed	E	Pass
CSH-081	P2	F	Expense memo "Fuel" twice same day	Two journals (no dedupe)	E	Pass
CSH-082	P2	F	Transfer 10,000.5	Stored	E	Pass
CSH-083	P2	F	Keyboard: Enter submits transfer form	Yes	B	Pass
CSH-084	P2	F	Escape closes dialogs	Yes	B	Pass
CSH-085	P2	F	Account select order	Bank first then cash or alphabetical; consistent	B	Pass
CSH-086	P2	F	Balance after company reset	0	—	Manual
CSH-087	P2	F	Account names with Urdu	Saved	E	Pass
CSH-088	P2	F	Record expense with date in previous open month	Posted there; P&L month correct	E	Pass
CSH-089	P2	D	P&L shows expense	Yes	E	Pass
CSH-090	P2	D	Balance sheet cash = sum of accounts	Yes	E	Pass
CSH-091	P2	D	Trial balance includes each account code	Yes	E	Pass
CSH-092	P2	X	50 accounts	Cards grid OK	E	Pass
CSH-093	P2	F	Hidden module then receipt → account still selectable	Yes	E	Pass
CSH-094	P2	F	Mobile card tap opens activity	Yes	B	Pass
CSH-095	P2	U	Card hover lift only with mouse	Yes	B	Fail
CSH-096	P2	F	Transfer confirmation copy	Names both accounts	E	Pass
CSH-097	P2	F	Activity for archived account	Viewable	E	Pass
CSH-098	P2	F	Attempt payment from archived account	Refused	E	Pass
CSH-099	P2	D	Reconciliation after 100 random operations (fuzz)	Still reconciled	E	Pass
CSH-100	P1	F	End to end: add account → receipts → transfer → expense → archive empty account; balances and reports agree	All pass	E	Pass

Commissions (COM): 100 pass · 0 fail · 0 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
COM-001	P1	F	Commissions → Salespeople → New employee "Hamza, Order Booker"	Saved; appears in invoice salesperson select	E	Pass
COM-002	P1	F	Plans: edit provisioned "Standard plan" to flat 2%	Saved	E	Pass
COM-003	P1	N	Plan effectiveTo before effectiveFrom	"Plan dates are invalid."	E	Pass
COM-004	P1	N	Second plan overlapping dates	"Commission plan dates overlap."	E	Pass
COM-005	P2	F	Adjacent plans (to = 2026-12-31, next from = 2027-01-01)	Allowed	E	Pass
COM-006	P2	N	Plan name 1 char	Refused	E	Pass
COM-007	P2	N	Rate −1	Refused	E	Pass
COM-008	P2	N	Rate 101 on category rates (pct)	Refused	E	Pass
COM-009	P2	N	Tiers empty	Validation min 1	E	Pass
COM-010	P2	N	Multiplier 0	Refused (pos)	E	Pass
COM-011	P2	F	Category plan: Lighting = 2.5	Lines in Lighting earn 2.5%, others 0 or base rate (assert)	E	Pass
COM-012	P2	F	Customer class plan: Wholesaler = 1	By customer class snapshot	E	Pass
COM-013	P2	F	Per-unit plan rate 10	Earned = qty × 10	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
COM-014	P2	F	Tiered plan 0 = 2, 500000 = 3	Rate by cumulative base (assert definition)	E	Pass
COM-015	P2	F	Target and multiplier	Applied when target met (assert definition)	E	Pass
COM-016	P1	F	Invoice 10,000 net with Hamza, flat 2%	Entitlement earned 200, released 0	E	Pass
COM-017	P1	F	Cash receipt 5,000	Released 100	E	Pass
COM-018	P1	F	Cheque cleared 5,000	Released 200 (full)	E	Pass
COM-019	P1	F	Cheque only deposited	No release	E	Pass
COM-020	P1	F	Cheque bounced after clear	Release reversed; movement release_reversed	E	Pass
COM-021	P1	F	Bounce after paid	Recovery recorded; paid > released; next payroll deducts	E	Pass
COM-022	P1	F	Approve released entitlement	approved=true; "Commission approved for payroll."	E	Pass
COM-023	P1	N	Approve with nothing released	"There is no released unpaid commission to approve."	E	Pass
COM-024	P2	N	Approve already fully paid	Same refusal	E	Pass
COM-025	P2	F	Approve several ids	All approved	E	Pass
COM-026	P2	N	Approve empty ids	Validation min 1	E	Pass
COM-027	P2	S	Branch-limited approver, invoice in other branch	Refused	E	Pass
COM-028	P1	D	Invariant earned ≥ released ≥ paid	Holds through all cases	E	Pass
COM-029	P1	D	Plan snapshot stored on entitlement	Changing the plan later does not alter earned	E	Pass
COM-030	P2	D	Movements immutable, one per event	accrued, released, release_reversed, paid, recovery_recorded	E	Pass
COM-031	P2	F	Invoice with no salesperson	No entitlement	E	Pass
COM-032	P2	F	Invoice line split 60/40 between two salespeople	Two entitlements by share	E	Pass
COM-033	P2	N	Splits total 90	"Commission splits must total 100%."	E	Pass
COM-034	P2	N	Split with unknown employee	"Employee not found."	E	Pass
COM-035	P2	F	Credit note (return) on commissioned invoice	Entitlement adjusted (entitlement_adjusted) proportionally	E	Pass
COM-036	P2	F	Return after release	Released capped at new earned; movement	E	Pass
COM-037	P2	F	Return after paid	Recovery	E	Pass
COM-038	P2	F	Discounted line base	Base = net after discount, before tax	E	Pass
COM-039	P2	F	Tax excluded from base	Yes	E	Pass
COM-040	P2	F	Entitlements list: columns earned/released/paid/approved	Correct	E	Pass
COM-041	P2	F	Filter by salesperson	Works	E	Pass
COM-042	P2	F	Entitlement links to invoice	Opens details	E	Pass
COM-043	P1	S	Salesperson sees only own entitlements	Yes	E	Pass
COM-044	P2	S	Sales Manager sees all; cannot see salaries	Employees shown with salary 0	E	Pass
COM-045	P2	S	Payroll Administrator sees entitlements	Yes	E	Pass
COM-046	P2	S	Recovery Officer sees Commissions	Hidden	E	Pass
COM-047	P2	S	Salesperson approves	Refused (Sales Manager, Accountant, Owner)	E	Pass
COM-048	P2	F	Employee hierarchy: Reports to	Saved; manager chain	E	Pass
COM-049	P2	N	Hierarchy cycle A→B→A	"The sales hierarchy cannot contain a cycle."	E	Pass
COM-050	P2	N	Employee as own manager	Cycle refused	E	Pass
COM-051	P2	F	Deactivate employee	Hidden from invoice select; history kept	E	Pass
COM-052	P2	F	Invoice with inactive employee via API	Allowed? assert; else refused	E	Pass
COM-053	P2	F	Employee import CSV	Created; manager by name	E	Pass
COM-054	P2	N	Import manager name not found	"manager X not found"	E	Pass
COM-055	P2	N	Import duplicate employee	"employee X appears more than once in the file"	E	Pass
COM-056	P2	N	Import existing employee create mode	"employee X already exists"	E	Pass
COM-057	P2	F	Export employees	Salary columns only for pay roles (assert projection)	E	Pass
COM-058	P2	E	Employee name 100 chars; 101 refused	Yes	E	Pass
COM-059	P2	E	Attendance 31	Refused (max 30)	E	Pass
COM-060	P2	E	Salary negative	Refused	E	Pass
COM-061	P2	F	Commission with Collections hidden	Cash still releases (engine) — assert	E	Pass
COM-062	P2	F	Commissions hidden then invoice submitted	No entitlement? assert gating of accrual	E	Pass
COM-063	P2	F	Turn Commissions on later	Past invoices do not accrue retroactively	E	Pass
COM-064	P2	D	Reconciliation: commission expense (accrued) = sum earned; payable = earned − paid	Reconciled	E	Pass
COM-065	P2	D	Journal on accrual: Dr commission expense / Cr commission payable	At submit	E	Pass
COM-066	P2	D	Journal on payroll pay	Payable cleared; no double expense	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
COM-067	P2	F	Plan effective dates vs invoice date	Invoice dated outside any plan → no entitlement	E	Pass
COM-068	P2	F	Two plans in different years, invoice in each	Correct plan each	E	Pass
COM-069	P2	E	Rate 0	Entitlement 0 (or none)	E	Pass
COM-070	P2	E	Rate 100	Earned = base	E	Pass
COM-071	P2	E	Per-unit on 0.5 qty	5	E	Pass
COM-072	P2	E	Tier threshold equal to base	Boundary inclusive? assert	E	Pass
COM-073	P2	F	Per-unit plan with quantity returned	Adjusted	E	Pass
COM-074	P2	F	Entitlement when invoice cancelled (draft)	Never created	E	Pass
COM-075	P2	F	Multiple receipts across many invoices	Each release attributed to its invoice	E	Pass
COM-076	P2	F	Overpayment prevented so release never > earned	Yes	E	Pass
COM-077	P2	D	Rounding: 2% of 33.33	0.6666 stored 4 dp; movements sum exactly	E	Pass
COM-078	P2	D	Sum of released across partial receipts = total when fully paid	Exact	E	Pass
COM-079	P2	F	Attention: released unapproved commissions	Listed	E	Pass
COM-080	P2	F	Sales analysis by salesperson includes commission	Yes	E	Pass
COM-081	P2	U	Entitlement badges earned/released/approved	Colours	B	Pass
COM-082	P2	U	Plans form pairs textarea "Lighting = 2.5"	Parsed; invalid lines ignored	E	Pass
COM-083	P2	E	Pairs with `=` in key	Split at first `=`	E	Pass
COM-084	P2	F	Plan form shows only fields for its type	Yes or all; consistent	B	Pass
COM-085	P2	F	Salesperson "View as" preview from Owner	Entitlements filtered	B	Pass
COM-086	P2	F	Employee territory field	Saved; filter	E	Pass
COM-087	P2	F	Search finds employee	Yes	E	Pass
COM-088	P2	X	10,000 entitlements	List < 2 s	E	Pass
COM-089	P2	F	LIVE label in sidebar	Present on Commissions	B	Pass
COM-090	P2	D	Idempotent approve retry	No change second time	E	Pass
COM-091	P2	F	Approve then invoice bounce	approved stays; released drops; payroll refuses "Commission changed…"	E	Pass
COM-092	P2	F	Walk-in customer invoice with salesperson	Commission accrues normally	E	Pass
COM-093	P2	F	Credit note on walk-in	Adjusts	E	Pass
COM-094	P2	F	Employee deactivated with unpaid commission	Still payable; shown	E	Pass
COM-095	P2	F	Payroll off: approvals still possible	Yes; "ready for payroll"	E	Pass
COM-096	P2	F	Hide Commissions with password	Counts show plans/entitlements	E	Pass
COM-097	P2	D	Audit log for approvals	Yes	E	Pass
COM-098	P2	S	`save_plan` as Salesperson	Refused	E	Pass
COM-099	P2	S	`approve_commission` for other tenant's id	Not found	E	Pass
COM-100	P1	F	End to end: plan → invoice with split → partial cash → cheque clear → approve → payroll pay → bounce → recovery in next payroll	All invariants hold	E	Pass

Payroll (PAY): 94 pass · 6 fail · 0 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PAY-001	P1	N	Create payroll with no statutory settings	"Enter EOBI and income-tax settings in Settings → Payroll before running payroll."	E	Pass
PAY-002	P1	F	Save settings: employer EOBI 5, employee EOBI 1, base 32,000, tax 5	Saved	E	Pass
PAY-003	P1	N	Create payroll with no active employees	"Add active employees before creating payroll."	E	Pass
PAY-004	P1	F	Create payroll 2026-10	Draft run with a line per active employee; gross, EOBI, tax, commission, net	E	Pass
PAY-005	P1	N	Create again for 2026-10	"A payroll run already exists for this month."	E	Pass
PAY-006	P2	N	Month `2026-13`	Validation	E	Pass
PAY-007	P2	N	Month `10-2026`	Validation	E	Pass
PAY-008	P2	F	Default date = month-28	Yes	E	Pass
PAY-009	P2	N	Date in locked period	Refused	E	Pass
PAY-010	P1	F	Gross = salary + allowance (attendance-prorated per rule)	Assert formula with attendance 30 vs 15	E	Pass
PAY-011	P2	E	Attendance 0	Gross 0 or base? assert rule; no negative	E	Pass
PAY-012	P1	F	Employee EOBI = base × 1%	320	E	Pass
PAY-013	P1	F	Employer EOBI = base × 5%	1,600 (expense, not deducted)	E	Pass
PAY-014	P1	F	Income tax = 5% of earnings	Computed on gross + commission	E	Pass
PAY-015	P1	N	Deductions exceed earnings	"Deductions exceed earnings for {name}."	E	Pass
PAY-016	P1	F	Approved released commission included	Commission column; commissionIds recorded	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PAY-017	P1	F	Unapproved commission excluded	Yes	E	Pass
PAY-018	P1	F	Released but unapproved → not paid; approve later → next run	Yes	E	Pass
PAY-019	P1	F	Commission held by an unpaid run is not in another run	Yes	E	Pass
PAY-020	P1	F	Review → Approve → Pay transitions	Each allowed in order	E	Pass
PAY-021	P1	N	Draft → Approved (skip)	"Invalid payroll transition."	E	Pass
PAY-022	P1	N	Paid → anything	Invalid	E	Pass
PAY-023	P1	N	Commission changed between calculate and approve (bounce)	"Commission changed since this payroll was calculated. Reconcile the run before approval."	E	Pass
PAY-024	P1	N	Recovery changed	"Commission recovery changed since this payroll was calculated. Reconcile the run before approval."	E	Pass
PAY-025	P2	F	Reconcile = recreate run (cancel draft and create)	Documented path works	E	Fail
PAY-026	P1	D	Pay posts: Dr salary expense / Dr employer EOBI / Cr payroll payable / Cr statutory payable / commission payable cleared	Balanced; commission not expensed twice	E	Pass
PAY-027	P1	D	Commission paid updates entitlement paid; movement paid	Yes	E	Pass
PAY-028	P1	F	Bounce recovery: paid 200, bounce → recovery 200 in next run	Net reduced by 200, floor 0	E	Pass
PAY-029	P2	F	Recovery larger than net	Partial recovery; remainder stays recoverable	E	Pass
PAY-030	P2	F	Attendance import CSV	Updates employees' attendance	E	Pass
PAY-031	P2	N	Attendance import unknown employee	"employee X not found"	E	Pass
PAY-032	P2	N	Attendance 31 in file	Row error	E	Fail
PAY-033	P2	F	Payroll print/payslip (if present)	Per employee	E	Pass
PAY-034	P2	F	Payroll list shows runs with status and total	Yes	E	Pass
PAY-035	P2	S	Payroll Administrator creates run	Allowed	E	Pass
PAY-036	P2	S	Sales Manager opens Payroll	Hidden; salaries zeroed in projection	E	Pass
PAY-037	P2	S	Salesperson API create_payroll	Refused	E	Pass
PAY-038	P2	S	Auditor views payroll	Read-only full	E	Pass
PAY-039	P2	S	Branch-limited Accountant	Payrolls not sent	E	Pass
PAY-040	P2	F	Payroll off in Features	Sidebar absent; API module-off message	E	Pass
PAY-041	P2	F	Turn Payroll on later	Accounts provisioned; settings still required	E	Pass
PAY-042	P2	E	Salary 0 employee	Line with 0 gross; commission still paid	E	Pass
PAY-043	P2	E	Employee with only commission	Net = commission − tax	E	Pass
PAY-044	P2	E	500 employees	Run computes < 5 s	E	Pass
PAY-045	P2	E	Settings: EOBI 100%	Allowed (pct ≤ 100)	E	Pass
PAY-046	P2	N	Settings tax 101	Refused	E	Pass
PAY-047	P2	E	Settings blank strings	Allowed to save but payroll refuses	E	Pass
PAY-048	P2	F	Numbering per year	PAY-2026-0001	E	Pass
PAY-049	P2	F	Run for a past month	Allowed if open	E	Pass
PAY-050	P2	F	Run for a future month	Allowed? assert	E	Pass
PAY-051	P2	D	Two runs different months with same commission	Second excludes paid ones	E	Pass
PAY-052	P2	D	Payroll totals = sum of lines	Yes	E	Pass
PAY-053	P2	D	Reconciliation: payroll payable	Reconciled	E	Pass
PAY-054	P2	F	Cancel/delete draft run	If offered; else document	E	Pass
PAY-055	P2	F	Employee deactivated after run created	Line stays	E	Pass
PAY-056	P2	F	Employee deactivated before run	Excluded	E	Pass
PAY-057	P2	F	Deduction field on employee	Deducted	E	Pass
PAY-058	P2	E	Deduction equal to earnings	Net 0 allowed	E	Pass
PAY-059	P2	E	Deduction > earnings	Refused message	E	Pass
PAY-060	P2	F	Review as Accountant, approve as Owner, pay as Accountant	Any permitted role per step	E	Pass
PAY-061	P2	S	Pay as Payroll Administrator	Allowed	E	Pass
PAY-062	P2	F	Settings → Payroll visible only with Payroll on	Yes	B	Fail
PAY-063	P2	U	Lines table columns readable on phone	Scroll	B	Fail
PAY-064	P2	U	Status badges	Colours	B	Fail
PAY-065	P2	F	Audit entries for each transition	Yes	E	Pass
PAY-066	P2	D	Idempotent pay retry	One journal	E	Pass
PAY-067	P2	F	Pay in locked period	Refused	E	Pass
PAY-068	P2	F	Lock period after pay	Allowed; run unchanged	E	Pass
PAY-069	P2	F	Commission from walk-in customer invoices	Included	E	Pass
PAY-070	P2	D	Entitlement paid ≤ released after pay	Yes	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PAY-071	P2	D	Movement paid amount per entitlement	Yes	E	Pass
PAY-072	P2	F	Employee salary change after draft run	Run keeps calculated values	E	Pass
PAY-073	P2	F	Settings change after draft	Run keeps values	E	Pass
PAY-074	P2	E	Base 0	EOBI 0	E	Pass
PAY-075	P2	E	Tax 0	No tax	E	Pass
PAY-076	P2	F	Export payroll CSV (if present)	Correct	E	Pass
PAY-077	P2	F	Search finds payroll run	Yes	E	Pass
PAY-078	P2	F	Overview no payroll numbers for non-pay roles	Correct	E	Pass
PAY-079	P2	D	Trial balance after pay	Balanced	E	Pass
PAY-080	P2	D	P&L shows salary and employer EOBI	Yes	E	Pass
PAY-081	P2	D	Balance sheet shows payables until settled	Yes	E	Pass
PAY-082	P2	F	Settle payroll payable via Cash & bank expense/transfer	Documented path	E	Pass
PAY-083	P2	F	Attendance 30 default	Yes	E	Pass
PAY-084	P2	E	Attendance 29.5	Allowed? number min 0 max 30 → allowed	E	Pass
PAY-085	P2	F	Reports: payroll summary (if present)	Correct	E	Pass
PAY-086	P2	S	`update_payroll` for other tenant id	Not found	E	Pass
PAY-087	P2	F	Hidden Payroll with existing runs	Kept; hidden	E	Pass
PAY-088	P2	F	Review button disabled while busy	Yes	B	Pass
PAY-089	P2	U	Line commission tooltip lists invoices	If present	B	Pass
PAY-090	P2	F	Multiple recoveries across employees	Each independent	E	Pass
PAY-091	P2	D	Recovery never exceeds paid − released	Yes	E	Pass
PAY-092	P2	D	After recovery, entitlement paid reduced	Yes	E	Pass
PAY-093	P2	F	Run with 0 total	Allowed; posting of zero lines skipped	E	Pass
PAY-094	P2	F	Error wording for prototype rules	No "prototype" banner in UI (removed); message text "Payroll calculated using prototype rules." still returned → flag for copy change	E	Fail
PAY-095	P2	X	12 runs in a year	List OK	E	Pass
PAY-096	P2	F	Keyboard flow through transitions	Buttons focusable	B	Pass
PAY-097	P2	F	Payroll Administrator sees sales invoices only (projection)	Yes	E	Pass
PAY-098	P2	F	Payroll for month with no invoices	Commission 0	E	Pass
PAY-099	P2	D	Sum of net pay + deductions + tax + employee EOBI = gross + commission − recovery	Exact	E	Pass
PAY-100	P1	F	End to end: settings → employees → invoices/receipts → approve → run → review → approve → pay → bounce → next run recovers	All pass	E	Pass

Accounting: journals, reversals, periods, invariants (ACC): 97 pass · 3 fail · 0 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
ACC-001	P1	F	Accounting view: journals list newest first with number, date, memo, status	Yes	E	Pass
ACC-002	P1	F	Open a journal	Lines with account names, debit, credit; totals equal	E	Pass
ACC-003	P1	F	Trial balance tab	Debits = credits	E	Pass
ACC-004	P1	F	Create manual journal Dr 5300 100 / Cr cash 100	"Balanced journal posted."	E	Pass
ACC-005	P1	N	Unbalanced journal	"Journal debits and credits must balance."	E	Pass
ACC-006	P1	N	Unknown account code	"Journal accounts and amounts must be valid."	E	Pass
ACC-007	P1	N	Line with both debit and credit	Same refusal	E	Pass
ACC-008	P1	N	Negative amount	Same	E	Pass
ACC-009	P2	N	One line only	Validation min 2	E	Pass
ACC-010	P2	E	Lines with zero on both sides	Dropped; if nothing left, no journal posted	E	Pass
ACC-011	P2	E	0.0001 amounts	Posted	E	Pass
ACC-012	P2	E	Memo empty	Allowed? assert; shows blank	E	Pass
ACC-013	P2	F	Journal with bank account code	Allowed (bank codes known)	E	Pass
ACC-014	P1	F	Reverse a manual journal	"Reversal · J-…" journal with swapped lines; reversalOf set	E	Pass
ACC-015	P1	N	Reverse twice	"Journal is already reversed."	E	Pass
ACC-016	P1	N	Reverse an invoice journal	"Use the source document's correction workflow for operational journals."	E	Pass
ACC-017	P1	N	Reverse an opening-balance journal	Same	E	Fail
ACC-018	P2	N	Reverse a receipt journal	Same	E	Pass
ACC-019	P2	F	Reverse with custom date	Dated accordingly; period check	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
ACC-020	P1	F	Lock period 2026-09	"Period locked."	E	Pass
ACC-021	P1	N	Post anything dated 2026-09	"Accounting period 2026-09 is locked."	E	Pass
ACC-022	P1	N	Accountant reopens	"Only the owner can reopen an accounting period."	E	Pass
ACC-023	P1	F	Owner reopens	"Period reopened."	E	Pass
ACC-024	P2	N	Lock month `2026-00`	Validation	E	Pass
ACC-025	P2	F	Lock current month, create invoice today	Refused	E	Pass
ACC-026	P2	F	Lock future month	Allowed; postings there refused	E	Pass
ACC-027	P2	F	Lock, then view reports for that month	Allowed	E	Pass
ACC-028	P2	F	Lock then submit draft dated in it	Refused; change date to open month → works	E	Pass
ACC-029	P2	S	Salesperson sets period	Refused	E	Pass
ACC-030	P1	D	Source uniqueness: post same source twice	"This source has already been posted."	E	Pass
ACC-031	P1	D	Every document submit yields exactly one journal	Checked across SAL/PUR/COL	E	Pass
ACC-032	P1	D	Posted journal immutable	No edit UI; API has no update	E	Pass
ACC-033	P1	D	Journal numbering sequential	J-2026-0001…	E	Pass
ACC-034	P2	D	Journal number gap after failed command	None (atomic)	E	Pass
ACC-035	P1	D	Receivable control = sum open customer balances	Reconciled	E	Pass
ACC-036	P1	D	Payable control = open supplier balances	Reconciled	E	Pass
ACC-037	P1	D	Inventory control = stock valuation	Reconciled	E	Pass
ACC-038	P1	D	Cash/bank control = account balances	Reconciled	E	Pass
ACC-039	P1	D	Commission payable = earned − paid	Reconciled	E	Pass
ACC-040	P1	D	Output tax = sum invoice tax − credit note tax	Reconciled	E	Pass
ACC-041	P2	D	Input tax = bills tax − returns	Reconciled	E	Pass
ACC-042	P2	D	Balance sheet: assets − liabilities − equity = 0	Exactly 0	E	Pass
ACC-043	P2	D	P&L net = revenue − COGS − expenses	Matches ledger	E	Pass
ACC-044	P2	D	Retained earnings / current-year profit on balance sheet	Equals P&L	E	Pass
ACC-045	P2	D	General ledger per account running balance	Correct	E	Pass
ACC-046	P2	F	Account names table complete (every code used has a name)	Yes	E	Pass
ACC-047	P2	F	Manual journal in locked period	Refused	E	Pass
ACC-048	P2	F	Manual journal to control accounts (1100/2000/1200)	Allowed? If allowed, reconciliation shows variance — assert design	E	Pass
ACC-049	P2	S	Auditor creates journal	Refused	E	Pass
ACC-050	P2	S	Accountant creates/reverses	Allowed	E	Pass
ACC-051	P2	S	Sales Manager sees Accounting	Hidden; journals not sent	E	Pass
ACC-052	P2	S	Branch-limited Owner sees journals	Not sent (company-wide) — per access rule	E	Pass
ACC-053	P2	F	Journal search by memo/number	Works	B	Fail
ACC-054	P2	F	Filter by date range	Works	B	Pass
ACC-055	P2	F	Journal print/export	CSV if present	E	Pass
ACC-056	P2	D	Rounding: 3 lines of 33.3333 vs 100	Journal balanced to 4 dp by allocation	E	Pass
ACC-057	P2	D	FX gain/loss postings balance	Yes	E	Pass
ACC-058	P2	D	Landed cost postings balance	Yes	E	Pass
ACC-059	P2	D	Payroll postings balance	Yes	E	Pass
ACC-060	P2	D	Bounce reversal exact	Yes	E	Pass
ACC-061	P2	D	Opening balance journal (3000) for opening stock	Yes	E	Pass
ACC-062	P2	D	Count correction to 5300	Yes	E	Pass
ACC-063	P2	D	Stock transfer no journal	Yes	E	Pass
ACC-064	P2	D	Cancelled draft no journal	Yes	E	Pass
ACC-065	P2	D	Quotation/order submit no journal	Yes	E	Pass
ACC-066	P2	D	Delivery journal COGS only	Yes	E	Pass
ACC-067	P2	D	Invoice from delivery no COGS	Yes	E	Pass
ACC-068	P2	D	Direct invoice COGS + revenue	Yes	E	Pass
ACC-069	P2	D	Credit note reverses proportionally	Yes	E	Pass
ACC-070	P2	D	Supplier return reverses	Yes	E	Pass
ACC-071	P2	D	Expense journal	Yes	E	Pass
ACC-072	P2	D	Transfer journal	Yes	E	Pass
ACC-073	P2	D	Receipt journal	Yes	E	Pass
ACC-074	P2	D	Cheque clear journal	Yes	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
ACC-075	P2	D	Commission accrual journal	Yes	E	Pass
ACC-076	P2	D	Allocation journal	Yes	E	Pass
ACC-077	P2	F	Period list shows locked months	Yes	E	Pass
ACC-078	P2	F	Lock then unlock then lock	State correct	E	Pass
ACC-079	P2	F	Audit entries for period changes	Yes	E	Pass
ACC-080	P1	D	Invariant sweep after full trading cycle (SAL-100, PUR-100, COL-100, COM-100, PAY-100)	All controls reconciled; trial balance balanced	E	Pass
ACC-081	P1	D	Invariant sweep after 100 random valid commands (fuzz)	Still balanced and reconciled	E	Pass
ACC-082	P1	D	Invariant sweep after retries/duplicates (idempotency keys reused)	No duplicate journals	E	Pass
ACC-083	P2	D	Journals dated out of order	Trial balance by date filter correct	E	Pass
ACC-084	P2	D	Year boundary: invoice 2026-12-31 and 2027-01-01	Numbering resets per year; P&L per year	E	Pass
ACC-085	P2	D	Database projection: journals table = snapshot journals	`private.sync_projection` consistent (count/sums)	E	Pass
ACC-086	P2	D	Projection rebuild (`p_rebuild=true`) idempotent	Same rows	E	Pass
ACC-087	P2	D	`commit_workspace` rejects unbalanced?	Snapshot-level: engine guarantees; DB CHECK not cross-line (documented)	E	Pass
ACC-088	P2	D	Snapshot revision conflict (expected revision stale)	Commit refused; client retries with fresh state	E	Pass
ACC-089	P2	F	Accounting hidden?	Not toggleable (always on)	E	Pass
ACC-090	P2	U	Journal detail on phone	Readable	B	Pass
ACC-091	P2	U	Debit/credit columns aligned, tabular numerals	Yes	B	Fail
ACC-092	P2	F	Export trial balance CSV	Balanced totals row	E	Pass
ACC-093	P2	E	Memo 2,000 chars	Saved/truncated consistently	E	Pass
ACC-094	P2	E	Journal with 200 lines	Posted	E	Pass
ACC-095	P2	X	100,000 journal lines	Trial balance < 3 s	E	Pass
ACC-096	P2	F	Reversal appears linked from original	Yes	E	Pass
ACC-097	P2	F	Attempt manual journal on 1100 to "fix" a customer balance	Allowed but reconciliation flags variance (assert)	E	Pass
ACC-098	P2	F	Period lock blocks cheque clear dated inside	Yes	E	Pass
ACC-099	P2	F	Period lock blocks payroll pay dated inside	Yes	E	Pass
ACC-100	P1	F	End to end: close month (lock), owner reopens for correction, reversal, re-lock; reports consistent	All pass	E	Pass

Reports and CSV export (RPT): 92 pass · 7 fail · 1 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
RPT-001	P1	F	Reports index lists all 11 reports	Yes	E	Pass
RPT-002	P1	F	Aged receivables default	Buckets current/1-30/31-60/61-90/90+ per customer; totals	E	Pass
RPT-003	P1	D	Aged total = receivable control	Equal	E	Pass
RPT-004	P2	F	Aged receivables as-of date in the past	Buckets recomputed for that date; receipts after date excluded	E	Fail
RPT-005	P2	E	As-of before any invoice	Empty with zero totals	E	Pass
RPT-006	P2	F	Walk-in customer in aging	One row "Walk-in customer"	E	Pass
RPT-007	P2	F	Archived customer with balance	Included	E	Pass
RPT-008	P1	F	Customer statement for a customer	Opening, each invoice/credit/receipt, running balance, closing	E	Pass
RPT-009	P1	D	Closing balance = customer outstanding	Equal	E	Pass
RPT-010	P2	F	Statement date range	Opening balance = prior activity	E	Pass
RPT-011	P2	F	Statement print	Branded; company header; period	B	Fail
RPT-012	P2	F	Supplier statement	Bills, returns, payments	E	Fail
RPT-013	P2	N	Statement without party selected	Prompt to choose; no crash	E	Pass
RPT-014	P1	F	Sales analysis by salesperson	Net, COGS, margin, commission per person	E	Pass
RPT-015	P2	F	Sales analysis by customer	Walk-in grouped under walk-in	E	Pass
RPT-016	P2	F	Sales analysis by item	Qty, net, margin	E	Pass
RPT-017	P2	F	Sales analysis date range	Filters by invoice date	E	Pass
RPT-018	P2	D	Sales analysis total = revenue in P&L for the range	Equal (net of credit notes)	E	Pass
RPT-019	P2	F	Aged payables	Buckets by bill due	E	Pass
RPT-020	P2	D	Aged payables total = payable control	Equal	E	Pass
RPT-021	P1	F	Stock valuation	Per item/warehouse qty, avg, value; total	E	Pass
RPT-022	P1	D	Valuation total = inventory control	Equal	E	Pass
RPT-023	P2	F	Stock movements with date range and item filter	Correct rows	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
RPT-024	P1	F	Profit & loss for a month	Revenue, COGS, gross profit, expenses, net	E	Pass
RPT-025	P2	F	P&L for a year	Sums months	E	Pass
RPT-026	P2	F	P&L empty range	Zeros	E	Pass
RPT-027	P1	F	Balance sheet as of date	Assets, liabilities, equity; difference 0	E	Pass
RPT-028	P2	F	Balance sheet earlier date	Excludes later postings	E	Pass
RPT-029	P1	F	Trial balance	Every account with balance; totals equal	E	Pass
RPT-030	P2	F	General ledger per account with running balance	Correct	E	Pass
RPT-031	P1	F	Control reconciliation	Each control "reconciled" with ledger vs subledger figures	E	Pass
RPT-032	P1	D	Introduce a manual journal to 1100	Reconciliation shows variance (not reconciled)	E	Pass
RPT-033	P1	F	Export CSV on every report	File downloads; headers; rows match screen	E	Pass
RPT-034	P2	E	Export with 10,000 rows	Completes; file size sane	E	Pass
RPT-035	P2	E	Export values with commas (Rs 1,234)	Raw numbers in CSV, not formatted strings (assert)	E	Pass
RPT-036	P2	E	Export Unicode names	UTF-8	E	Pass
RPT-037	P2	E	Export empty report	Headers only	E	Pass
RPT-038	P2	F	Date pickers default to current month	Yes	E	Pass
RPT-039	P2	N	From date after to date	Swapped or validation; no crash	E	Pass
RPT-040	P2	E	Date far future	Empty	E	Pass
RPT-041	P2	S	Salesperson opens Reports	Limited set or hidden; no ledger reports	E	Pass
RPT-042	P2	S	Sales Manager sees sales reports, not P&L	Per role	E	Pass
RPT-043	P2	S	Auditor sees all	Yes	E	Pass
RPT-044	P2	S	Branch-limited Accountant: P&L	Not available or marked company-wide (journals not sent)	E	Fail
RPT-045	P2	F	Branch filter on operational reports	Works for stock/sales	E	Pass
RPT-046	P2	F	Report link from Overview "Needs attention"	Opens correct report	B	Pass
RPT-047	P2	U	Tables on phone	Horizontal scroll; totals visible	B	Pass
RPT-048	P2	U	Print a report	Clean print CSS	B	Fail
RPT-049	P2	F	Reports hidden module	Sidebar gone; Overview unaffected	B	Pass
RPT-050	P2	D	Aging after partial receipt	Remaining in original bucket	E	Pass
RPT-051	P2	D	Aging with credit note	Reduced	E	Pass
RPT-052	P2	D	Aging with overpayment (customer credit)	Negative shown in current or separate	E	Pass
RPT-053	P2	D	Statement with bounced cheque	Receipt and reversal lines	E	Fail
RPT-054	P2	D	Statement ordering by date then number	Stable	E	Pass
RPT-055	P2	D	Sales analysis margin after landed cost	Uses COGS actually posted	E	Pass
RPT-056	P2	D	Sales analysis excludes drafts/cancelled	Yes	E	Pass
RPT-057	P2	D	Stock movements sum = valuation delta	Yes	E	Pass
RPT-058	P2	D	P&L includes expense classification from imports	Yes	E	Pass
RPT-059	P2	D	P&L includes payroll	Yes	E	Pass
RPT-060	P2	D	P&L FX gains/losses	Yes	E	Pass
RPT-061	P2	D	Balance sheet includes cheques in hand (if asset)	Per design	E	Pass
RPT-062	P2	D	Balance sheet deposits (imports)	Yes	E	Pass
RPT-063	P2	D	Trial balance excludes zero accounts?	Consistent option	E	Pass
RPT-064	P2	D	GL running balance across date filter	Opening balance line	E	Pass
RPT-065	P2	F	Reconciliation after company reset	All zero reconciled	E	Pass
RPT-066	P2	F	Report parameters persist while switching tabs	Acceptable either way; no crash	B	Pass
RPT-067	P2	F	Reports keyboard accessible	Yes	B	Pass
RPT-068	P2	X	P&L on 100k journals	< 3 s	E	Pass
RPT-069	P2	X	Aging with 10k customers	< 3 s	E	Pass
RPT-070	P2	F	Customer statement for customer with no activity	"No activity"	E	Pass
RPT-071	P2	F	Supplier statement for walk-in supplier	Grouped	E	Pass
RPT-072	P2	F	Sales analysis by salesperson with splits	Each gets share	E	Fail
RPT-073	P2	F	Sales analysis "no salesperson" bucket	Present	E	Pass
RPT-074	P2	F	Export file name includes report and date	Yes	E	Pass
RPT-075	P2	F	Export CSV quoting of notes with newlines	Correct	E	Pass
RPT-076	P2	F	Report headings and currency label PKR	Yes	E	Pass
RPT-077	P2	F	Number formatting with tabular numerals	Yes	B	Pass
RPT-078	P2	F	Negative numbers formatting	`−` or `(…)` consistent	E	Pass
RPT-079	P2	F	Totals row sticky/bold	Yes	B	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
RPT-080	P2	F	Statement period with locked months	Works	E	Pass
RPT-081	P2	D	Aging vs statement consistency per customer	Equal closing	E	Pass
RPT-082	P2	D	Trial balance vs balance sheet + P&L	Consistent	E	Pass
RPT-083	P2	D	Reconciliation after bounce sequence	Reconciled	E	Pass
RPT-084	P2	D	Reconciliation after period reopen and reversal	Reconciled	E	Pass
RPT-085	P2	F	Reports in demo company	Match seeded data	E	Pass
RPT-086	P2	F	Reports with branch picker (2 branches)	Operational reports filter; ledger reports say company-wide	E	Pass
RPT-087	P2	S	Direct API export endpoint (if any) auth	401 signed out	E	Pass
RPT-088	P2	U	Empty states text	Helpful, no jargon	B	Pass
RPT-089	P2	U	Report selector on phone	Dropdown/tabs usable	B	Pass
RPT-090	P2	F	Deep link to a report (if routed)	Opens	B	Pass
RPT-091	P2	F	Stock valuation includes archived items with stock	Yes	E	Pass
RPT-092	P2	F	Stock valuation excludes zero-stock items?	Consistent toggle	E	Pass
RPT-093	P2	F	Aged payables for USD bills	Shown in PKR at bill rate; currency column	E	Pass
RPT-094	P2	F	Statement for USD supplier	PKR amounts with FX lines	E	Pass
RPT-095	P2	F	Sales analysis by item with returns	Net of returns	E	Pass
RPT-096	P2	D	P&L month boundaries use Asia/Karachi dates	Invoice dated 2026-10-31 stays in October	E	Pass
RPT-097	P2	F	Report refresh after a new command in another tab	Reflects on next load	B	Pass
RPT-098	P2	F	CSV opens in Excel with correct columns	Yes	—	Manual
RPT-099	P2	F	Reports accessible screen-reader table headers	`th` scope	E	Pass
RPT-100	P1	F	End to end: after full trading cycle, every report reconciles to the ledger and to each other	All pass	E	Pass

Printing, PDF and document view (PRN): 85 pass · 13 fail · 2 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PRN-001	P1	F	Print a submitted sales invoice	Header company name/address/phone/email, "SALES INVOICE", number, date, due	E	Pass
PRN-002	P1	F	Logo uploaded	Logo at top-left, sized ≤ 80px high, aspect kept	E	Pass
PRN-003	P1	F	No logo	Company initial tile in accent colour	E	Pass
PRN-004	P2	F	Letterhead uploaded	Replaces whole header; no duplicate company name	E	Pass
PRN-005	P1	F	Accent colour applied to heading and table header	Yes	E	Pass
PRN-006	P1	F	Footer text printed	Yes; multi-line kept	E	Pass
PRN-007	P1	F	NTN / STRN line when set	"NTN / STRN: …"	E	Pass
PRN-008	P2	F	Bill-to block: customer name, city, email, tax id	Only present fields	E	Pass
PRN-009	P1	F	Walk-in invoice	Typed name; no address	E	Pass
PRN-010	P2	F	Terms line	"30 days" from customer terms; blank for walk-in? (assert)	E	Pass
PRN-011	P2	F	Salesperson line when set	Yes	E	Pass
PRN-012	P1	F	Lines: description, SKU small, qty + unit, rate, disc %, tax %, amount	Correct	E	Pass
PRN-013	P2	E	Unit blank	Qty only	E	Pass
PRN-014	P2	E	Unit "1000 pcs"	Printed verbatim (documented)	E	Pass
PRN-015	P2	E	Qty 0.5	"0.5 kg"	E	Pass
PRN-016	P2	E	Qty 1,234,567	Thousands separators	E	Fail
PRN-017	P1	F	Totals: subtotal, tax, total, paid, returns, balance due	Correct arithmetic	E	Pass
PRN-018	P2	F	Fully paid invoice	Balance due 0	E	Pass
PRN-019	P2	F	Overpaid/credit	"Customer credit" line	E	Pass
PRN-020	P2	F	Draft invoice print	"DRAFT" status visible	E	Fail
PRN-021	P2	F	Cancelled invoice print	"CANCELLED" visible	E	Fail
PRN-022	P2	F	Credit note print	Title "Credit note", reference to invoice	E	Pass
PRN-023	P2	F	Quotation print	Title, validity? no due; no payment lines	E	Pass
PRN-024	P2	F	Sales order print	Correct title	E	Pass
PRN-025	P2	F	Delivery note print	Quantities; prices per design	E	Pass
PRN-026	P2	F	Purchase order print	Currency USD and rate; totals in both	E	Fail
PRN-027	P2	F	GRN print	Received quantities	E	Pass
PRN-028	P2	F	Supplier bill print	Due by terms	E	Pass
PRN-029	P2	F	Supplier return print	Title	E	Pass
PRN-030	P2	F	Receipt print	"Receipt", "Received from", method, cheque no, allocations	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PRN-031	P2	F	Payment voucher print	"Payment voucher", "Paid to"	E	Pass
PRN-032	P2	F	Customer statement print	Period, running balance	E	Pass
PRN-033	P1	S	Print another tenant's document id	401 / not found	E	Pass
PRN-034	P1	S	Print while signed out	Login/401	E	Pass
PRN-035	P2	S	Salesperson prints another salesperson's invoice	Not found (scoped)	E	Pass
PRN-036	P2	S	Branch-limited user prints other branch document	Not found	E	Pass
PRN-037	P2	S	Logo URL signed and short-lived	Expired link fails cleanly	E	Pass
PRN-038	P2	F	"Print / Save as PDF" button	Opens browser print; button itself hidden in print output	B	Pass
PRN-039	P2	F	Print preview A4 portrait	Fits one page for 10 lines; page breaks for 60 lines without splitting a row	B	Fail
PRN-040	P2	F	Print preview Letter	Fits	B	Fail
PRN-041	P2	U	Print CSS: no sidebar, no top bar, no buttons	Correct	E	Pass
PRN-042	P2	U	Colours in print: accent preserved (print-color-adjust)	Yes	E	Pass
PRN-043	P2	U	Fonts: Geist loads in print; fallback OK offline	Yes	E	Pass
PRN-044	P2	E	100-line invoice	Paginates; totals on last page	B	Fail
PRN-045	P2	E	Very long product name (120 chars)	Wraps within column	B	Fail
PRN-046	P2	E	Notes 2,000 chars	Wraps; not clipped	B	Fail
PRN-047	P2	E	Footer 500 chars	Wraps	B	Pass
PRN-048	P2	E	Company name 120 chars	Wraps; header intact	B	Pass
PRN-049	P2	E	Urdu/Arabic script names	Renders right-to-left glyphs correctly	B	Pass
PRN-050	P2	E	Emoji in notes	Renders or boxes; no crash	E	Pass
PRN-051	P2	E	Logo 1×1 px	Scales; no layout break	B	Pass
PRN-052	P2	E	Logo very wide (2000×100)	Constrained	B	Pass
PRN-053	P2	E	Logo transparent PNG on accent background	Readable	B	Pass
PRN-054	P2	F	Print on phone (iOS Share → Print)	Layout fits; no horizontal clipping	—	Manual
PRN-055	P2	U	Document view in-app (details) matches print content	Same numbers	E	Pass
PRN-056	P2	F	Document chain shown in details (QUO → SO → DN → INV)	Links work	B	Fail
PRN-057	P2	F	Details: payments list with cheque status	Yes	B	Pass
PRN-058	P2	F	Details: returns list	Yes	B	Pass
PRN-059	P2	F	Details: Submit/Return/Convert/Print buttons by state and role	Correct	B	Pass
PRN-060	P2	F	Details for walk-in invoice	Typed name	B	Pass
PRN-061	P2	F	Details for cancelled draft	Reason in notes	B	Pass
PRN-062	P2	U	Details dialog on phone	Bottom sheet; scroll	B	Pass
PRN-063	P2	F	Print URL with wrong tenant param but your own doc	Doc looked up in the given tenant → not found; with correct tenant works	E	Pass
PRN-064	P2	F	Print URL without tenant param	Falls back to member's company; works if doc there	E	Pass
PRN-065	P2	E	Print id of a receipt via document route	Not found; receipt route separate	E	Pass
PRN-066	P2	E	Unknown kind `/print/foo/<id>`	404	E	Pass
PRN-067	P2	F	Print in a company still in setup	Allowed? assert; sample doc only	E	Pass
PRN-068	P2	F	Branding changed after invoice	Reprint shows new branding (not snapshotted) — documented	E	Pass
PRN-069	P2	F	Tax snapshot shows registered status at invoice time	Yes	E	Fail
PRN-070	P2	F	Amount in words (if present)	Correct Pakistani style	E	Pass
PRN-071	P2	F	"amounts in PKR" footer	Yes; USD docs state currency	E	Pass
PRN-072	P2	F	Rate column formatting `Rs 2,400` / `$10.00`	Correct	E	Fail
PRN-073	P2	F	Discount column shows `10%` only when non-zero	Yes	E	Pass
PRN-074	P2	F	Tax column `0%` vs blank	Consistent	E	Pass
PRN-075	P2	F	Save as PDF filename suggestion	Document number (browser default acceptable)	B	Pass
PRN-076	P2	X	Print page load < 1.5 s	Yes	B	Pass
PRN-077	P2	F	Print page has no console errors	Yes	B	Pass
PRN-078	P2	F	Print page title = document number	Yes	E	Pass
PRN-079	P2	F	Back from print returns to app state	Yes	B	Pass
PRN-080	P2	U	Table header repeats on each printed page	`thead` repeat	E	Pass
PRN-081	P2	U	Page margins 15–20 mm	Yes	E	Pass
PRN-082	P2	U	Dark mode OS	Print stays light	E	Pass
PRN-083	P2	F	Print via keyboard Ctrl/Cmd+P	Same output	B	Pass
PRN-084	P2	F	Print a document while its module is hidden	Allowed? assert (document kept)	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
PRN-085	P2	F	Statement print for walk-in customer	Grouped	E	Pass
PRN-086	P2	F	Receipt print for cheque bounced	Status bounced	E	Pass
PRN-087	P2	F	Two tabs printing different docs	Independent	B	Pass
PRN-088	P2	F	Print of invoice from delivery shows chain	Yes	E	Pass
PRN-089	P2	F	Print of partially returned invoice	Returns line	E	Pass
PRN-090	P2	F	Invoice number barcode/QR (if present)	Encodes number	E	Pass
PRN-091	P2	F	Address with newlines in header	Line breaks	E	Fail
PRN-092	P2	F	Email/phone separators "·"	Only between present values	E	Pass
PRN-093	P2	F	Letterhead + NTN	NTN still printed below letterhead or omitted consistently	E	Pass
PRN-094	P2	E	Document with 0 total	Prints "Rs 0"	E	Pass
PRN-095	P2	E	Document with negative balance due	Credit wording	E	Pass
PRN-096	P2	S	Print page does not expose other documents in HTML	Only this doc's data	E	Pass
PRN-097	P2	S	Print page XSS via notes `<img onerror>`	Escaped	E	Pass
PRN-098	P2	F	Print page without JS	Static HTML renders	E	Pass
PRN-099	P2	F	Print page in Safari	Matches Chrome	—	Manual
PRN-100	P1	F	End to end: branded invoice with logo, NTN, footer, walk-in name, partial payment and return prints correctly on A4	Pass	E	Pass

Shell: navigation, search, notifications, workspace switcher, Overview (NAV): 87 pass · 10 fail · 3 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
NAV-001	P1	F	Sidebar lists enabled modules in order with coloured icon tiles	Overview, Sales, Purchasing, Imports, Inventory, Collections, Cash & bank, Commissions (LIVE), Payroll, Accounting, Reports, Settings	E	Pass
NAV-002	P1	F	Click each item	View changes; breadcrumb "Workspace › {Module}"; URL unchanged (SPA state)	B	Pass
NAV-003	P1	F	Active item has lilac fill and violet bar	Yes	E	Pass
NAV-004	P2	F	Nav disabled until hydrated	Click before hydration not lost	E	Pass
NAV-005	P2	F	Keyboard: Tab through nav, Enter activates	Yes	B	Pass
NAV-006	P1	F	Role hides modules (Salesperson: no Purchasing, Cash & bank, Accounting, Payroll)	Per canView	E	Pass
NAV-007	P2	F	Hidden module (Settings) removes item	Yes	E	Pass
NAV-008	P1	F	Workspace switcher lists memberships with demo tag	Yes; current highlighted	B	Pass
NAV-009	P1	F	Switch company	Navigates with `?tenant=`; data of the other company	B	Pass
NAV-010	P2	F	Switcher with one company	Shows company; menu has only that	B	Pass
NAV-011	P2	S	Switcher never lists companies you are not a member of	Correct	E	Pass
NAV-012	P1	F	Top bar: date (Asia/Karachi), search, bell, avatar	Yes	E	Pass
NAV-013	P2	F	Date shows "Wed, 14 Oct" style	Yes; hidden on phone	E	Pass
NAV-014	P1	F	Search ⌘K / Ctrl+K opens	Instantly, no animation	B	Pass
NAV-015	P1	F	Search "INV-2026"	Documents listed; Enter opens the first	B	Fail
NAV-016	P2	F	Search customer name	Customers and their documents	B	Pass
NAV-017	P2	F	Search walk-in typed name	Document found	B	Pass
NAV-018	P2	F	Search product SKU	Product	B	Pass
NAV-019	P2	F	Search employee	Only for roles allowed	B	Pass
NAV-020	P2	F	Search respects module gating	Hidden module records absent	B	Pass
NAV-021	P2	F	Search respects role scoping (Salesperson sees own docs)	Yes	B	Pass
NAV-022	P2	E	Search empty string	Prompt text	B	Pass
NAV-023	P2	E	Search 2 chars	Results or "type more"	B	Pass
NAV-024	P2	E	Search with regex chars `(.*`	Literal; no crash	E	Pass
NAV-025	P2	E	Search Urdu	Matches	B	Pass
NAV-026	P2	F	Arrow keys move selection, Escape closes	Yes	B	Fail
NAV-027	P2	F	Result click navigates to module and opens record	Yes	B	Fail
NAV-028	P2	X	Search with 20k records	< 100 ms per keystroke	B	Fail
NAV-029	P1	F	Bell badge count = attention items	Yes	E	Pass
NAV-030	P2	F	Bell menu lists items grouped; click goes to module	Yes	B	Pass
NAV-031	P2	F	Attention: draft sales to submit, cheques due, low stock, follow-ups due, unallocated shipments, released commissions, overdue invoices/bills	Each appears when relevant	E	Pass
NAV-032	P2	F	Attention clears after action	Yes	E	Pass
NAV-033	P2	F	Attention respects role/module	Yes	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
NAV-034	P2	F	Avatar initials	From name	E	Pass
NAV-035	P2	F	Sign out from profile	Login	E	Pass
NAV-036	P1	F	Overview metrics: Total sales, Collections, Outstanding receivables, Inventory value	Correct values	E	Pass
NAV-037	P2	F	Metric cards stagger in	Yes; reduced motion no stagger	E	Fail
NAV-038	P2	F	Sales & collections chart last 6 months	Bars violet/cyan; legend	E	Pass
NAV-039	P2	F	Chart empty state	"Submit an invoice or clear a receipt to see activity here."	E	Pass
NAV-040	P2	F	Needs your attention card	Queues with counts; "Nothing needs attention."	E	Pass
NAV-041	P2	F	Recent sales table last 5	Walk-in names; View buttons	E	Pass
NAV-042	P2	F	Welcome strip copy	"A clear view of your day. Every figure comes from your posted records." (ledger roles) / role-limited copy	E	Pass
NAV-043	P2	F	No "Design partner", "Revision", "prototype" copy anywhere	Confirmed	E	Fail
NAV-044	P2	F	"All changes saved / Saved as you work" in sidebar	Yes	E	Pass
NAV-045	P2	F	Page heading description per module	Yes	E	Pass
NAV-046	P2	F	Header action button per module (New sales invoice, New purchase order, Record receipt)	Only with edit permission	E	Pass
NAV-047	P2	F	Branch picker only with ≥2 branches	Yes	E	Pass
NAV-048	P2	F	Branch picker filters views	Yes	B	Pass
NAV-049	P2	F	Notice bar success (neutral) vs error (red, role alert)	Yes	E	Pass
NAV-050	P2	F	Notice clears on next command	Yes	B	Pass
NAV-051	P1	F	Phone: hamburger opens drawer with scrim	Drawer slides; scrim blurred; close X	B	Pass
NAV-052	P2	F	Tap scrim closes	Yes	B	Pass
NAV-053	P2	F	Tap item closes drawer and navigates	Yes	B	Pass
NAV-054	P2	F	Escape closes drawer	Yes	B	Fail
NAV-055	P2	F	Drawer safe-area padding	Yes	E	Pass
NAV-056	P2	F	Toggle has aria-expanded and aria-controls	Yes; labels "Open navigation"/"Close navigation" unique	E	Pass
NAV-057	P2	F	Tabbed modules (Sales: Documents/Customers; Inventory: Stock/Products/Warehouses)	Tabs work; aria	B	Pass
NAV-058	P2	F	Tab state resets when switching module	Acceptable; no crash	B	Pass
NAV-059	P2	F	View transition fade/slide on module change	Yes	B	Pass
NAV-060	P2	F	Browser back/forward	No broken state (SPA; stays on app)	B	Pass
NAV-061	P2	F	Reload keeps company (via cookie) but returns to Overview	Yes	B	Pass
NAV-062	P2	F	Deep link `?tenant` reload	Same company	B	Pass
NAV-063	P2	F	Preview as role chip and Exit	Only demo + Owner	B	Pass
NAV-064	P2	F	Footer "SAHULAT ERP · Made for the way you do business · PKR · Asia/Karachi"	Present	E	Pass
NAV-065	P2	F	Modal auto-close only for follow-up and stock transfer	Other dialogs stay open on unrelated commands	E	Pass
NAV-066	P2	F	Two dialogs (editor + confirm) stacking	Focus trapped in top one	B	Pass
NAV-067	P2	F	Dialog close X labelled "Close dialog"	Yes	E	Pass
NAV-068	P2	F	Dialog Escape	Closes	B	Pass
NAV-069	P2	F	Dialog overlay click	Closes (or not for editor — consistent)	B	Pass
NAV-070	P2	X	First load TTI < 3 s on 4G	Yes	—	Manual
NAV-071	P2	X	Module switch < 100 ms	Yes	B	Pass
NAV-072	P2	F	Console free of errors on every module	Yes	B	Pass
NAV-073	P2	F	Hydration warnings none	Yes	B	Pass
NAV-074	P2	F	Workspace menu animation from trigger origin	Yes	B	Pass
NAV-075	P2	F	Hover motion only with mouse	Touch no stuck hover	—	Manual
NAV-076	P2	F	Attention item "Draft sales documents to submit" opens Sales filtered to drafts	Yes	B	Fail
NAV-077	P2	F	Metric subtext "From posted records"	Yes	E	Pass
NAV-078	P2	F	Overview for role-limited user says figures cover their scope	Yes	E	Pass
NAV-079	P2	F	Overview after company reset	All zero; welcome strip	E	Pass
NAV-080	P2	F	Search dialog on phone	Full-width; keyboard 16px input	B	Fail
NAV-081	P2	F	Bell on phone	Menu fits	B	Pass
NAV-082	P2	F	Long company name in switcher	Truncated with ellipsis	B	Fail
NAV-083	P2	F	Many companies (10) in switcher	Scrollable	B	Pass
NAV-084	P2	F	Sidebar scroll with all modules on small height	Scrolls; bottom profile visible	B	Pass
NAV-085	P2	F	Module label "Imports & costing" wraps on narrow sidebar	OK	B	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
NAV-086	P2	F	LIVE label on Commissions	Gradient pill	E	Pass
NAV-087	P2	S	Nav for Auditor: all read-only	No action buttons	E	Pass
NAV-088	P2	S	Salesperson header action "New sales invoice"	Present (can create)	E	Pass
NAV-089	P2	F	Period chip removed	Yes	E	Pass
NAV-090	P2	F	Breadcrumb "Workspace" not a link	Acceptable	E	Pass
NAV-091	P2	F	Title tag constant	"SahulatERP · Your business, connected"	E	Pass
NAV-092	P2	F	Theme-color meta	#f6f7fb	E	Pass
NAV-093	P2	F	Demo company tag in heading and switcher	"Demo workspace" only there	E	Pass
NAV-094	P2	F	Non-demo company has no tag	Correct	E	Pass
NAV-095	P2	F	Idle for 1 hour then act	Works or clean re-login	—	Manual
NAV-096	P2	F	Two tabs, command in one	Other tab updates on next command (revision) without data loss	B	Pass
NAV-097	P2	F	Offline then command	Error notice; retry works	B	Pass
NAV-098	P2	F	Dev compile race (500 HTML)	Client retries once; no duplicate	E	Pass
NAV-099	P2	F	Zoom 200% desktop	Layout usable; no overlap	B	Pass
NAV-100	P1	F	End to end tour (docs/DEMO-TOUR.md) at desktop and phone	Completes without console errors	B	Pass

Security, tenancy, roles, API (SEC): 81 pass · 8 fail · 11 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
SEC-001	P1	S	POST command for a tenant you are not a member of	401/403; no change	E	Pass
SEC-002	P1	S	POST with missing `x-tenant-id`	400	E	Pass
SEC-003	P1	S	POST with another member's tenant and your valid session	Refused	E	Pass
SEC-004	P1	S	Read workspace of another tenant via page `?tenant=`	Falls back; no data	E	Pass
SEC-005	P1	S	RLS: query `workspace_snapshots` with anon key and user JWT for other tenant	0 rows	E	Pass
SEC-006	P1	S	RLS: `memberships` lists only own	Yes	E	Pass
SEC-007	P1	S	RLS: `tenants` only member tenants	Yes	E	Pass
SEC-008	P1	S	Storage: list `branding/<other tenant>/` with user JWT	Denied	E	Pass
SEC-009	P1	S	Storage: download other tenant logo path	Denied	E	Pass
SEC-010	P1	S	Service-role key absent from client bundle	grep built JS: none	E	Pass
SEC-011	P1	S	`NEXT_PUBLIC_*` contains only URL and publishable key	Yes	E	Pass
SEC-012	P1	S	Commands never executed with service role before membership check	Code review + test: unauthenticated POST never touches DB	E	Pass
SEC-013	P1	S	Role gate: Auditor POST any mutating command	"Your role does not permit this action."	E	Pass
SEC-014	P1	S	Salesperson POST `save_item`	Refused	E	Pass
SEC-015	P1	S	Salesperson POST `create_document` purchase	"Your role cannot create purchase documents."	E	Pass
SEC-016	P1	S	Warehouse Manager POST `create_receipt`	Refused	E	Pass
SEC-017	P1	S	Recovery Officer POST supplier payment	"Recovery officers cannot pay suppliers."	E	Pass
SEC-018	P1	S	Payroll Administrator POST `submit_document`	Refused	E	Pass
SEC-019	P1	S	Sales Manager POST `set_period`	Refused	E	Pass
SEC-020	P1	S	Accountant POST `set_period` reopen	"Only the owner can reopen an accounting period."	E	Pass
SEC-021	P1	S	Module gate: Payroll off → `create_payroll`	400 module-off message	E	Pass
SEC-022	P1	S	Module gate: Sales off → `create_document` sales_invoice	Refused	E	Pass
SEC-023	P1	S	Module gate: Collections off → `update_cheque`	Refused	E	Pass
SEC-024	P1	S	Password gate: `set_modules` removing a module without password	401	E	Pass
SEC-025	P1	S	Password gate with wrong password	401 "Password incorrect."	E	Pass
SEC-026	P1	S	Password gate adding a module	No password needed	E	Pass
SEC-027	P2	S	Password verified against the caller's own account only	Another member's password fails	E	Pass
SEC-028	P1	S	Branch scoping: member with branchIds [A] POSTs for warehouse in B	"This warehouse is outside your permitted branches."	E	Pass
SEC-029	P1	S	Branch scoping on page load	Only branch A documents/stock returned	E	Pass
SEC-030	P1	S	Salesperson page load	Only own sales docs, customers, own commissions; employees salaries zero; no journals/audit/payroll	E	Fail
SEC-031	P1	S	Warehouse Manager page load	Warehouse docs, suppliers; no receipts/journals	E	Pass
SEC-032	P1	S	Recovery Officer page load	Sales docs, receipts, follow-ups; no stock moves	E	Pass
SEC-033	P1	S	Payroll Administrator page load	Payrolls, employees with salaries, sales invoices only	E	Pass
SEC-034	P1	S	Auditor page load	Everything read-only	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
SEC-035	P2	S	`bankAccounts` sent to all roles (names only)	Balances not derivable without journals	E	Pass
SEC-036	P1	S	Idempotency: same key twice	Second returns same result; no duplicate	E	Pass
SEC-037	P1	S	Idempotency key < 8 chars	Validation	E	Pass
SEC-038	P2	S	Idempotency key 101 chars	Validation	E	Pass
SEC-039	P2	S	Same key, different payload	Original result returned (or conflict) — assert	E	Pass
SEC-040	P2	S	Idempotency across tenants (same key)	Independent	E	Pass
SEC-041	P1	S	Revision conflict: two clients commit from same revision	One succeeds; other retried by client; final state has both	E	Pass
SEC-042	P2	S	`commit_workspace` with non-object data	Rejected by RPC	E	Pass
SEC-043	P2	S	Unknown command type	400 validation	E	Pass
SEC-044	P2	S	Payload 5 MB	413 or validation; no crash	E	Pass
SEC-045	P2	S	Malformed JSON	400	E	Pass
SEC-046	P2	S	Content-Type text/plain	400	E	Fail
SEC-047	P2	S	GET /api/workspace	405 or 401	E	Pass
SEC-048	P2	S	CSRF: cross-site POST with cookies	Blocked by SameSite / origin check (assert)	E	Pass
SEC-049	P2	S	CORS from another origin	No permissive headers	E	Pass
SEC-050	P2	S	Security headers (X-Frame-Options/CSP)	Present or documented	E	Pass
SEC-051	P2	S	XSS via every free-text field (name, notes, memo, footer, reason)	Escaped in lists, dialogs, print	E	Pass
SEC-052	P2	S	HTML in CSV import cells	Escaped	E	Pass
SEC-053	P2	S	CSV injection `=cmd()` in export	Prefixed/quoted on export (assert)	E	Fail
SEC-054	P2	S	Path traversal in branding upload	Ignored	E	Fail
SEC-055	P2	S	Upload executable renamed .png	Type sniffed/refused	E	Pass
SEC-056	P2	S	Logo signed URL reuse after expiry	Fails	—	Manual
SEC-057	P2	S	`/api/branding/logo?tenant=other`	401	E	Pass
SEC-058	P2	S	`/api/demo-reset` on non-demo	Refused	E	Pass
SEC-059	P2	S	`/api/demo-reset` as non-owner	Refused	E	Pass
SEC-060	P2	S	Rate limiting on commands (200/min)	Throttled gracefully or documented	E	Pass
SEC-061	P2	S	Session fixation: sign in doesn't reuse pre-login session id	Yes	—	Manual
SEC-062	P2	S	Logout invalidates refresh token	Yes	—	Manual
SEC-063	P2	S	Password reset token single-use	Yes	—	Manual
SEC-064	P2	S	Secrets in repo	None (`.env*` ignored; scan)	E	Pass
SEC-065	P2	S	Audit log cannot be edited via API	No command	E	Pass
SEC-066	P2	S	Audit records actor id, type, time, not payload secrets	Yes	E	Fail
SEC-067	P2	S	Error responses don't leak stack traces	Yes	E	Pass
SEC-068	P2	S	500 on engine exception returns generic message	And logs server-side	E	Fail
SEC-069	P2	S	Dependency audit `npm audit`	No high/critical	—	Manual
SEC-070	P2	S	Supabase RLS policies present on every table (migration review)	Yes	E	Pass
SEC-071	P2	S	`private.sync_projection` not callable by anon	Yes	E	Pass
SEC-072	P2	S	`commit_workspace` callable only with service role	Yes	E	Pass
SEC-073	P2	S	Demo reset guard trigger prevents tenant delete without flag	Yes	E	Pass
SEC-074	P2	S	Member removed → immediate effect on next request	Yes	E	Pass
SEC-075	P2	S	Role changed → next request uses new role	Yes	E	Pass
SEC-076	P2	S	View-as-role preview cannot elevate	Server uses real role	E	Pass
SEC-077	P2	S	Branch ids changed → scoping updates	Yes	E	Pass
SEC-078	P2	S	Document id of another tenant in `submit_document`	"Document not found."	E	Pass
SEC-079	P2	S	Party id of another tenant	Not found	E	Pass
SEC-080	P2	S	Enumerate ids by guessing (uuid)	Infeasible; still scoped	E	Pass
SEC-081	P2	S	Print route with other tenant id	Not found	E	Pass
SEC-082	P2	S	Scripts (`member:create` , `company:reset`) require env secrets	Fail without	E	Pass
SEC-083	P2	S	`company:reset` never touches demo or others	Verified by tenant filter	E	Pass
SEC-084	P2	S	`qa:tenant` only touches QA tenant id	Yes	E	Pass
SEC-085	P2	S	Vercel env has no service key exposed as NEXT_PUBLIC	Yes	—	Manual
SEC-086	P2	S	HTTPS only; HSTS	Yes	—	Manual
SEC-087	P2	S	Cookie secure flags in prod	Yes	—	Manual
SEC-088	P2	S	Clickjacking: app in iframe	Blocked or acceptable	E	Pass
SEC-089	P2	S	Open redirect in `next` param	Blocked	E	Fail

ID	P	T	CASE	EXPECTED	LAYER	RESULT
SEC-090	P2	S	Large number of memberships (100)	Page still loads	E	Pass
SEC-091	P2	S	Unicode normalisation attack on names (duplicate check)	Documented behaviour	E	Pass
SEC-092	P2	S	Command replay from old revision	Rejected/retried; no double posting	E	Pass
SEC-093	P2	S	Timeout mid-commit	Either committed or not; retry with key safe	E	Pass
SEC-094	P2	S	Server clock vs Asia/Karachi dates	`today()` uses Karachi	E	Pass
SEC-095	P2	S	Log review: no PII in logs	Yes	E	Pass
SEC-096	P2	S	Backup/restore of snapshot table	Restore reproduces projection via rebuild	—	Manual
SEC-097	P2	S	Penetration smoke with OWASP ZAP baseline	No high alerts	—	Manual
SEC-098	P2	S	Auth brute force	Supabase limits	—	Manual
SEC-099	P2	S	Role matrix document vs code	Matches `access.ts`	E	Pass
SEC-100	P1	S	End to end: eight roles each run a scripted set of allowed/denied commands	Matrix matches	E	Fail

Responsive, mobile, accessibility, motion (UIX): 71 pass · 14 fail · 15 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
UIX-001	P1	U	Every module at 390×844	No horizontal page scroll (`scrollWidth ≤ innerWidth`)	B	Fail
UIX-002	P1	U	Every module at 1440×900	Layout as designed	B	Pass
UIX-003	P2	U	768×1024 tablet	Sidebar visible or drawer; readable	B	Fail
UIX-004	P2	U	1920×1080	Content max width; no stretching	B	Pass
UIX-005	P2	U	320px width	Usable; wraps	B	Fail
UIX-006	P1	U	Inputs 16px on touch devices	iOS does not zoom on focus	E	Pass
UIX-007	P1	U	Tap targets ≥ 44px on touch	Buttons, nav, switches	E	Pass
UIX-008	P1	U	Dialogs on phone are bottom sheets with sticky actions	Yes	B	Pass
UIX-009	P2	U	Sheet content scrolls with overscroll contained	No page rubber-band	—	Manual
UIX-010	P2	U	Safe-area insets top/bottom (notch devices)	Padding applied	E	Pass
UIX-011	P2	U	Keyboard open on phone with sheet	Actions remain reachable (interactive-widget resizes)	—	Manual
UIX-012	P2	U	No tap highlight flash	Transparent	E	Pass
UIX-013	P2	U	Press feedback on buttons (opacity/scale)	Yes	E	Pass
UIX-014	P2	U	Long-press on button doesn't select text	user-select none on controls	E	Pass
UIX-015	P2	U	Body text selectable	Yes	E	Pass
UIX-016	P2	U	Hover styles not stuck after tap	Gated by hover media	E	Pass
UIX-017	P2	U	Theme-color matches page background	Yes	E	Pass
UIX-018	P2	U	100dvh app shell	No cut-off with browser bars	E	Pass
UIX-019	P2	U	Landscape phone	Usable	—	Manual
UIX-020	P2	U	Pinch zoom allowed	Yes (no user-scalable=no)	E	Pass
UIX-021	P1	U	Contrast: body text ≥ 4.5:1	Graphite on white/grey passes	E	Pass
UIX-022	P1	U	Contrast: muted/subtle text ≥ 4.5:1	#646464 passes	E	Pass
UIX-023	P1	U	Contrast: violet buttons white text ≥ 4.5:1	#6b3df5 passes	E	Pass
UIX-024	P2	U	Contrast: badges (green/amber/red/violet/cyan on pastels) ≥ 4.5:1	Each passes	E	Fail
UIX-025	P2	U	Contrast: links violet on white	Passes	E	Pass
UIX-026	P2	U	Contrast: placeholder text ≥ 3:1	Passes or documented	E	Pass
UIX-027	P2	U	Contrast: setup footer white on tinted scrim	Passes	E	Fail
UIX-028	P1	U	Keyboard: full app operable without mouse	Yes	B	Pass
UIX-029	P1	U	Focus visible everywhere (violet ring)	Yes	B	Fail
UIX-030	P2	U	Focus trapped in dialogs; returns to trigger on close	Yes	B	Fail
UIX-031	P2	U	Skip to content link (if present)	Works	E	Pass
UIX-032	P2	U	Landmarks: nav, main, header	Present	E	Pass
UIX-033	P2	U	Headings order h1→h2→h3	No skips	E	Pass
UIX-034	P2	U	All inputs labelled (aria-label or label)	axe clean	B	Pass
UIX-035	P2	U	Icon-only buttons have aria-label	Close, menu, search	E	Pass
UIX-036	P2	U	Switches `role=switch` + checked state	Yes	E	Pass
UIX-037	P2	U	Tabs `role=tablist/tab` + aria-selected	Yes	E	Pass
UIX-038	P2	U	Dialogs `role=dialog` + aria-labelledby	Yes (setup card too)	E	Pass
UIX-039	P2	U	Alerts `role=alert` for errors, `status` for notices	Yes	E	Pass
UIX-040	P2	U	Tables have `th` headers	Yes	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
UIX-041	P2	U	Images alt text (logo "{company} logo")	Yes	E	Pass
UIX-042	P2	U	Decorative icons aria-hidden	Yes	E	Pass
UIX-043	P2	U	Screen reader reads setup steps and progress	Current step announced	—	Manual
UIX-044	P2	U	Provisioning screen `aria-live=polite`	Announced	E	Pass
UIX-045	P2	U	Reduced motion: no slide/stagger; fades only	Yes	E	Pass
UIX-046	P2	U	Reduced motion: provisioning ring slow	Yes	E	Pass
UIX-047	P2	U	Animations ≤ 320ms UI, ease-out	Yes	E	Pass
UIX-048	P2	U	No `transition: all`	grep clean	E	Pass
UIX-049	P2	U	No layout-animating properties (width/height) except accordions	grep clean	E	Pass
UIX-050	P2	U	Dialog enter/exit symmetric	Yes	E	Pass
UIX-051	P2	U	Drawer uses drawer easing	Yes	E	Pass
UIX-052	P2	U	Search dialog no animation	Yes	E	Pass
UIX-053	P2	U	Metric stagger 40ms	Yes	E	Fail
UIX-054	P2	U	Hover lift only pointer:fine	Yes	E	Pass
UIX-055	P2	U	Fonts: Geist loads; fallback system sans	No FOIT beyond swap	E	Pass
UIX-056	P2	U	Tabular numerals in tables	Yes	E	Pass
UIX-057	P2	U	Money format `Rs 1,234` consistent	Everywhere	E	Pass
UIX-058	P2	U	Dates `YYYY-MM-DD` in tables, friendly in top bar	Consistent	B	Pass
UIX-059	P2	U	Empty states have a next action	Yes	B	Fail
UIX-060	P2	U	Error copy plain English, names the fix	Yes	B	Pass
UIX-061	P2	U	No raw hex in components (tokens only)	`validate-tokens` clean	E	Pass
UIX-062	P2	U	Single radius scale 6/10/14/20	No stray radii	B	Fail
UIX-063	P2	U	Shadows indigo-tinted	Yes	E	Pass
UIX-064	P2	U	Sidebar tiles colours per module consistent with picker	Yes	E	Pass
UIX-065	P2	U	Gradient only on brand mark, welcome strip border, LIVE, final setup button, provisioning ring/progress	No gradient on data surfaces	E	Pass
UIX-066	P2	U	Badge pill style	Yes	E	Pass
UIX-067	P2	U	Button heights 38/30	Yes	E	Pass
UIX-068	P2	U	Input height 40	Yes	E	Pass
UIX-069	P2	U	Card hairline borders	Yes	E	Pass
UIX-070	P2	U	Top bar blur	Yes	E	Pass
UIX-071	P2	U	Zoom 200% desktop	No overlap	B	Pass
UIX-072	P2	U	Windows high-contrast mode	Borders visible	—	Manual
UIX-073	P2	U	Safari desktop rendering	Matches Chrome	—	Manual
UIX-074	P2	U	Firefox rendering	Matches	—	Manual
UIX-075	P2	U	iOS Safari bottom bar overlap	Sticky actions above bar	—	Manual
UIX-076	P2	U	Android Chrome keyboard resize	Layout resizes	—	Manual
UIX-077	P2	U	Datalist on iOS	Suggestions or free text OK	—	Manual
UIX-078	P2	U	Date input on iOS	Native picker	—	Manual
UIX-079	P2	U	Number inputs show numeric keyboard	`type=number`	E	Pass
UIX-080	P2	U	Select elements styled consistently	Yes	B	Pass
UIX-081	P2	U	Scrollbars styled (desktop)	Thin	E	Pass
UIX-082	P2	U	Table scroll hint text on phone	"Swipe sideways to see every column."	E	Pass
UIX-083	P2	U	Long names truncate with ellipsis where fixed width	Yes	B	Fail
UIX-084	P2	U	Urdu text rendering	Correct font fallback	B	Pass
UIX-085	P2	U	RTL text in LTR UI	No layout break	B	Pass
UIX-086	P2	U	Print styles don't affect screen	Yes	E	Pass
UIX-087	P2	U	Loading spinners on busy buttons only	Yes	B	Pass
UIX-088	P2	U	Skeleton shimmer where used	Reduced-motion static	E	Pass
UIX-089	P2	U	Toast/notice readable on phone	Yes	B	Fail
UIX-090	P2	U	Lighthouse accessibility ≥ 95 on login, overview, sales	Yes	—	Manual
UIX-091	P2	U	Lighthouse performance ≥ 80 mobile	Yes	—	Manual
UIX-092	P2	U	axe-core no serious violations on every module	Yes	B	Fail
UIX-093	P2	U	Colour-blind simulation: status not colour-only	Text labels present	E	Pass
UIX-094	P2	U	Voice control names match visible labels	Yes	—	Manual
UIX-095	P2	U	Dialog title ids unique with nested dialogs	Yes	B	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
UIX-096	P2	U	`inert` backdrop during setup	Not focusable	E	Pass
UIX-097	P2	U	Preset pills wrap on phone	Descriptions hidden	B	Pass
UIX-098	P2	U	Module tiles single column on phone	Yes	B	Pass
UIX-099	P2	U	Setup card full-screen on phone	Yes	B	Fail
UIX-100	P1	U	Full tour on iPhone hardware	No blockers	—	Manual

Data integrity, concurrency, failure recovery, migrations (DAT): 92 pass · 5 fail · 3 manual

ID	P	T	CASE	EXPECTED	LAYER	RESULT
DAT-001	P1	D	Money stored as 4-dp decimal strings everywhere	No floats in snapshot	E	Pass
DAT-002	P1	D	Sum of 0.1 ten times	Exactly 1.0000	E	Pass
DAT-003	P1	D	Allocation of 100 across 3 lines	33.3334/33.3333/33.3333 (last-line remainder); sum exact	E	Pass
DAT-004	P1	D	Quantity precision 4 dp	0.00005 rounds consistently	E	Fail
DAT-005	P1	D	FX precision 4 dp	280.12345 → 280.1235 (assert rounding mode)	E	Pass
DAT-006	P1	D	Tax rounding per line then summed	Totals consistent with print	E	Pass
DAT-007	P2	D	Negative zero never stored	"-0.0000" absent	E	Pass
DAT-008	P2	D	Very large values (1e11)	Stored; one-trillion guard	E	Pass
DAT-009	P1	D	Idempotency: retry create_document same key	One document	E	Pass
DAT-010	P1	D	Retry submit same key	One journal/stock move	E	Pass
DAT-011	P1	D	Retry with network error after server commit	Client retry returns original result	E	Fail
DAT-012	P1	D	Two users commit concurrently	Revision conflict handled; both applied in sequence	E	Pass
DAT-013	P1	D	10 concurrent invoice creations	10 distinct numbers; no gaps	E	Pass
DAT-014	P1	D	Concurrent receipt and return on same invoice	Outstanding never negative; one refused if exceeding	E	Pass
DAT-015	P1	D	Concurrent stock out from two tabs	Second refused if insufficient	E	Pass
DAT-016	P2	D	Concurrent set_stock_quantity with stale expected	"Stock changed since you opened this item…"	E	Pass
DAT-017	P2	D	Concurrent module toggle and command in gated module	Deterministic: refused or applied by order	E	Pass
DAT-018	P1	D	commit_workspace rejects stale `p_expected`	Yes	E	Pass
DAT-019	P1	D	commit_workspace is one transaction (snapshot + projection)	Failure in projection rolls back snapshot	E	Pass
DAT-020	P1	D	Projection tables after every command equal snapshot (counts, sums)	Verified by SQL	E	Pass
DAT-021	P2	D	`p_rebuild=true` after manual projection corruption	Restored	E	Pass
DAT-022	P2	D	Snapshot stored as JSON string (legacy)	Migration normalises; RPC rejects non-object	E	Pass
DAT-023	P2	D	Migration dry-run on hosted copy	Succeeds	—	Manual
DAT-024	P2	D	Migrations idempotent (run twice)	No error	E	Pass
DAT-025	P2	D	Codec defaults for old snapshots (no modules/setup/branding/bankAccounts)	Filled; setup.complete = demo or hasData	E	Pass
DAT-026	P2	D	Snapshot with unknown extra keys	Preserved or dropped consistently	E	Pass
DAT-027	P2	D	Snapshot schema_version bump path	Documented	E	Pass
DAT-028	P2	D	Backup/restore of one tenant	Restored tenant reconciles	E	Pass
DAT-029	P2	D	Restore then rebuild projection	Equal	E	Pass
DAT-030	P1	D	Invariant sweep after restore	Balanced and reconciled	E	Pass
DAT-031	P2	D	Audit entries append-only; count equals commands	Yes	E	Pass
DAT-032	P2	D	Audit includes failed commands?	Documented (no)	E	Pass
DAT-033	P2	D	Revision monotonic +1 per commit	Yes	E	Pass
DAT-034	P2	D	Timestamps ISO with timezone	Yes	E	Pass
DAT-035	P2	D	`today()` in Asia/Karachi at 23:30 local vs UTC	Karachi date used	E	Pass
DAT-036	P2	D	Document dates stored as YYYY-MM-DD	Yes	E	Pass
DAT-037	P2	D	Numbering per year per prefix	Resets 1 Jan	E	Pass
DAT-038	P2	D	Numbering with cancelled docs	Number kept; no reuse	E	Pass
DAT-039	P2	D	Numbering width 4 then 10,000th doc	5 digits; still sorts	E	Pass
DAT-040	P2	D	IDs unique (uuid)	No collisions in 100k	E	Pass
DAT-041	P2	D	Party rename does not alter document references	ids stable	E	Pass
DAT-042	P2	D	Item rename	Lines keep itemId; print shows new name (document)	E	Pass
DAT-043	P2	D	Warehouse rename	Stable	E	Pass
DAT-044	P2	D	Archived flags survive edits	Yes	E	Pass
DAT-045	P2	D	Hidden module data survives on/off cycles	Yes	E	Pass
DAT-046	P2	D	Company reset script leaves other tenants untouched	Verified counts	E	Pass
DAT-047	P2	D	QA tenant reset leaves others	Verified	E	Pass

ID	P	T	CASE	EXPECTED	LAYER	RESULT
DAT-048	P2	D	Demo reset restores exact seed (hash)	Yes	E	Pass
DAT-049	P2	D	Snapshot size growth 10k documents	< 20 MB; page load acceptable or documented limit	E	Pass
DAT-050	P2	X	Command latency with 10k docs	< 1 s	E	Pass
DAT-051	P2	X	Page load with 10k docs	< 4 s	B	Pass
DAT-052	P2	D	Engine purity: same input twice	Same output	E	Pass
DAT-053	P2	D	Engine never mutates input on failure	Workspace unchanged after thrown error	E	Pass
DAT-054	P2	D	Partial failure in multi-step UI flows (save_party ok, create_document fails)	Party exists; user told; no orphan document	E	Pass
DAT-055	P2	D	Product saved, opening stock fails	Product exists; message; stock 0	E	Pass
DAT-056	P2	D	Submit fails after create (one-step)	Draft kept; retry updates same draft	E	Pass
DAT-057	P2	D	Walk-in account creation then document fails	Walk-in account exists (acceptable); documented	E	Pass
DAT-058	P2	D	Browser closed mid-command	Server completes; reload shows result	B	Fail
DAT-059	P2	D	Server restart mid-commit	Transaction atomic	E	Pass
DAT-060	P2	D	DB unreachable	500 with generic message; retry later works	E	Fail
DAT-061	P2	D	Storage unreachable on logo upload	Error; branding unchanged	E	Pass
DAT-062	P2	D	Supabase auth slow	UI waits; no duplicate	—	Manual
DAT-063	P2	D	Clock skew client vs server	Dates from client date fields; server validates format only	E	Pass
DAT-064	P2	D	Period lock enforced server-side regardless of client date	Yes	E	Pass
DAT-065	P2	D	Role enforced server-side regardless of UI	Yes	E	Pass
DAT-066	P2	D	Module gate server-side	Yes	E	Pass
DAT-067	P2	D	Journal source uniqueness server-side	Yes	E	Pass
DAT-068	P2	D	Stock non-negative server-side	Yes	E	Pass
DAT-069	P2	D	Credit limit server-side	Yes	E	Pass
DAT-070	P2	D	Allocation sums server-side	Yes	E	Pass
DAT-071	P2	D	Validation errors return message and no change	Yes	E	Pass
DAT-072	P2	D	Unknown field in payload ignored	Yes	E	Pass
DAT-073	P2	D	Null vs empty string normalisation	Trimmed strings	E	Pass
DAT-074	P2	D	Unicode normalisation of names	NFC consistent (or documented)	E	Pass
DAT-075	P2	D	Whitespace-only inputs	Rejected where required	E	Fail
DAT-076	P2	D	Extremely long strings capped by schema	Yes	E	Pass
DAT-077	P2	D	Deep JSON nesting attack	Rejected by schema	E	Pass
DAT-078	P2	D	Prototype pollution keys `__proto__`	Ignored	E	Pass
DAT-079	P2	D	Duplicate lines in arrays caught (allocations, returns, lines)	Yes	E	Pass
DAT-080	P2	D	Order of lines preserved	Yes	E	Pass
DAT-081	P2	D	Snapshot JSON stable key order (diffability)	Not required; documented	E	Pass
DAT-082	P2	D	Projection `companies.setup_complete` mirrors snapshot	Yes	E	Pass
DAT-083	P2	D	Projection `enabled_modules` mirrors	Yes	E	Pass
DAT-084	P2	D	Projection journals balanced per journal (SQL check)	Yes	E	Pass
DAT-085	P2	D	Projection documents totals = snapshot	Yes	E	Pass
DAT-086	P2	D	Unit tests run against PGlite with real migrations	Pass	E	Pass
DAT-087	P2	D	pgcrypto available in test DB	Yes	E	Pass
DAT-088	P2	D	Playwright suites independent of order	Pass in any order	B	Pass
DAT-089	P2	D	Playwright global warm-up prevents compile 500s	Yes	B	Pass
DAT-090	P2	D	Client retries once on ≥500 with same key	Yes	E	Pass
DAT-091	P2	D	No retry on 4xx	Yes	E	Pass
DAT-092	P2	D	Error notice wording from server passes through	Yes	E	Pass
DAT-093	P2	D	Workspace in client refreshed from command result	Yes	E	Pass
DAT-094	P2	D	Stale tab commits from old revision	Conflict → retry with fresh workspace → applied	E	Pass
DAT-095	P2	D	1,000 commands sequentially	All applied; revision 1000	E	Pass
DAT-096	P2	D	Fuzz: 500 random valid commands	Invariants hold	E	Pass
DAT-097	P2	D	Fuzz: 500 random invalid commands	All refused; state unchanged	E	Pass
DAT-098	P2	D	Vercel deploy rollback	Previous build works with current snapshots (codec backward compatible)	—	Manual
DAT-099	P2	D	Forward compatibility: new field unknown to old client	Ignored safely	E	Pass
DAT-100	P1	D	Full invariant suite after everything in files 01–19	Balanced, reconciled, no duplicates, no negatives	E	Pass